MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3230 GRAVITY SEPARATION
DESIGN CALCULATION OF FOUNDATION

Document Number: **MD-216-3230-EG-CV-CAL-1167**

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1.0 INTENT

The Purpose of this document is to analysis & design of Gravity Seperation (3230) - Structure Foundation.
The dimensions and reinforcement are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah- Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3.0 SCOPE

The analysis has been carried out in STAAD Pro. RCC cloumn added below the structural STAAD model.
Stability check against Overturning & Sliding are done in the excel sheet. Base Pressure check has been done in the excel sheet
For strength calculation Bending moment & Shear Force values are taken from STAAD output & designed in the excel sheet.
Isometric views of STAAD model are attached in Appendix-B

4.0 DESIGN REQUIREMENTS

4.1 Reference Codes & Books

Reference is made in this standard as indicated in STRUCTURAL ENGINEERING DESIGN CRITERIA Doc No. : MD-216-1000-EG-ST-CRT-1001-A02 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

4.2 Materials Property

A. Density

4.2.B.(i) Density of Dry Soil : 1.9 T/m^3
4.2.B.(ii) Density of Concrete : 2.5 T/m^3
4.2.B.(iii) Density of Water : 1.0 T/m^3

B. Concrete Grade

4.2.B.(i) Foundation and Structural Members: 35 Mpa Strength of 28 days
4.2.B.(ii) Blinding Concrete below foundation bases: 15 Mpa Strength of 28 days

C. Reinforcement Grade

4.2.C.(i) Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
4.2.C.(ii) Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

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D. Grout Grade

4.2.D.(i) Grout shall be 30 mm cementitious of Strength 40 MPa.

4.3 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

3.4 Concrete Clear Cover

Concrete Exposure	Member	MinimumCove (mm)
Exposed to Soil	Foundation Bottom & Side	75
	Foundation Top	50
	Column	50

5.0 LOAD

5.1 Primary Loads

The primary load values are calculated in the excel sheet. The primary loads are mention as given below.

1 DEAD LOAD (D)
2 LIVE LOAD (L)
3 OPERATING FLUID LOAD (FL1)
4 OPERATING FLUID LOAD (FL2)
5 EQP DL (EQDL)
6 EQP LL (EQLL)
7 WIND LOAD (+X) WL(+X)
8 WIND LOAD (-X) WL(-X)
9 WIND LOAD (+Z) WL(+Z)
10 WIND LOAD (-Z) WL(-Z)
11 SEISMIC LOAD SL(+X)
12 SEISMIC LOAD SL(-X)
13 SEISMIC LOAD SL(+Z)
14 SEISMIC LOAD SL(-Z)
15 TEMPERATURE LOAD (TL)
16 PIPE LATERAL LOAD (PL)
17 EQP TEST LOAD (EQTL)

5.2 Load Combinations for Stress Calculation

LOAD CASE	LOAD COMBINATION
101	DL+EQDL+FL1
102	DL+EQDL+FL2
103	DL+EQDL+FL1+LL+EQLL+TL
104	DL+EQDL+FL2+LL+EQLL+TL
105	DL+EQDL+FL1+0.75(LL+EQLL)+0.75TL
106	DL+EQDL+FL2+0.75(LL+EQLL)+0.75TL
107	DL+EQDL+FL1+PL
108	DL+EQDL+FL2+PL
109	DL+EQDL+FL1+PL+LL+EQLL+TL
110	DL+EQDL+FL2+PL+LL+EQLL+TL
111	DL+EQDL+FL1+PL+0.75(LL+EQLL)+0.75TL
112	DL+EQDL+FL2+PL+0.75(LL+EQLL)+0.75TL
113	DL+EQDL+FL1+WL(+X)
114	DL+EQDL+FL1+WL(-X)
115	DL+EQDL+FL1+WL(+Z)
116	DL+EQDL+FL1+WL(-Z)

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LOAD CASE	LOAD COMBINATION
117	DL+EQDL+FL2+WL(+X)
118	DL+EQDL+FL2+WL(-X)
119	DL+EQDL+FL2+WL(+Z)
120	DL+EQDL+FL2+WL(-Z)
121	DL+EQDL+FL1+0.7SL(+X)
122	DL+EQDL+FL1+0.7SL(-X)
123	DL+EQDL+FL1+0.7SL(+Z)
124	DL+EQDL+FL1+0.7SL(-Z)
125	DL+EQDL+FL2+0.7SL(+X)
126	DL+EQDL+FL2+0.7SL(-X)
127	DL+EQDL+FL2+0.7SL(+Z)
128	DL+EQDL+FL2+0.7SL(-Z)
129	DL+EQDL+FL1+0.75WL(+X)+0.75(LL+EQLL)
130	DL+EQDL+FL1+0.75WL(-X)+0.75(LL+EQLL)
131	DL+EQDL+FL1+0.75WL(+Z)+0.75(LL+EQLL)
132	DL+EQDL+FL1+0.75WL(-Z)+0.75(LL+EQLL)
133	DL+EQDL+FL2+0.75WL(+X)+0.75(LL+EQLL)
134	DL+EQDL+FL2+0.75WL(-X)+0.75(LL+EQLL)
135	DL+EQDL+FL2+0.75WL(+Z)+0.75(LL+EQLL)
136	DL+EQDL+FL2+0.75WL(-Z)+0.75(LL+EQLL)
137	DL+EQDL+FL1+0.525SL(+X)+0.75(LL+EQLL)
138	DL+EQDL+FL1+0.525SL(-X)+0.75(LL+EQLL)
139	DL+EQDL+FL1+0.525SL(+Z)+0.75(LL+EQLL)
140	DL+EQDL+FL1+0.525SL(-Z)+0.75(LL+EQLL)
141	DL+EQDL+FL2+0.525SL(+X)+0.75(LL+EQLL)
142	DL+EQDL+FL2+0.525SL(-X)+0.75(LL+EQLL)
143	DL+EQDL+FL2+0.525SL(+Z)+0.75(LL+EQLL)
144	DL+EQDL+FL2+0.525SL(-Z)+0.75(LL+EQLL)
145	0.6(DL+EQDL)+WL(+X)
146	0.6(DL+EQDL)+WL(-X)
147	0.6(DL+EQDL)+WL(+Z)
148	0.6(DL+EQDL)+WL(-Z)
149	0.6(DL+EQDL)+0.7SL(+X)
150	0.6(DL+EQDL)+0.7SL(-X)
151	0.6(DL+EQDL)+0.7SL(+Z)
152	0.6(DL+EQDL)+0.7SL(-Z)
153	DL+EQTL

5.3 Load Combinations for Strength Calculation

LOAD CASE	LOAD COMBINATION
201	1.4(DL+EQDL+FL1)
202	1.4(DL+EQDL+FL2)
203	1.4(DL+EQDL+FL1)+1.7(LL+EQLL)
204	1.4(DL+EQDL+FL2)+1.7(LL+EQLL)
205	1.4(DL+EQDL+FL1+TL)+1.7(LL+EQLL)
206	1.4(DL+EQDL+FL2+TL)+1.7(LL+EQLL)
207	1.4(DL+EQDL+FL1+PL)
208	1.4(DL+EQDL+FL2+PL)
209	1.4(DL+EQDL+FL1+PL)+1.7(LL+EQLL)
210	1.4(DL+EQDL+FL2+PL)+1.7(LL+EQLL)
211	1.4(DL+EQDL+FL1+PL+TL)+1.7(LL+EQLL)
212	1.4(DL+EQDL+FL2+PL+TL)+1.7(LL+EQLL)
213	1.2(DL+EQDL)+0.8WL(+X)
214	1.2(DL+EQDL)+0.8WL(-X)
215	1.2(DL+EQDL)+0.8WL(+Z)

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LOAD CASE	LOAD COMBINATION
216	1.2(DL+EQDL)+0.8WL(-Z)
217	1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
218	1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
219	1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
220	1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
221	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL1
222	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL1
223	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL1
224	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL1
225	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL2
226	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL2
227	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL2
228	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL2
229	0.9(DL+EQDL)+1.6WL(+X)
230	0.9(DL+EQDL)+1.6WL(-X)
231	0.9(DL+EQDL)+1.6WL(+Z)
232	0.9(DL+EQDL)+1.6WL(-Z)
233	0.9(DL+EQDL)+1.6SL(+X)
234	0.9(DL+EQDL)+1.6SL(-X)
235	0.9(DL+EQDL)+1.6SL(+Z)
236	0.9(DL+EQDL)+1.6SL(-Z)
237	1.2DL+EQTL

6.0 SAFETY FACTOR AGAINST STABILITY

5.1 Overturning : For foundation design a minimum factor of safety against overturning is consider 2.0
5.2 Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

7.0 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design as obtained from page 39 of Doc. No. MD-216-0000-EM-GM-RPT-0003 is 670 KN/m2 at 1.2m depth below FGL
The net safe bearing capacity of soil as per approved L&T soil test report (Table-6 of Doc. No. MD-216-3000-EG-CV-RPT-1008-A02) is 980 KN/m2 for foundation size 2mX2m & 1120 KN/m2 for foundation size 3mX3m at 1.5m depth below FGL
However, for safe side we have considered Net Safe Bearing Capacity at founding level as **670 KN/m2**.

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Reaction Summary (FOUNDATION F1A)

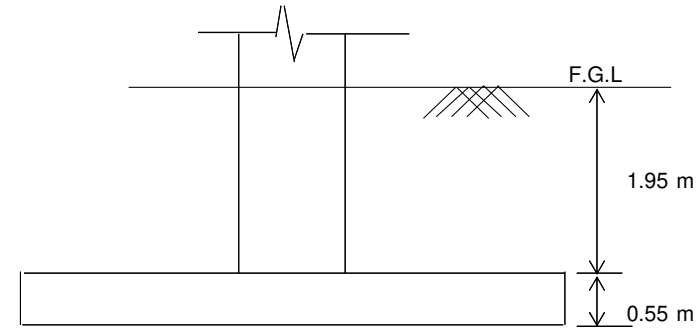
			Horizontal	Vertical	Horizontal	Moment		
	Node	L/C	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
Max FX	410	134:DL+EQDL-	**10.696**	60.138	1.408	4.434	0	-33.691
Min FX	411	129:DL+EQDL-	**-9.658**	100.613	-1.015	-3.197	0	30.422
Max FY	411	136:DL+EQDL-	-6.337	**113.595**	10.813	34.060	0	19.963
Min FY	410	147:0.6(DL+EC	0.514	**-8.578**	-10.140	-31.940	0	-1.620
Max FZ	411	120:DL+EQDL-	-3.819	78.426	**14.179**	44.665	0	12.031
Min FZ	411	115:DL+EQDL-	-2.139	29.752	**-14.510**	-45.707	0	6.739
Max MX	411	120:DL+EQDL-	-3.819	78.426	14.179	**44.665**	0	12.031
Min MX	411	115:DL+EQDL-	-2.139	29.752	-14.510	**-45.707**	0	6.739
Max MY	410	101:DL+EQDL-	2.970	23.567	-0.616	-1.939	**0**	-9.357
Min MY	410	101:DL+EQDL-	2.970	23.567	-0.616	-1.939	**0**	-9.357
Max MZ	411	129:DL+EQDL-	-9.658	100.613	-1.015	-3.197	0	**30.422**
Min MZ	410	134:DL+EQDL-	10.696	60.138	1.408	4.434	0	**-33.691**

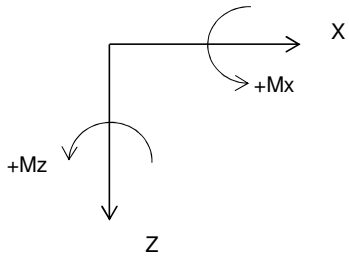
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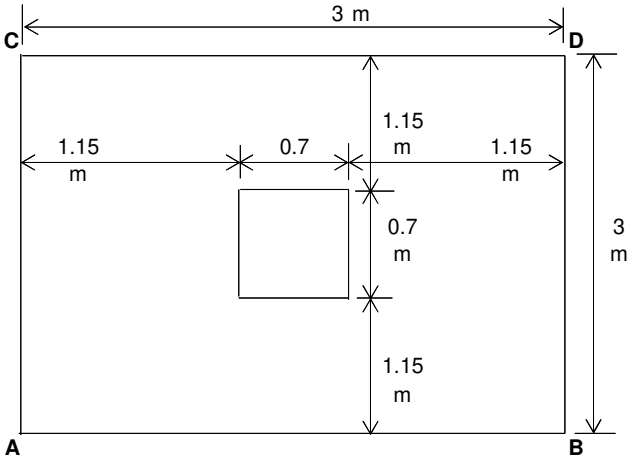
8.1 DESIGN OF FOUNDATION MKD. F1A :

(NODE NO. : 410 To 412)

Net Safe Bearing Capacity of Soil =	67 Mton/m^2	*(For BH-2)*
Depth of Foundation from FGL =	2.5 m	
Gross Allowable Bearing Capacity of Soil =	67 + 2.5 X 1.9 =	71.75 Mton/m^2
Compressive Strength of Concrete , fc' =	35 N/mm^2	
Yield Strength of Main Reinf. , fy =	420 N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420 N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	75 mm	
Density of Concrete =	2.5 Mton/m^3	
Density of Dry Soil =	1.9 Mton/m^3	
Density of Saturated Soil =	2 Mton/m^3	
Density of Water =	1 Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	3 m
Width of Foundation (Lz) =	3 m
Thickness of Base Raft =	0.55 m
Length of Column (lx) =	0.7 m
Width of Column (lz) =	0.7 m

Area , A = Lx*Lz
= 9 m^2
Zxx = Lx*Lz^2/6
= 4.5 m^3
Zzz = Lz*Lx^2/6
= 4.5 m^3

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WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	3X3X0.55X2.5 =	12.375
Column =	0.7X0.7X1.95X2.5 =	2.389
Soil =	((3X3)-(0.7X0.7))X1.95X1.9 =	31.53
	Total =	46.294

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

(Staad Group : F1A)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	410	134 DL+EC	**10.696**	60.138	1.408	4.434	0	-33.691
Min Fx	411	129 DL+EC	**-9.658**	100.613	-1.015	-3.197	0	30.422
Max Fy	411	136 DL+EC	-6.337	**113.595**	10.813	34.06	0	19.963
Min Fy	410	147 0.6(DL	0.514	**-8.578**	-10.14	-31.94	0	-1.62
Max Fz	411	120 DL+EC	-3.819	78.426	**14.179**	44.665	0	12.031
Min Fz	411	115 DL+EC	-2.139	29.752	**-14.51**	-45.707	0	6.739
Max Mx	411	120 DL+EC	-3.819	78.426	14.179	**44.665**	0	12.031
Min Mx	411	115 DL+EC	-2.139	29.752	-14.51	**-45.707**	0	6.739
Max My	410	101 DL+EC	2.97	23.567	-0.616	-1.939	**0**	-9.357
Min My	410	101 DL+EC	2.97	23.567	-0.616	-1.939	**0**	-9.357
Max Mz	411	129 DL+EC	-9.658	100.613	-1.015	-3.197	0	**30.422**
Min Mz	410	134 DL+EC	10.696	60.138	1.408	4.434	0	**-33.691**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
134	106.432	10.788	0.5	53.216	4.93	1.5	O.K.
129	146.907	9.711	0.5	73.454	7.56	1.5	O.K.
136	159.889	12.533	0.5	79.945	6.38	1.5	O.K.
147	37.716	10.153	0.5	18.858	1.86	1.5	O.K.
120	124.72	14.684	0.5	62.36	4.25	1.5	O.K.
115	76.046	14.667	0.5	38.023	2.59	1.5	O.K.
120	124.72	14.684	0.5	62.36	4.25	1.5	O.K.
115	76.046	14.667	0.5	38.023	2.59	1.5	O.K.
101	69.861	3.033	0.5	34.931	11.52	1.5	O.K.
101	69.861	3.033	0.5	34.931	11.52	1.5	O.K.
129	146.907	9.711	0.5	73.454	7.56	1.5	O.K.
134	106.432	10.788	0.5	53.216	4.93	1.5	O.K.

OVERTURNING CHECK:

Max Upword Force =	8.578 Mton			
Self Wt. of Foundation =	46.294 Mton			
Overturning Moment =	8.578 X 3/2 =	12.867 Mton-m		
Resisting Moment =	46.294 X 0.9 X 3/2 =	62.497 Mton-m		
F.O.S =	4.86	>.	2	O.K.

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zxx (Mton/m^2)	Mz/Zzz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D	Remarks
134	11.826	0.985	7.487	**20.298**	5.324	**18.328**	3.354	O.K.
129	16.323	0.71	6.76	**23.793**	**10.273**	**22.373**	8.853	O.K.
136	17.765	7.569	4.436	**29.77**	**20.898**	**14.632**	5.76	O.K.
147	4.191	7.098	0.36	**11.649**	**10.929**	-2.547	-3.267	O.K.
120	13.858	9.926	2.674	**26.458**	**21.11**	6.606	1.258	O.K.
115	8.45	10.157	1.498	**20.105**	**17.109**	-0.209	-3.205	O.K.
120	13.858	9.926	2.674	**26.458**	**21.11**	6.606	1.258	O.K.
115	8.45	10.157	1.498	**20.105**	**17.109**	-0.209	-3.205	O.K.
101	7.762	0.431	2.079	**10.272**	6.114	**9.41**	5.252	O.K.
101	7.762	0.431	2.079	**10.272**	6.114	**9.41**	5.252	O.K.
129	16.323	0.71	6.76	**23.793**	**10.273**	**22.373**	8.853	O.K.
134	11.826	0.985	7.487	**20.298**	5.324	**18.328**	3.354	O.K.
			MAX =	29.77	21.11	22.373	8.853	

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

(Staad Group : F1A)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	410	DL+FL2+TL	**15.161**	91.715	1.731	5.451	0	-47.758
Min Fx	411	+FL1+PL+T	**-14.308**	162.066	-3.746	-11.801	0	45.071
Max Fy	411	QDL+FL2)+	-10.216	**170.739**	1.007	3.173	0	32.182
Min Fy	410	L+EQDL)+	-2.951	**-15.747**	-9.77	-30.532	0	9.296
Max Fz	411	L+EQDL)+	-3.465	71.218	**21.307**	67.117	0	10.915
Min Fz	411	)+1.6WL(+	-2.712	38.049	**-21.826**	-68.752	0	8.542
Max Mx	411	L+EQDL)+	-3.465	71.218	21.307	**67.117**	0	10.915
Min Mx	411	)+1.6WL(+	-2.712	38.049	-21.826	**-68.752**	0	8.542
Max My	410	4(DL+EQDL	4.159	32.993	-0.862	-2.715	**0**	-13.1
Min My	410	4(DL+EQDL	4.159	32.993	-0.862	-2.715	**0**	-13.1
Max Mz	411	+FL1+PL+T	-14.308	162.066	-3.746	-11.801	0	**45.071**
Min Mz	410	DL+FL2+TL	15.161	91.715	1.731	5.451	0	**-47.758**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zx (Mton/m^2)	Mz/Zz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D
206	15.334	1.211	10.613	27.158	5.932	24.736	3.51
211	23.151	2.622	10.016	35.789	15.757	30.545	10.513
204	24.115	0.705	7.152	31.972	17.668	30.562	16.258
235	3.394	6.785	2.066	12.245	8.113	-1.325	-5.457
232	13.057	14.915	2.426	30.398	25.546	0.568	-4.284
219	9.371	15.278	1.898	26.547	22.751	-4.009	-7.805
232	13.057	14.915	2.426	30.398	25.546	0.568	-4.284
219	9.371	15.278	1.898	26.547	22.751	-4.009	-7.805
201	8.81	0.603	2.911	12.324	6.502	11.118	5.296
201	8.81	0.603	2.911	12.324	6.502	11.118	5.296
211	23.151	2.622	10.016	35.789	15.757	30.545	10.513
206	15.334	1.211	10.613	27.158	5.932	24.736	3.51

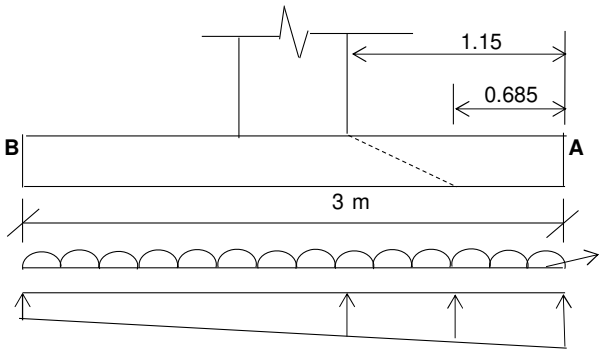
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PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point B (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
206	27.158	5.932	19.02	22.31	18.187	5.938
211	35.789	15.757	28.11	31.22	23.691	7.919
204	31.972	17.668	26.49	28.71	20.199	6.819
235	12.245	8.113	10.66	11.3	5.437	1.829
232	30.398	25.546	28.54	29.29	17.561	6.113
219	26.547	22.751	25.09	25.68	14.837	5.172
232	30.398	25.546	28.54	29.29	17.561	6.113
219	26.547	22.751	25.09	25.68	14.837	5.172
201	12.324	6.502	10.09	10.99	5.775	1.908
201	12.324	6.502	10.09	10.99	5.775	1.908
211	35.789	15.757	28.11	31.22	23.691	7.919
206	27.158	5.932	19.02	22.31	18.187	5.938
				MAX =	23.691	7.919

ALONG X - DIRECTION :

Mu = 23.691 Mton-m

Φ*Kn = Mu/bd^2 deff. = 465 mm

Mu/bd^2 = 23.691X10^7/(1000X465^2) = 1.096 N/mm^2 < 6.9 N/mm^2

Pt = 0.0030

Min. Pt = 0.0018 Reqd. Ast = 0.003X1000X465/100 = 13.95 cm^2

Provided **20** mm @ **200** C/C Provide Ast = 15.71 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0034

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ONE-WAY SHEAR CHECK

Vu = 7.919 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0034X7.919X10^4X465/23.691/10^7)X1000X465/10^4

= 44.433 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0034)X1000X465/10^4

= 46.703 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X465/10^4

= 79.778 Mton

HENCE, Vc = 44.433 Mton

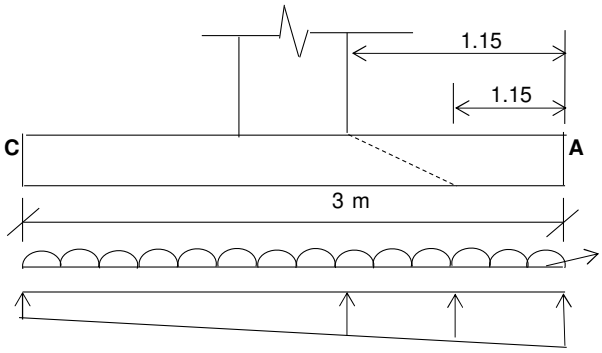
Φt*Vc = 0.75X44.433 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 33.325 Mton **> 7.919Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(44.433X10^4+0.66X35^0.5X1000X465)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 139.506 Mton **> Vu= 7.919Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point C (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
206	27.158	24.736	26.23	26.23	15.008	12.339
211	35.789	30.545	33.78	33.78	21.192	15.413
204	31.972	30.562	31.43	31.43	18.021	16.467
235	12.245	-1.325	7.04	7.04	7.032	-7.923
232	30.398	0.568	18.96	18.96	21.784	-11.091
219	26.547	-4.009	14.83	14.83	19.36	-14.315
232	30.398	0.568	18.96	18.96	21.784	-11.091
219	26.547	-4.009	14.83	14.83	19.36	-14.315
201	12.324	11.118	11.86	11.86	4.995	3.666
201	12.324	11.118	11.86	11.86	4.995	3.666
211	35.789	30.545	33.78	33.78	21.192	15.413
206	27.158	24.736	26.23	26.23	15.008	12.339
				MAX =	21.784	16.467

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ALONG Z - DIRECTION :

Mu = 21.784 T-m

Φ*Kn = Mu/bd^2 deff. = 465 mm

Mu/bd^2 = 21.784X10^7/(1000X465^2) = 1.007 N/mm^2 < 6.9 N/mm^2

Pt = 0.0027

Min. Pt = 0.0018 Reqd. Ast = 0.0027X1000X465/100 = 12.56 cm^2

Provided 20 mm @ 200 C/C Provide Ast = 15.71 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0034

ONE-WAY SHEAR CHECK

Vu = 16.467 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0034X16.467X10^4X465/21.784/10^7)X1000X465/10^4

= 44.96 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0034)X1000X465/10^4

= 46.703 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X465/10^4

= 79.778 Mton

HENCE, Vc = 44.96 Mton

Φt*Vc = 0.75X44.96 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 33.72 Mton **> 16.467Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(44.96X10^4+0.66X35^0.5X1000X465)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 139.545 Mton **> Vu= 16.467Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 9 m^2

Perimeter (bo) = 2 X ((0.7 + 0.465) + (0.7 + 0.465)) = 4.66 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 29.77 Mton/m^2

Vu = ((9 - (0.7X0.7))X29.77

= 253.343 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X4.66X10^3X465/10^4

= 423.045 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X4.66X10^3X465/10^4

= 653.797 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X465/(4.66X10^3))X1X35^0.5X(4.66X10^3)X465/10^4

= 531.326 Mton

HENCE, Vc = 423.045 Mton

Φt*Vn = Φt*Vc = 0.75X423.045 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

317.284 Mton **> 253.343Mton , Hence O.K.**

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Reaction Summary (FOUNDATION F1B)

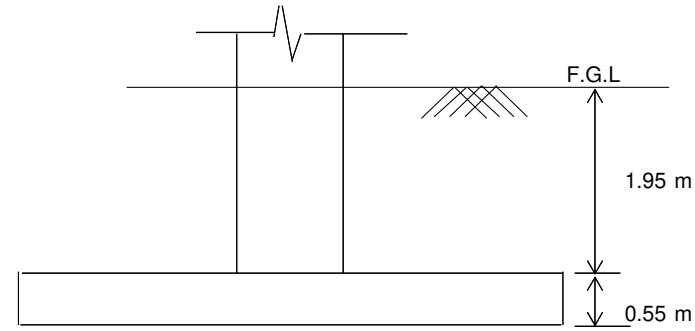
			Horizontal	Vertical	Horizontal	Moment		
	Node	L/C	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
Max FX	408	146:0.6(DL+EC	**3.331**	1.447	0.061	0.194	0	-10.492
Min FX	408	109:DL+EQDL	**-9.459**	94.280	0.190	0.599	0	29.797
Max FY	408	109:DL+EQDL	-9.459	**94.280**	0.190	0.599	0	29.797
Min FY	408	146:0.6(DL+EC	3.331	**1.447**	0.061	0.194	0	-10.492
Max FZ	408	120:DL+EQDL	-2.123	40.092	**0.957**	3.014	0	6.689
Min FZ	408	147:0.6(DL+EC	-0.782	12.013	**-0.717**	-2.259	0	2.465
Max MX	408	120:DL+EQDL	-2.123	40.092	0.957	**3.014**	0	6.689
Min MX	408	147:0.6(DL+EC	-0.782	12.013	-0.717	**-2.259**	0	2.465
Max MY	408	101:DL+EQDL	-3.108	42.637	0.157	0.494	**0**	9.790
Min MY	408	101:DL+EQDL	-3.108	42.637	0.157	0.494	**0**	9.790
Max MZ	408	109:DL+EQDL	-9.459	94.280	0.190	0.599	0	**29.797**
Min MZ	408	146:0.6(DL+EC	3.331	1.447	0.061	0.194	0	**-10.492**

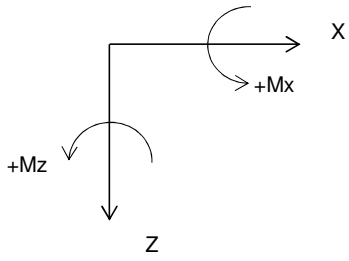
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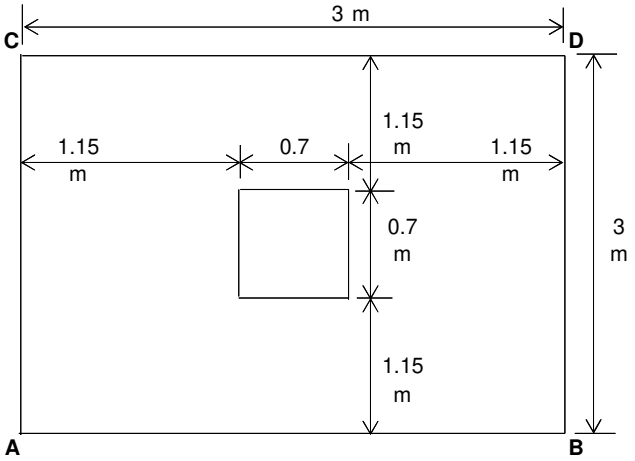
8.2 DESIGN OF FOUNDATION MKD. F1B :

(NODE NO. : 408)

Net Safe Bearing Capacity of Soil =	67 Mton/m^2	*(For BH-2)*
Depth of Foundation from FGL =	2.5 m	
Gross Allowable Bearing Capacity of Soil =	67 + 2.5 X 1.9 =	71.75 Mton/m^2
Compressive Strength of Concrete , fc' =	35 N/mm^2	
Yield Strength of Main Reinf. , fy =	420 N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420 N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	75 mm	
Density of Concrete =	2.5 Mton/m^3	
Density of Dry Soil =	1.9 Mton/m^3	
Density of Saturated Soil =	2 Mton/m^3	
Density of Water =	1 Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	3 m
Width of Foundation (Lz) =	3 m
Thickness of Base Raft =	0.55 m
Length of Column (lx) =	0.7 m
Width of Column (lz) =	0.7 m

Area , A = Lx*Lz
= 9 m^2
Zxx = Lx*Lz^2/6
= 4.5 m^3
Zzz = Lz*Lx^2/6
= 4.5 m^3

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WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	3X3X0.55X2.5 =	12.375
Column =	0.7X0.7X1.95X2.5 =	2.389
Soil =	((3X3)-(0.7X0.7))X1.95X1.9 =	31.53
	Total =	46.294

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

(Staad Group : F1B)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	408	146 0.6(DL	**3.331**	1.447	0.061	0.194	0	-10.492
Min Fx	408	109 DL+EC	**-9.459**	94.28	0.19	0.599	0	29.797
Max Fy	408	109 DL+EC	-9.459	**94.28**	0.19	0.599	0	29.797
Min Fy	408	146 0.6(DL	3.331	**1.447**	0.061	0.194	0	-10.492
Max Fz	408	120 DL+EC	-2.123	40.092	**0.957**	3.014	0	6.689
Min Fz	408	147 0.6(DL	-0.782	12.013	**-0.717**	-2.259	0	2.465
Max Mx	408	120 DL+EC	-2.123	40.092	0.957	**3.014**	0	6.689
Min Mx	408	147 0.6(DL	-0.782	12.013	-0.717	**-2.259**	0	2.465
Max My	408	101 DL+EC	-3.108	42.637	0.157	0.494	**0**	9.79
Min My	408	101 DL+EC	-3.108	42.637	0.157	0.494	**0**	9.79
Max Mz	408	109 DL+EC	-9.459	94.28	0.19	0.599	0	**29.797**
Min Mz	408	146 0.6(DL	3.331	1.447	0.061	0.194	0	**-10.492**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
146	47.741	3.332	0.5	23.871	7.16	1.5	O.K.
109	140.574	9.461	0.5	70.287	7.43	1.5	O.K.
109	140.574	9.461	0.5	70.287	7.43	1.5	O.K.
146	47.741	3.332	0.5	23.871	7.16	1.5	O.K.
120	86.386	2.329	0.5	43.193	18.55	1.5	O.K.
147	58.307	1.061	0.5	29.154	27.48	1.5	O.K.
120	86.386	2.329	0.5	43.193	18.55	1.5	O.K.
147	58.307	1.061	0.5	29.154	27.48	1.5	O.K.
101	88.931	3.112	0.5	44.466	14.29	1.5	O.K.
101	88.931	3.112	0.5	44.466	14.29	1.5	O.K.
109	140.574	9.461	0.5	70.287	7.43	1.5	O.K.
146	47.741	3.332	0.5	23.871	7.16	1.5	O.K.

OVERTURNING CHECK:

Max Upword Force =	0 Mton		
Self Wt. of Foundation =	46.294 Mton		
Overturning Moment =	0 X 3/2 =	0	Mton-m
Resisting Moment =	46.294 X 0.9 X 3/2 =	62.497	Mton-m
F.O.S =	>.	2	O.K.

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zxx (Mton/m^2)	Mz/Zzz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D	Remarks
146	5.305	0.043	2.332	7.68	3.016	7.594	2.93	O.K.
109	15.619	0.133	6.622	**22.374**	**9.13**	**22.108**	8.864	O.K.
109	15.619	0.133	6.622	**22.374**	**9.13**	**22.108**	8.864	O.K.
146	5.305	0.043	2.332	7.68	3.016	7.594	2.93	O.K.
120	9.598	0.67	1.486	**11.754**	8.782	**10.414**	7.442	O.K.
147	6.479	0.502	0.548	7.529	6.433	6.525	5.429	O.K.
120	9.598	0.67	1.486	**11.754**	8.782	**10.414**	7.442	O.K.
147	6.479	0.502	0.548	7.529	6.433	6.525	5.429	O.K.
101	9.881	0.11	2.176	**12.167**	7.815	**11.947**	7.595	O.K.
101	9.881	0.11	2.176	**12.167**	7.815	**11.947**	7.595	O.K.
109	15.619	0.133	6.622	**22.374**	**9.13**	**22.108**	8.864	O.K.
146	5.305	0.043	2.332	7.68	3.016	7.594	2.93	O.K.
			MAX =	22.374	9.13	22.108	8.864	

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

(Staad Group : F1B)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	408	)L+EQDL)+	**5.64**	-4.157	0.201	0.634	0	-17.449
Min Fx	408	+FL1+PL+	**-14.068**	145.643	0.275	0.867	0	44.313
Max Fy	408	+FL1+PL+	-14.068	**145.643**	0.275	0.867	0	44.313
Min Fy	408	)L+EQDL)+	5.64	**-4.157**	0.201	0.634	0	-17.449
Max Fz	408	L)+1.6WL(-	-3.201	54.04	**1.356**	4.27	0	10.083
Min Fz	408	L+EQDL)+1	-1.166	17.817	**-1.153**	-3.632	0	3.674
Max Mx	408	L)+1.6WL(-	-3.201	54.04	1.356	**4.27**	0	10.083
Min Mx	408	L+EQDL)+1	-1.166	17.817	-1.153	**-3.632**	0	3.674
Max My	408	4(DL+EQDL	-4.351	59.692	0.219	0.691	**0**	13.706
Min My	408	4(DL+EQDL	-4.351	59.692	0.219	0.691	**0**	13.706
Max Mz	408	+FL1+PL+	-14.068	145.643	0.275	0.867	0	**44.313**
Min Mz	408	)L+EQDL)+	5.64	-4.157	0.201	0.634	0	**-17.449**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zx (Mton/m^2)	Mz/Zz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D
234	4.682	0.141	3.878	8.701	0.945	8.419	0.663
211	21.326	0.193	9.847	31.366	11.672	30.98	11.286
211	21.326	0.193	9.847	31.366	11.672	30.98	11.286
234	4.682	0.141	3.878	8.701	0.945	8.419	0.663
220	11.148	0.949	2.241	14.338	9.856	12.44	7.958
231	7.123	0.807	0.816	8.746	7.114	7.132	5.5
220	11.148	0.949	2.241	14.338	9.856	12.44	7.958
231	7.123	0.807	0.816	8.746	7.114	7.132	5.5
201	11.776	0.154	3.046	14.976	8.884	14.668	8.576
201	11.776	0.154	3.046	14.976	8.884	14.668	8.576
211	21.326	0.193	9.847	31.366	11.672	30.98	11.286
234	4.682	0.141	3.878	8.701	0.945	8.419	0.663

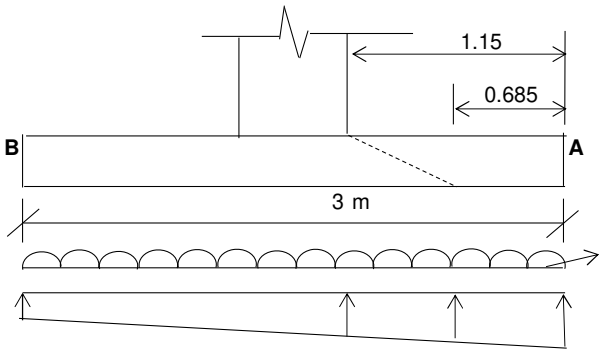
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PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point B (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
234	8.701	0.945	5.73	6.93	3.704	1.127
211	31.366	11.672	23.82	26.87	20.708	6.87
211	31.366	11.672	23.82	26.87	20.708	6.87
234	8.701	0.945	5.73	6.93	3.704	1.127
220	14.338	9.856	12.62	13.31	6.879	2.333
231	8.746	7.114	8.12	8.37	2.7	0.919
220	14.338	9.856	12.62	13.31	6.879	2.333
231	8.746	7.114	8.12	8.37	2.7	0.919
201	14.976	8.884	12.64	13.58	7.574	2.54
201	14.976	8.884	12.64	13.58	7.574	2.54
211	31.366	11.672	23.82	26.87	20.708	6.87
234	8.701	0.945	5.73	6.93	3.704	1.127
				MAX =	20.708	6.87

ALONG X - DIRECTION :

Mu = 20.708 Mton-m

Φ*Kn = Mu/bd^2 deff. = 465 mm

Mu/bd^2 = 20.708X10^7/(1000X465^2) = 0.958 N/mm^2 < 6.9 N/mm^2

Pt = 0.0026

Min. Pt = 0.0018 Reqd. Ast = 0.0026X1000X465/100 = 12.09 cm^2

Provided **20** mm @ **200** C/C Provide Ast = 15.71 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0034

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ONE-WAY SHEAR CHECK

Vu = 6.87 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0034X6.87X10^4X465/20.708/10^7)X1000X465/10^4

= 44.43 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0034)X1000X465/10^4

= 46.703 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X465/10^4

= 79.778 Mton

HENCE, Vc = 44.43 Mton

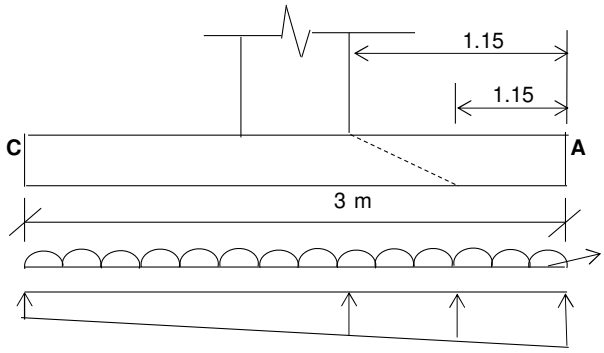
Φt*Vc = 0.75X44.43 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 33.323 Mton **> 6.87Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(44.43X10^4+0.66X35^0.5X1000X465)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 139.506 Mton **> Vu= 6.87Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point C (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
234	8.701	8.419	8.59	8.59	2.443	2.133
211	31.366	30.98	31.22	31.22	17.446	17.021
211	31.366	30.98	31.22	31.22	17.446	17.021
234	8.701	8.419	8.59	8.59	2.443	2.133
220	14.338	12.44	13.61	13.61	6.443	4.351
231	8.746	7.132	8.13	8.13	2.696	0.917
220	14.338	12.44	13.61	13.61	6.443	4.351
231	8.746	7.132	8.13	8.13	2.696	0.917
201	14.976	14.668	14.86	14.86	6.595	6.255
201	14.976	14.668	14.86	14.86	6.595	6.255
211	31.366	30.98	31.22	31.22	17.446	17.021
234	8.701	8.419	8.59	8.59	2.443	2.133
				MAX =	17.446	17.021

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ALONG Z - DIRECTION :

Mu = 17.446 T-m

Φ*Kn = Mu/bd^2 deff. = 465 mm

Mu/bd^2 = 17.446X10^7/(1000X465^2) = 0.807 N/mm^2 < 6.9 N/mm^2

Pt = 0.0022

Min. Pt = 0.0018 Reqd. Ast = 0.0022X1000X465/100 = 10.23 cm^2

Provided 20 mm @ 200 C/C Provide Ast = 15.71 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0034

ONE-WAY SHEAR CHECK

Vu = 17.021 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0034X17.021X10^4X465/17.446/10^7)X1000X465/10^4

= 45.235 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0034)X1000X465/10^4

= 46.703 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X465/10^4

= 79.778 Mton

HENCE, Vc = 45.235 Mton

Φt*Vc = 0.75X45.235 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 33.926 Mton **> 17.021Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(45.235X10^4+0.66X35^0.5X1000X465)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 139.566 Mton **> Vu= 17.021Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 9 m^2

Perimeter (bo) = 2 X ((0.7 + 0.465) + (0.7 + 0.465)) = 4.66 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 22.374 $Mton/m^2$

Vu = ((9 - (0.7X0.7))X22.374

= 190.403 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X4.66X10^3X465/10^4

= 423.045 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X4.66X10^3X465/10^4

= 653.797 Mton

[c] = $0.083*(2+\alpha s*d.eff/b_o)*\lambda*fc'^{0.5}*bo*d.eff$

= 0.083X(2+30X465/(4.66X10^3))X1X35^0.5X(4.66X10^3)X465/10^4

= 531.326 Mton

HENCE, Vc = 423.045 Mton

Φt*Vn = Φt*Vc = 0.75X423.045 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

317.284 Mton **> 190.403Mton , Hence O.K.**

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Reaction Summary (FOUNDATION F2)

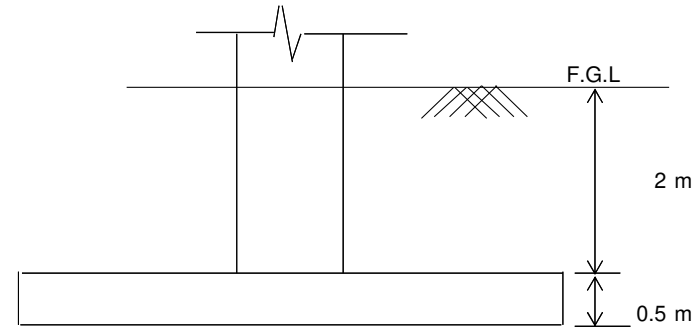
	Node	L/C	Horizontal FX (Mton)	Vertical FY (Mton)	Horizontal FZ (Mton)	Moment MX (MTon-m)	Moment MY (MTon-m)	Moment MZ (MTon-m)
Max FX	407	134:DL+EQDL-	**8.415**	50.435	0.056	0.175	0	-26.507
Min FX	414	109:DL+EQDL-	**-6.161**	57.315	-0.315	-0.992	0	19.407
Max FY	414	131:DL+EQDL-	-4.249	**64.094**	-0.555	-1.750	0	13.386
Min FY	413	148:0.6(DL+EC	-0.698	**-13.980**	0.296	0.932	0	2.198
Max FZ	407	116:DL+EQDL-	1.938	20.075	**0.486**	1.530	0	-6.105
Min FZ	414	115:DL+EQDL-	-3.080	49.723	**-0.603**	-1.900	0	9.702
Max MX	407	116:DL+EQDL-	1.938	20.075	0.486	**1.530**	0	-6.105
Min MX	414	115:DL+EQDL-	-3.080	49.723	-0.603	**-1.900**	0	9.702
Max MY	407	101:DL+EQDL-	1.856	19.928	0.082	0.259	**0**	-5.846
Min MY	407	101:DL+EQDL-	1.856	19.928	0.082	0.259	**0**	-5.846
Max MZ	414	109:DL+EQDL-	-6.161	57.315	-0.315	-0.992	0	**19.407**
Min MZ	407	134:DL+EQDL-	8.415	50.435	0.056	0.175	0	**-26.507**

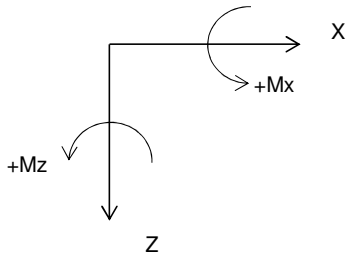
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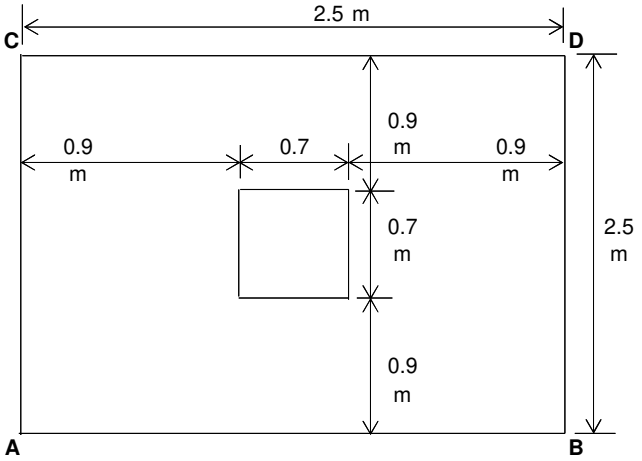
8.3 DESIGN OF FOUNDATION MKD. F2 :

(NODE NO. : 407 409 413 To 415)

Net Safe Bearing Capacity of Soil =	67 Mton/m^2	*(For BH-2)*
Depth of Foundation from FGL =	2.5 m	
Gross Allowable Bearing Capacity of Soil =	67 + 2.5 X 1.9 =	71.75 Mton/m^2
Compressive Strength of Concrete , fc' =	35 N/mm^2	
Yield Strength of Main Reinf. , fy =	420 N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420 N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	75 mm	
Density of Concrete =	2.5 Mton/m^3	
Density of Dry Soil =	1.9 Mton/m^3	
Density of Saturated Soil =	2 Mton/m^3	
Density of Water =	1 Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	2.5 m
Width of Foundation (Lz) =	2.5 m
Thickness of Base Raft =	0.5 m
Length of Column (lx) =	0.7 m
Width of Column (lz) =	0.7 m

Area , A = Lx*Lz
= 6.25 m^2
Zxx = Lx*Lz^2/6
= 2.604 m^3
Zzz = Lz*Lx^2/6
= 2.604 m^3

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WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	2.5X2.5X0.5X2.5 =	7.813
Column =	0.7X0.7X2X2.5 =	2.45
Soil =	((2.5X2.5)-(0.7X0.7))X2X1.9 =	21.888
	Total =	32.151

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

(Staad Group : F2)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	407	134 DL+EC	**8.415**	50.435	0.056	0.175	0	-26.507
Min Fx	414	109 DL+EC	**-6.161**	57.315	-0.315	-0.992	0	19.407
Max Fy	414	131 DL+EC	-4.249	**64.094**	-0.555	-1.75	0	13.386
Min Fy	413	148 0.6(DL	-0.698	**-13.98**	0.296	0.932	0	2.198
Max Fz	407	116 DL+EC	1.938	20.075	**0.486**	1.53	0	-6.105
Min Fz	414	115 DL+EC	-3.08	49.723	**-0.603**	-1.9	0	9.702
Max Mx	407	116 DL+EC	1.938	20.075	0.486	**1.53**	0	-6.105
Min Mx	414	115 DL+EC	-3.08	49.723	-0.603	**-1.9**	0	9.702
Max My	407	101 DL+EC	1.856	19.928	0.082	0.259	**0**	-5.846
Min My	407	101 DL+EC	1.856	19.928	0.082	0.259	**0**	-5.846
Max Mz	414	109 DL+EC	-6.161	57.315	-0.315	-0.992	0	**19.407**
Min Mz	407	134 DL+EC	8.415	50.435	0.056	0.175	0	**-26.507**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
134	82.586	8.415	0.5	41.293	4.91	1.5	O.K.
109	89.466	6.169	0.5	44.733	7.25	1.5	O.K.
131	96.245	4.285	0.5	48.123	11.23	1.5	O.K.
148	18.171	0.758	0.5	9.086	11.99	1.5	O.K.
116	52.226	1.998	0.5	26.113	13.07	1.5	O.K.
115	81.874	3.138	0.5	40.937	13.05	1.5	O.K.
116	52.226	1.998	0.5	26.113	13.07	1.5	O.K.
115	81.874	3.138	0.5	40.937	13.05	1.5	O.K.
101	52.079	1.858	0.5	26.04	14.02	1.5	O.K.
101	52.079	1.858	0.5	26.04	14.02	1.5	O.K.
109	89.466	6.169	0.5	44.733	7.25	1.5	O.K.
134	82.586	8.415	0.5	41.293	4.91	1.5	O.K.

OVERTURNING CHECK:

Max Upword Force =	13.98	Mton		
Self Wt. of Foundation =	32.151	Mton		
Overturning Moment =	13.98 X 2.5/2 =	17.475	Mton-m	
Resisting Moment =	32.151 X 0.9 X 2.5/2 =	36.17	Mton-m	
F.O.S =	2.07	>.	2	O.K.

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A	Mx/Zxx	Mz/Zzz	BASE PRESSURE AT POINT				Remarks
	(Mton/m^2)	(Mton/m^2)	(Mton/m^2)	A	B	C	D	
134	13.214	0.067	10.179	**23.46**	3.102	**23.326**	2.968	O.K.
109	14.315	0.381	7.453	**22.149**	7.243	**21.387**	6.481	O.K.
131	15.399	0.672	5.141	**21.212**	**10.93**	**19.868**	**9.586**	O.K.
148	2.907	0.358	0.844	4.109	2.421	3.393	1.705	O.K.
116	8.356	0.588	2.344	**11.288**	6.6	**10.112**	5.424	O.K.
115	13.1	0.73	3.726	**17.556**	**10.104**	**16.096**	8.644	O.K.
116	8.356	0.588	2.344	**11.288**	6.6	**10.112**	5.424	O.K.
115	13.1	0.73	3.726	**17.556**	**10.104**	**16.096**	8.644	O.K.
101	8.333	0.099	2.245	**10.677**	6.187	**10.479**	5.989	O.K.
101	8.333	0.099	2.245	**10.677**	6.187	**10.479**	5.989	O.K.
109	14.315	0.381	7.453	**22.149**	7.243	**21.387**	6.481	O.K.
134	13.214	0.067	10.179	**23.46**	3.102	**23.326**	2.968	O.K.
			MAX =	23.46	10.93	23.326	9.586	

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

(Staad Group : F1B)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	407	DL+FL2+TL	**12.27**	70.093	0.064	0.2	0	-38.65
Min Fx	414	+FL1+PL+T	**-9.219**	88.446	-0.467	-1.471	0	29.041
Max Fy	414	+FL1+PL+T	-9.219	**88.446**	-0.467	-1.471	0	29.041
Min Fy	413	L+EQDL)+	-1.17	**-23.053**	0.475	1.496	0	3.686
Max Fz	407	L+EQDL)+	1.238	12.203	**0.644**	2.028	0	-3.898
Min Fz	414	)+1.6WL(+	-4.381	75.21	**-0.891**	-2.807	0	13.8
Max Mx	407	L+EQDL)+	1.238	12.203	0.644	**2.028**	0	-3.898
Min Mx	414	)+1.6WL(+	-4.381	75.21	-0.891	**-2.807**	0	13.8
Max My	407	4(DL+EQDL	2.598	27.899	0.115	0.363	**0**	-8.185
Min My	407	4(DL+EQDL	2.598	27.899	0.115	0.363	**0**	-8.185
Max Mz	414	+FL1+PL+T	-9.219	88.446	-0.467	-1.471	0	**29.041**
Min Mz	407	DL+FL2+TL	12.27	70.093	0.064	0.2	0	**-38.65**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A	Mx/Zx	Mz/Zz	BASE PRESSURE AT POINT			
	(Mton/m^2)	(Mton/m^2)	(Mton/m^2)	A	B	C	D
206	16.359	0.077	14.843	31.279	1.593	31.125	1.439
211	19.296	0.565	11.152	31.013	8.709	29.883	7.579
211	19.296	0.565	11.152	31.013	8.709	29.883	7.579
232	1.456	0.575	1.416	3.447	0.615	2.297	-0.535
232	7.097	0.779	1.497	9.373	6.379	7.815	4.821
219	17.178	1.078	5.3	23.556	12.956	21.4	10.8
232	7.097	0.779	1.497	9.373	6.379	7.815	4.821
219	17.178	1.078	5.3	23.556	12.956	21.4	10.8
201	9.608	0.139	3.143	12.89	6.604	12.612	6.326
201	9.608	0.139	3.143	12.89	6.604	12.612	6.326
211	19.296	0.565	11.152	31.013	8.709	29.883	7.579
206	16.359	0.077	14.843	31.279	1.593	31.125	1.439

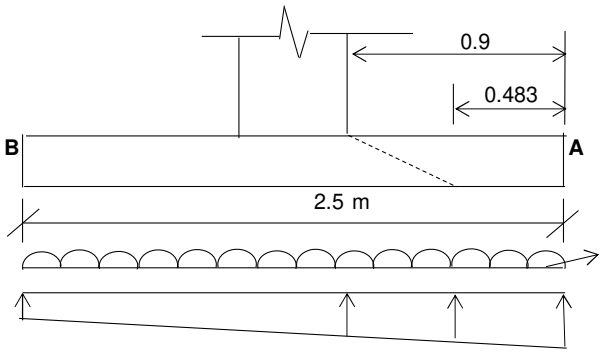
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PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point B (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
206	31.279	1.593	20.59	25.54	13.509	3.506
211	31.013	8.709	22.98	26.7	12.684	3.364
211	31.013	8.709	22.98	26.7	12.684	3.364
232	3.447	0.615	2.43	2.9	-0.375	-0.144
232	9.373	6.379	8.3	8.79	2.041	0.55
219	23.556	12.956	19.74	21.51	8.525	2.318
232	9.373	6.379	8.3	8.79	2.041	0.55
219	23.556	12.956	19.74	21.51	8.525	2.318
201	12.89	6.604	10.63	11.68	3.785	1.009
201	12.89	6.604	10.63	11.68	3.785	1.009
211	31.013	8.709	22.98	26.7	12.684	3.364
206	31.279	1.593	20.59	25.54	13.509	3.506
				MAX =	13.509	3.506

ALONG X - DIRECTION :

Mu = 13.509 Mton-m

Φ*Kn = Mu/bd^2 deff. = 417 mm

Mu/bd^2 = 13.509X10^7/(1000X417^2) = 0.777 N/mm^2 < 6.9 N/mm^2

Pt = 0.0021

Min. Pt = 0.0018 Reqd. Ast = 0.0021X1000X417/100 = 8.76 cm^2

Provided **16** mm @ **200** C/C Provide Ast = 10.05 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0024

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ONE-WAY SHEAR CHECK

Vu = 3.506 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0024X3.506X10^4X417/13.509/10^7)X1000X417/10^4

= 39.656 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0024)X1000X417/10^4

= 41.173 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X417/10^4

= 71.543 Mton

HENCE, Vc = 39.656 Mton

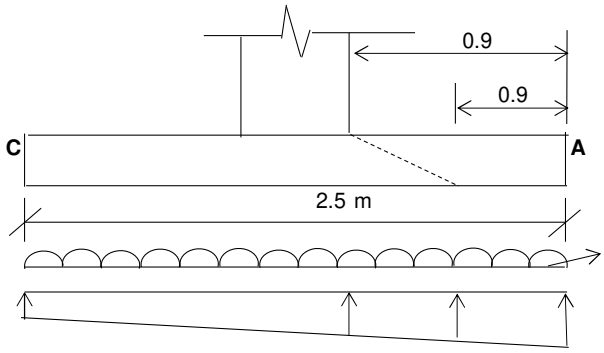
Φt*Vc = 0.75X39.656 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 29.742 Mton **> 3.506Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(39.656X10^4+0.66X35^0.5X1000X417)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 125.091 Mton **> Vu= 3.506Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point C (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
206	31.279	31.125	31.22	31.22	10.639	10.535
211	31.013	29.883	30.61	30.61	10.624	9.861
211	31.013	29.883	30.61	30.61	10.624	9.861
232	3.447	2.297	3.03	3.03	-0.537	-1.313
232	9.373	7.815	8.81	8.81	1.903	0.851
219	23.556	21.4	22.78	22.78	7.704	6.249
232	9.373	7.815	8.81	8.81	1.903	0.851
219	23.556	21.4	22.78	22.78	7.704	6.249
201	12.89	12.612	12.79	12.79	3.202	3.015
201	12.89	12.612	12.79	12.79	3.202	3.015
211	31.013	29.883	30.61	30.61	10.624	9.861
206	31.279	31.125	31.22	31.22	10.639	10.535
				MAX =	10.639	10.535

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ALONG Z - DIRECTION :

Mu = 10.639 T-m

Φ*Kn = Mu/bd^2 deff. = 417 mm

Mu/bd^2 = 10.639X10^7/(1000X417^2) = 0.612 N/mm^2 < 6.9 N/mm^2

Pt = 0.0016

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X417/100 = 7.51 cm^2

Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0024

ONE-WAY SHEAR CHECK

Vu = 10.535 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0024X10.535X10^4X417/10.639/10^7)X1000X417/10^4

= 40.175 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0024)X1000X417/10^4

= 41.173 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X417/10^4

= 71.543 Mton

HENCE, Vc = 40.175 Mton

Φt*Vc = 0.75X40.175 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 30.131 Mton **> 10.535Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(40.175X10^4+0.66X35^0.5X1000X417)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 125.13 Mton **> Vu= 10.535Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 6.25 m^2

Perimeter (bo) = 2 X ((0.7 + 0.417) + (0.7 + 0.417)) = 4.468 m

β = 1

αs = 20 (Corner Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 23.46 $Mton/m^2$

Vu = ((6.25 - (0.7X0.7))X23.46

= 135.13 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X4.468X10^3X417/10^4

= 363.745 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X4.468X10^3X417/10^4

= 562.152 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+20X417/(4.468X10^3))X1X35^0.5X(4.468X10^3)X417/10^4

= 353.746 Mton

HENCE, Vc = 353.746 Mton

Φt*Vn = Φt*Vc = 0.75X353.746 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

265.31 Mton **> 135.13Mton , Hence O.K.**

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Reaction Summary (FOUNDATION F3)

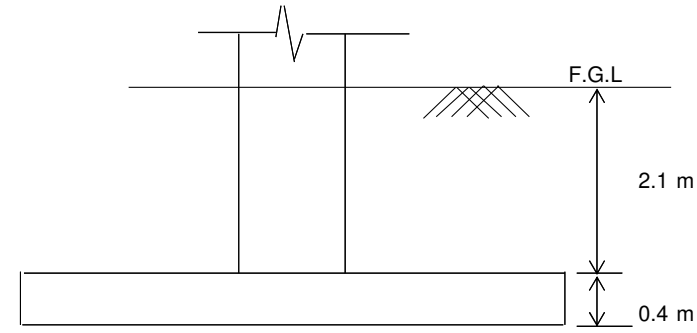
			Horizontal	Vertical	Horizontal	Moment		
	Node	L/C	FX (Mton)	FY (Mton)	FZ (Mton)	MX (MTon-m)	MY (MTon-m)	MZ (MTon-m)
Max FX	405	126:DL+EQDL	**0.080**	3.744	0.115	0.362	0	-0.170
Min FX	405	149:0.6(DL+EC	**-0.059**	2.246	0.069	0.217	0	0.103
Max FY	406	131:DL+EQDL	0.001	**6.733**	-0.564	-1.777	0	-0.004
Min FY	405	147:0.6(DL+EC	0.005	**0.858**	-0.382	-1.202	0	-0.017
Max FZ	405	109:DL+EQDL	0.053	6.264	**0.697**	2.197	0	-0.167
Min FZ	406	109:DL+EQDL	0.009	6.322	**-0.697**	-2.197	0	-0.028
Max MX	405	109:DL+EQDL	0.053	6.264	0.697	**2.197**	0	-0.167
Min MX	406	109:DL+EQDL	0.009	6.322	-0.697	**-2.197**	0	-0.028
Max MY	405	101:DL+EQDL	0.011	3.744	0.115	0.362	**0**	-0.03443
Min MY	405	101:DL+EQDL	0.011	3.744	0.115	0.362	**0**	-0.03443
Max MZ	405	149:0.6(DL+EC	-0.059	2.246	0.069	0.217	0	**0.103**
Min MZ	405	104:DL+EQDL	0.062	6.264	0.697	2.197	0	**-0.194**

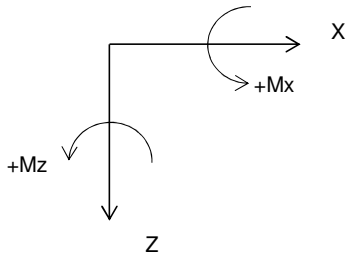
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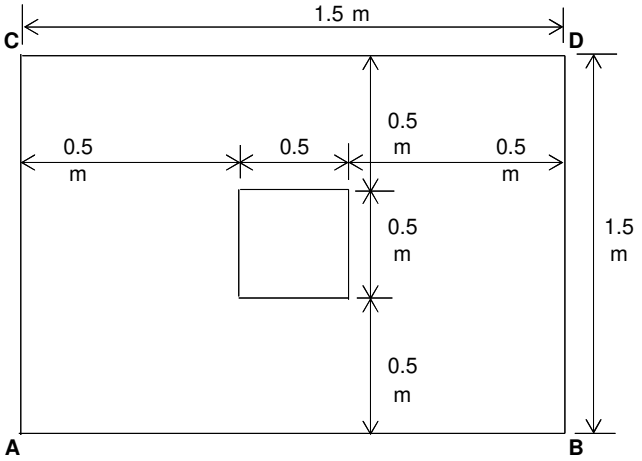
8.4 DESIGN OF FOUNDATION MKD. F3 :

(NODE NO. : 407 409 413 To 415)

Net Safe Bearing Capacity of Soil =	67 Mton/m^2	*(For BH-2)*
Depth of Foundation from FGL =	2.5 m	
Gross Allowable Bearing Capacity of Soil =	67 + 2.5 X 1.9 =	71.75 Mton/m^2
Compressive Strength of Concrete , fc' =	35 N/mm^2	
Yield Strength of Main Reinf. , fy =	420 N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420 N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	75 mm	
Density of Concrete =	2.5 Mton/m^3	
Density of Dry Soil =	1.9 Mton/m^3	
Density of Saturated Soil =	2 Mton/m^3	
Density of Water =	1 Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	1.5 m
Width of Foundation (Lz) =	1.5 m
Thickness of Base Raft =	0.4 m
Length of Column (lx) =	0.5 m
Width of Column (lz) =	0.5 m

Area , A = Lx*Lz
= 2.25 m^2
Zxx = Lx*Lz^2/6
= 0.563 m^3
Zzz = Lz*Lx^2/6
= 0.563 m^3

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WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	1.5X1.5X0.4X2.5 =	2.25
Column =	0.5X0.5X2.1X2.5 =	1.313
Soil =	((1.5X1.5)-(0.5X0.5))X2.1X1.9 =	7.98
	Total =	11.543

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

(Staad Group : F3)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	405	126 DL+EC	**0.08**	3.744	0.115	0.362	0	-0.17
Min Fx	405	149 0.6(DL	**-0.059**	2.246	0.069	0.217	0	0.103
Max Fy	406	131 DL+EC	0.001	**6.733**	-0.564	-1.777	0	-0.004
Min Fy	405	147 0.6(DL	0.005	**0.858**	-0.382	-1.202	0	-0.017
Max Fz	405	109 DL+EC	0.053	6.264	**0.697**	2.197	0	-0.167
Min Fz	406	109 DL+EC	0.009	6.322	**-0.697**	-2.197	0	-0.028
Max Mx	405	109 DL+EC	0.053	6.264	0.697	**2.197**	0	-0.167
Min Mx	406	109 DL+EC	0.009	6.322	-0.697	**-2.197**	0	-0.028
Max My	405	101 DL+EC	0.011	3.744	0.115	0.362	**0**	-0.034
Min My	405	101 DL+EC	0.011	3.744	0.115	0.362	**0**	-0.034
Max Mz	405	149 0.6(DL	-0.059	2.246	0.069	0.217	0	**0.103**
Min Mz	405	104 DL+EC	0.062	6.264	0.697	2.197	0	**-0.194**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
126	15.287	0.14	0.5	7.644	54.6	1.5	O.K.
149	13.789	0.091	0.5	6.895	75.77	1.5	O.K.
131	18.276	0.564	0.5	9.138	16.2	1.5	O.K.
147	12.401	0.382	0.5	6.201	16.23	1.5	O.K.
109	17.807	0.699	0.5	8.904	12.74	1.5	O.K.
109	17.865	0.697	0.5	8.933	12.82	1.5	O.K.
109	17.807	0.699	0.5	8.904	12.74	1.5	O.K.
109	17.865	0.697	0.5	8.933	12.82	1.5	O.K.
101	15.287	0.116	0.5	7.644	65.9	1.5	O.K.
101	15.287	0.116	0.5	7.644	65.9	1.5	O.K.
149	13.789	0.091	0.5	6.895	75.77	1.5	O.K.
104	17.807	0.7	0.5	8.904	12.72	1.5	O.K.

OVERTURNING CHECK:

Max Upword Force =	0	Mton		
Self Wt. of Foundation =	11.543	Mton		
Overturning Moment =	0 X 1.5/2 =		0	Mton-m
Resisting Moment =	11.543 X 0.9 X 1.5/2 =		7.792	Mton-m
F.O.S =	>.		2	O.K.

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BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zxx (Mton/m^2)	Mz/Zzz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D	Remarks
126	6.794	0.643	0.302	7.739	7.135	6.453	5.849	O.K.
149	6.128	0.385	0.183	6.696	6.33	5.926	5.56	O.K.
131	8.123	3.156	0.007	**11.286**	**11.272**	4.974	4.96	O.K.
147	5.512	2.135	0.03	7.677	7.617	3.407	3.347	O.K.
109	7.914	3.902	0.297	**12.113**	**11.519**	4.309	3.715	O.K.
109	7.94	3.902	0.05	**11.892**	**11.792**	4.088	3.988	O.K.
109	7.914	3.902	0.297	**12.113**	**11.519**	4.309	3.715	O.K.
109	7.94	3.902	0.05	**11.892**	**11.792**	4.088	3.988	O.K.
101	6.794	0.643	0.06	7.497	7.377	6.211	6.091	O.K.
101	6.794	0.643	0.06	7.497	7.377	6.211	6.091	O.K.
149	6.128	0.385	0.183	6.696	6.33	5.926	5.56	O.K.
104	7.914	3.902	0.345	**12.161**	**11.471**	4.357	3.667	O.K.
			MAX =	12.161	11.792	6.453	6.091	

SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

(Staad Group : F3)

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	405	234 0.9(DL	**0.157**	3.369	0.103	0.326	0	-0.303
Min Fx	405	233 0.9(DL	**-0.14**	3.369	0.103	0.326	0	0.249
Max Fy	406	211 1.4(DL	0.013	**9.606**	-1.021	-3.216	0	-0.04
Min Fy	405	231 0.9(DL	0.008	**1.148**	-0.618	-1.945	0	-0.025
Max Fz	405	211 1.4(DL	0.078	9.525	**1.021**	3.216	0	-0.245
Min Fz	406	211 1.4(DL	0.013	9.606	**-1.021**	-3.216	0	-0.04
Max Mx	405	211 1.4(DL	0.078	9.525	1.021	**3.216**	0	-0.245
Min Mx	406	211 1.4(DL	0.013	9.606	-1.021	**-3.216**	0	-0.04
Max My	405	201 1.4(DL	0.015	5.241	0.161	0.507	**0**	-0.048
Min My	405	201 1.4(DL	0.015	5.241	0.161	0.507	**0**	-0.048
Max Mz	405	233 0.9(DL	-0.14	3.369	0.103	0.326	0	**0.249**
Min Mz	405	234 0.9(DL	0.157	3.369	0.103	0.326	0	**-0.303**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A (Mton/m^2)	Mx/Zx (Mton/m^2)	Mz/Zz (Mton/m^2)	BASE PRESSURE AT POINT A	B	C	D
234	6.628	0.579	0.538	7.745	6.669	6.587	5.511
233	6.628	0.579	0.442	7.649	6.765	6.491	5.607
211	9.4	5.712	0.071	15.183	15.041	3.759	3.617
231	5.64	3.455	0.044	9.139	9.051	2.229	2.141
211	9.364	5.712	0.435	15.511	14.641	4.087	3.217
211	9.4	5.712	0.071	15.183	15.041	3.759	3.617
211	9.364	5.712	0.435	15.511	14.641	4.087	3.217
211	9.4	5.712	0.071	15.183	15.041	3.759	3.617
201	7.46	0.901	0.085	8.446	8.276	6.644	6.474
201	7.46	0.901	0.085	8.446	8.276	6.644	6.474
233	6.628	0.579	0.442	7.649	6.765	6.491	5.607
234	6.628	0.579	0.538	7.745	6.669	6.587	5.511

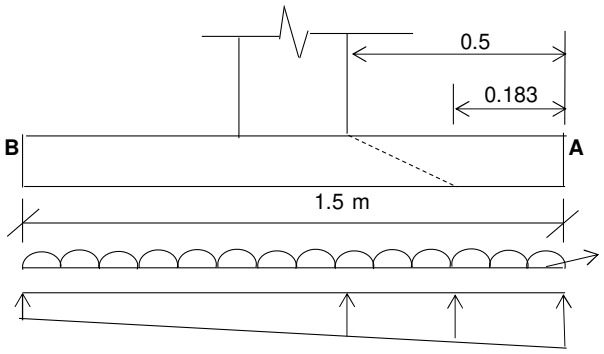
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PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m^2)	Pressure at Point B (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
234	7.745	6.669	7.39	7.61	0.374	0.048
233	7.649	6.765	7.35	7.54	0.357	0.046
211	15.183	15.041	15.14	15.17	1.278	0.171
231	9.139	9.051	9.11	9.13	0.521	0.07
211	15.511	14.641	15.22	15.4	1.339	0.177
211	15.183	15.041	15.14	15.17	1.278	0.171
211	15.511	14.641	15.22	15.4	1.339	0.177
211	15.183	15.041	15.14	15.17	1.278	0.171
201	8.446	8.276	8.39	8.43	0.437	0.058
201	8.446	8.276	8.39	8.43	0.437	0.058
233	7.649	6.765	7.35	7.54	0.357	0.046
234	7.745	6.669	7.39	7.61	0.374	0.048
				MAX =	1.339	0.177

ALONG X - DIRECTION :

Mu = 1.339 Mton-m

Φ*Kn = Mu/bd^2 deff. = 317 mm

Mu/bd^2 = 1.339X10^7/(1000X317^2) = 0.133 N/mm^2 < 6.9 N/mm^2

Pt = 0.0004

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X317/100 = 5.71 cm^2

Provided 16 mm @ 200 C/C Provide Ast = 10.05 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0032

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ONE-WAY SHEAR CHECK

Vu = 0.177 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0032X0.177X10^4X317/1.339/10^7)X1000X317/10^4

= 30.079 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0032)X1000X317/10^4

= 31.731 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X317/10^4

= 54.387 Mton

HENCE, Vc = 30.079 Mton

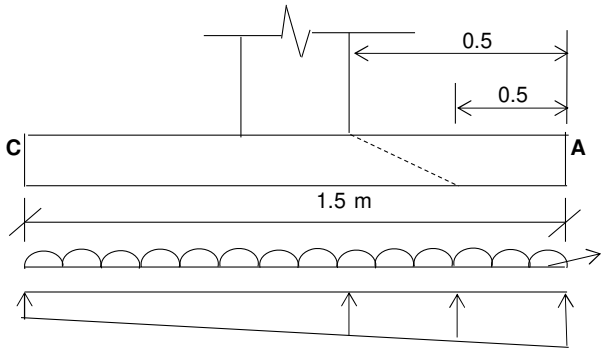
Φt*Vc = 0.75X30.079 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 22.559 Mton **> 0.177Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(30.079X10^4+0.66X35^0.5X1000X317)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 95.088 Mton **> Vu= 0.177Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



4.99 (Soil+Concrete Downward Pressure)

L/C	Pressure at Point A (Mton/m^2)	Pressure at Point C (Mton/m^2)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
234	7.745	6.587	7.36	7.36	0.376	0.135
233	7.649	6.491	7.26	7.26	0.365	0.124
211	15.183	3.759	11.38	11.38	1.591	-0.789
231	9.139	2.229	6.84	6.84	0.71	-0.729
211	15.511	4.087	11.7	11.7	1.633	-0.747
211	15.183	3.759	11.38	11.38	1.591	-0.789
211	15.511	4.087	11.7	11.7	1.633	-0.747
211	15.183	3.759	11.38	11.38	1.591	-0.789
201	8.446	6.644	7.85	7.85	0.482	0.106
201	8.446	6.644	7.85	7.85	0.482	0.106
233	7.649	6.491	7.26	7.26	0.365	0.124
234	7.745	6.587	7.36	7.36	0.376	0.135
				MAX =	1.633	0.135

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ALONG Z - DIRECTION :

Mu = 1.633 T-m

$\Phi*Kn$ = Mu/bd^2 deff. = 317 mm

Mu/bd^2 = 1.633X10^7/(1000X317^2) = 0.163 N/mm^2 < 6.9 N/mm^2

Pt = 0.0004

Min. Pt = 0.0018 Reqd. Ast = 0.0018X1000X317/100 = 5.71 cm^2

Provided **16** mm @ **200** C/C Provide Ast = 10.05 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0032

ONE-WAY SHEAR CHECK

Vu = 0.135 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0032X0.135X10^4X317/1.633/10^7)X1000X317/10^4

= 30.052 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0032)X1000X317/10^4

= 31.731 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X317/10^4

= 54.387 Mton

HENCE, Vc = 30.052 Mton

$\Phi t*Vc$ = 0.75X30.052 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 22.539 Mton **> 0.135Mton , Hence O.K.**

$\Phi t*(Vc+0.66*fc'^{0.5}*b*d.eff)$ = 0.75X(30.052X10^4+0.66X35^0.5X1000X317)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 95.086 Mton **> Vu= 0.135Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 2.25 m^2

Perimeter (bo) = 2 X ((0.5 + 0.317) + (0.5 + 0.317)) = 3.268 m

β = 1

αs = 20 (Corner Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 12.161 $Mton/m^2$

Vu = ((2.25 - (0.5X0.5))X12.161

= 24.322 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X3.268X10^3X317/10^4

= 202.25 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X3.268X10^3X317/10^4

= 312.569 Mton

[c] = $0.083*(2+\alpha s*d.eff/b_o)*\lambda*fc'^{0.5}*bo*d.eff$

= 0.083X(2+20X317/(3.268X10^3))X1X35^0.5X(3.268X10^3)X317/10^4

= 200.425 Mton

HENCE, Vc = 200.425 Mton

$\Phi t*Vn$ = $\Phi t*Vc$ = 0.75X200.425 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

150.319 Mton **> 24.322Mton , Hence O.K.**

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Beam Force Detail Summary (COLUMN C1)

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

				Axial	Shear		Torsion	Bending	
	Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
Max Fx	3065	204:1.4(DL+EC	0	**1.67E+3**	100.189	9.879	0	-31.119	315.596
Min Fx	3067	232:0.9(DL+EC	3.150	**-260.438**	11.475	4.658	-0	-0	-0
Max Fy	3065	211:1.4(DL+EC	0	1.59E+3	**140.317**	-36.738	0	115.725	441.997
Min Fy	3064	206:1.4(DL+EC	0	899.419	**-148.682**	16.971	0	-53.460	-468.349
Max Fz	3065	232:0.9(DL+EC	0	698.414	33.980	**208.950**	0	-658.191	107.035
Min Fz	3065	219:1.2(DL+EC	0	373.138	26.593	**-214.041**	0	674.228	83.769
Max Mx	3061	201:1.4(DL+EC	0	273.599	-25.480	1.130	**0**	-3.558	-80.263
Min Mx	3061	201:1.4(DL+EC	0	273.599	-25.480	1.130	**0**	-3.558	-80.263
Max My	3065	219:1.2(DL+EC	0	373.138	26.593	-214.041	0	**674.228**	83.769
Min My	3065	232:0.9(DL+EC	0	698.414	33.980	208.950	0	**-658.191**	107.035
Max Mz	3065	211:1.4(DL+EC	0	1.59E+3	140.317	-36.738	0	115.725	**441.997**
Min Mz	3064	206:1.4(DL+EC	0	899.419	-148.682	16.971	0	-53.460	**-468.349**

MA'ADEN Saudi Arabian Mining Company		Outotec / LARSEN & TOUBRO
MD-216-3230-EG-CV-CAL-1167	Mansourah Massarah Gold Project	Page:
900707_GAU01_ZA01_CED01_32301_00671093_R2	CALCULATION GRAVITY SEPARATION DESIGN CALCULATION OF FOUNDATION	Date 09.11.2023
O19045_450_C_K_24082_001_R2	AS BUILT	Rev. D01

9.1 Design of column C1

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 700 mm

Depth parallel to Z-Z axis, b = 700 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.8

Y_b = (b-d')/b = 0.8

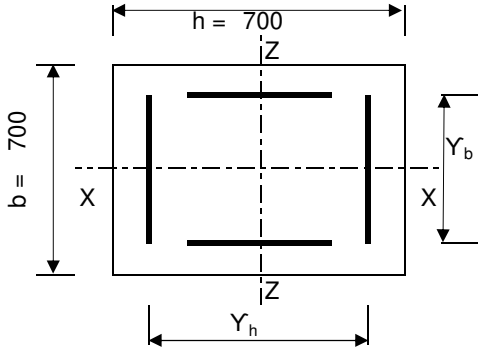
Unsupported length of column parallel to X-X axis, l_{ux} = 3.88 m kl_{ux}/b = 5.543 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 3.88 m kl_{uz}/h = 5.543

$P_{n,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 13232.1 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx} = M_{ux}/φ$	$M_{nz} = M_{uz}/φ$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
204	1670	31.1	315.6	2569	47.9	485.5	0.15	0.100	1201	0.04	0.100	1201	0.40		0.24	safe
232	-260.4	0.0	0.0	-401	0.0	0.0	-0.02	0.100	1201	0.00	0.100	1201	0.00		0	safe
211	1590	115.7	442	2446	178.0	680.0	0.143	0.100	1201	0.15	0.100	1201	0.57		0.45	safe
206	899.4	53.5	468	1384	82.2	720.5	0.081	0.100	1201	0.07	0.100	1201	0.60		0.45	safe
232	698.4	658.2	107.0	1074	1013	164.7	0.063	0.100	1201	0.84	0.100	1201	0.14		0.8	safe
219	373.1	674.2	83.8	574	1037	128.9	0.033	0.100	1201	0.86	0.100	1201	0.11		0.82	safe
201	273.6	3.6	80.3	421	5.5	123.5	0.025	0.100	1201	0.00	0.100	1201	0.10	1.61	0.03	safe
201	273.6	3.6	80.3	421	5.5	123.5	0.025	0.100	1201	0.00	0.100	1201	0.10		0.03	safe
219	373.1	674.2	83.8	574	1037	128.9	0.033	0.100	1201	0.86	0.100	1201	0.11		0.82	safe
232	698.4	658.2	107.0	1074	1013	164.7	0.063	0.100	1201	0.84	0.100	1201	0.14		0.8	safe
211	1590	115.7	442	2446	178.0	680.0	0.143	0.100	1201	0.15	0.100	1201	0.57		0.45	safe
206	899.4	53.5	468	1384	82.2	720.5	0.081	0.100	1201	0.07	0.100	1201	0.60		0.45	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db				A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
4900	16	20	+		5029	1.03	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.7

ACI DESIGN HANDBOOK-SP-17M(09)

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Beam Force Detail Summary (COLUMN C2)

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

				Axial	Shear		Torsion	Bending	
	Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
Max Fx	3060	211:1.4(DL+EC	0	**94.204**	-0.125	-10.013	0	31.542	-0.395
Min Fx	3059	231:0.9(DL+EC	3.150	**-6.277**	-0.077	-6.056	-0	-0	-0
Max Fy	3059	233:0.9(DL+EC	0	33.040	**0.776**	1.014	0	-3.195	2.443
Min Fy	3059	234:0.9(DL+EC	0	33.043	**-0.944**	1.014	0	-3.195	-2.973
Max Fz	3059	211:1.4(DL+EC	0	93.410	-0.764	**10.013**	0	-31.542	-2.405
Min Fz	3060	211:1.4(DL+EC	0	94.204	-0.125	**-10.013**	0	31.542	-0.395
Max Mx	3059	201:1.4(DL+EC	0	51.398	-0.150	1.578	**0**	-4.970	-0.473
Min Mx	3059	201:1.4(DL+EC	0	51.398	-0.150	1.578	**0**	-4.970	-0.473
Max My	3060	211:1.4(DL+EC	0	94.204	-0.125	-10.013	0	**31.542**	-0.395
Min My	3059	211:1.4(DL+EC	0	93.410	-0.764	10.013	0	**-31.542**	-2.405
Max Mz	3059	233:0.9(DL+EC	0	33.040	0.776	1.014	0	-3.195	**2.443**
Min Mz	3059	234:0.9(DL+EC	0	33.043	-0.944	1.014	0	-3.195	**-2.973**

MA'ADEN Saudi Arabian Mining Company		Outotec / LARSEN & TOUBRO
MD-216-3230-EG-CV-CAL-1167	Mansourah Massarah Gold Project	Page:
900707_GAU01_ZA01_CED01_32301_00671093_R2	CALCULATION GRAVITY SEPARATION DESIGN CALCULATION OF FOUNDATION	Date 09.11.2023
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9.2 Design of column C2

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.72

Y_b = (b-d')/b = 0.72

Unsupported length of column parallel to X-X axis, l_{ux} = 3.88 m kl_{ux}/b = 7.76

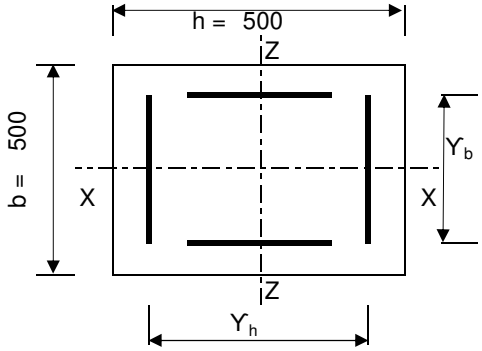
Unsupported length of column parallel to Z-Z axis, l_{uz} = 3.88 m kl_{uz}/h = 7.76

< 22 neglect slenderness effect

$P_{n,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6735.18 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx} = M_{ux}/φ$	$M_{nz} = M_{uz}/φ$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
211	94	31.5	0.4	145	48.5	0.6	0.017	0.110	481	0.10	0.110	481.3	0.00		0.03	safe
231	-6.3	0.0	0.0	-10	0.0	0.0	-0	0.110	481	0.00	0.110	481.3	0.00		0	safe
233	33	3.2	2	51	4.9	3.8	0.006	0.110	481	0.01	0.110	481.3	0.01		0	safe
234	33.0	3.2	3	51	4.9	4.6	0.006	0.110	481	0.01	0.110	481.3	0.01		0	safe
211	93.4	31.5	2.4	144	49	3.7	0.016	0.110	481	0.10	0.110	481.3	0.01		0.03	safe
211	94.2	31.5	0.4	145	49	0.6	0.017	0.110	481	0.10	0.110	481.3	0.00		0.03	safe
201	51.4	5.0	0.5	79	7.6	0.7	0.009	0.110	481	0.02	0.110	481.3	0.00	1.61	0	safe
201	51.4	5.0	0.5	79	7.6	0.7	0.009	0.110	481	0.02	0.110	481.3	0.00		0	safe
211	94.2	31.5	0.4	145	49	0.6	0.017	0.110	481	0.10	0.110	481.3	0.00		0.03	safe
211	93.4	31.5	2.4	144	49	3.7	0.016	0.110	481	0.10	0.110	481.3	0.01		0.03	safe
233	33	3.2	2	51	4.9	3.8	0.006	0.110	481	0.01	0.110	481.3	0.01		0	safe
234	33.0	3.2	3	51	4.9	4.6	0.006	0.110	481	0.01	0.110	481.3	0.01		0	safe

Reinforcement Calculation						Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db				A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	8	20	+		2515	1.01	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.7

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ATTACHMENT - 1

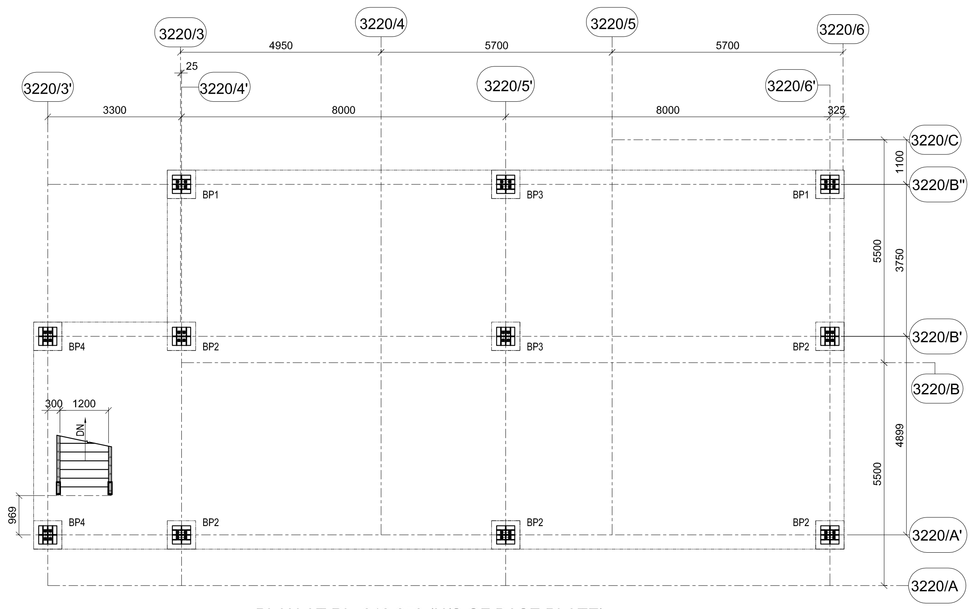
Mechanical & Structural Input Drawing.

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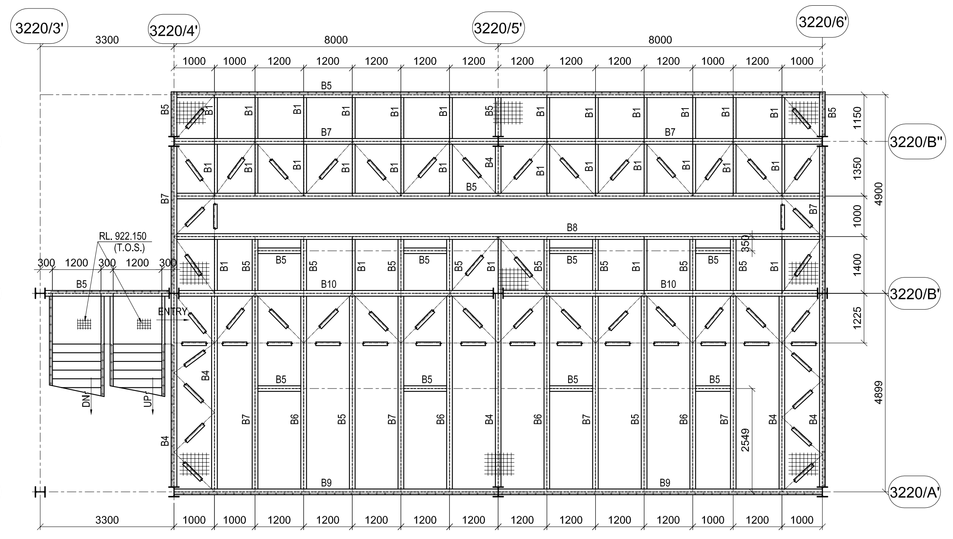
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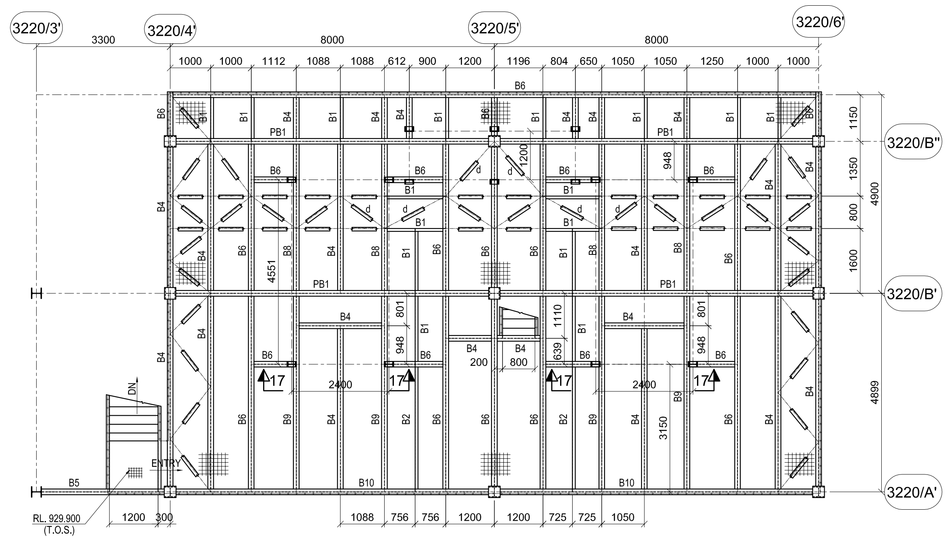
DRAWING NUMBER: MD - 216 - 3230 - EG - ST - GAD - 2150 | REV: A01



PLAN AT RL. 913.650 (U/S OF BASE PLATE)
(SCALE 1:75)



FLOOR PLAN AT RL. 922.150(T.O.S)
(ALL FLOOR BRCG. ARE 65x65x6 & 100MM. BELOW T.O.S.)
(SCALE 1:75)

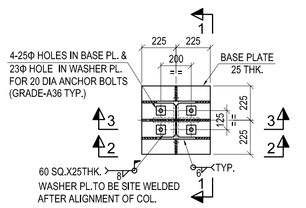


FLOOR PLAN AT RL. 929.900(T.O.S)
(ALL FLOOR BRCG. ARE 65x65x6 & 100MM. BELOW T.O.S.)
(SCALE 1:75)

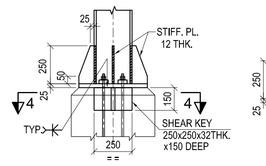
SCHEDULE OF MEMBERS						
MEMBER MKD.	MEMBER	SECTION	FACTORED AXIAL FORCE(T)	FACTORED BENDING MOMENT(T-m)	FACTORED SHEAR FORCE(T)	REMARKS
a	L 50x50x6	⌐	-			MATERIAL GRADE : S275 JR
b	L 65x65x6	⌐	8			
d	L 90x90x8	⌐	18			
k	L 150x150x13	⌐	33			
sa	L 50x50x6	⌐	2			
sb	L 65x65x6	⌐	15			
sd	L 90x90x8	⌐	10			
sf	L 100x100x8	⌐	2			
sh	L 130x130x10	⌐	41			
B1	UPN100	C	16	0.3	1	
B2	UPN120	C	6	0.8	1	
B3	UPN240	C	1	0.5	1	
B4	IPE200	I	15	2.9	4.8	
B5	IPE240	I	13	6.7	9.1	
B6	IPE300	I	13	9.1	28	
B7	IPE360	I	32	11.5	5.4	
B8	IPE400	I	6	22.1	29.7	
B9	IPE450	I	33	29.1	18	
B10	IPE500	I	23	42.1	17.2	
C1	HEA260	H	31	7.6	5.1	
PB1	1- WEB PL.760x10 2- FLG. PL.250x20	② ① ②	27	97.1	39.4	

ALL GUSSET PLATE ARE 8 MILLIMETER THK (UNO)
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6 MILLIMETER FILLET WELD OF 150 MILLIMETER LENGTH SHALL BE PROVIDED.

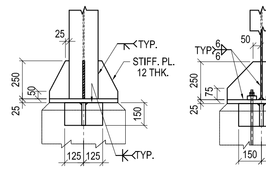
SCHEDULE OF COLUMN		
COLUMN MKD.	COLUMN SECTION	
3220/A'-3220/3'	RL 913.650 TO RL 929.900	
	H IPE200	
3220/B'-3220/3'	RL 913.650 TO RL 927.410	
	H IPE200	
3220/A'-3220/4' 3220/B'-3220/4' 3220/A'-3220/6' 3220/B'-3220/6'	RL 913.650 TO RL 922.150	ABOVE RL 922.150
	H HEA260	H HEA200
3220/B'-3220/4' 3220/A'-3220/5' 3220/B'-3220/5' 3220/B'-3220/6'	RL 913.650 TO RL 929.900	
	H HEA260	
3220/B'-3220/5'	RL 913.650 TO RL 925.850	ABOVE RL 925.850
	H HEA260	H HEA200 +2/FLG. PL. 200x12



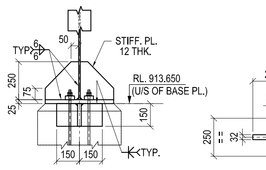
DETAIL OF BASE PLATE MKD. BP1
(FOR COLUMN MKD. 3220/B"-3220/4', 3220/B'-3220/6')
(SCALE 1:20)
(FOR EXACT ORIENTATION OF COLUMN REFER PLAN AT RL. 913.650)



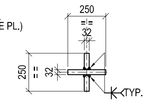
SECTION 1-1
(SCALE 1:20)



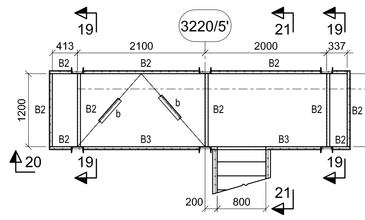
SECTION 2-2
(SCALE 1:20)



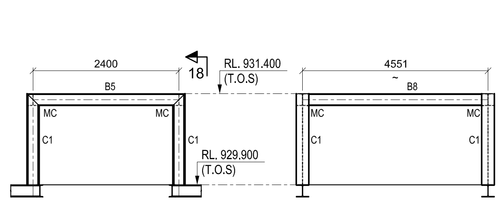
SECTION 3-3
(SCALE 1:20)



SECTION 4-4
(SCALE 1:20)

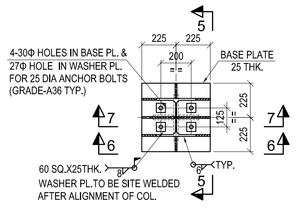


FLOOR PLAN AT RL. 932.843(T.O.S)
(SCALE 1:50)

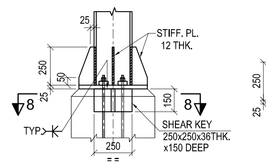


SECTION 17-17
(SCALE 1:50)

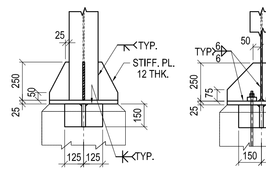
SECTION 18-18
(SCALE 1:50)



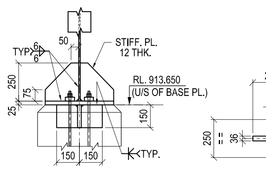
DETAIL OF BASE PLATE MKD. BP2
(FOR COLUMN MKD. 3220/A'-3220/4', 3220/A'-3220/5', 3220/A'-3220/6', 3220/B'-3220/4' & 3220/B'-3220/6')
(SCALE 1:20)
(FOR EXACT ORIENTATION OF COLUMN REFER PLAN AT RL. 913.650)



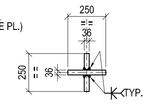
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(SCALE 1:20)



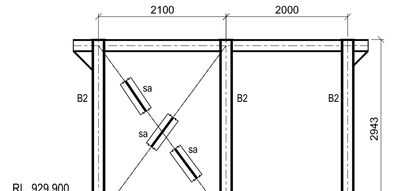
SECTION 6-6
(SCALE 1:20)



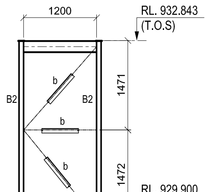
SECTION 7-7
(SCALE 1:20)



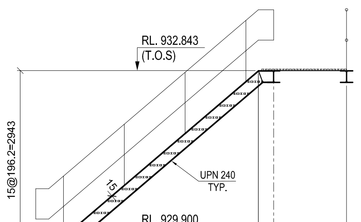
SECTION 8-8
(SCALE 1:20)



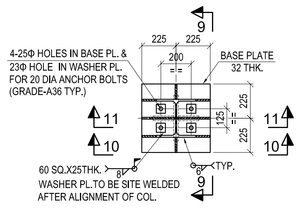
SECTION 20-20
(SCALE 1:50)



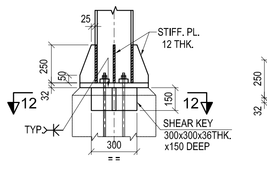
SECTION 19-19
(SCALE 1:50)



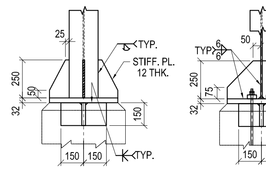
SECTION 21-21
(SCALE 1:50)



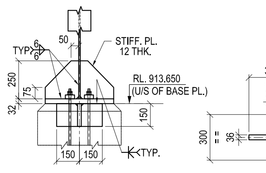
DETAIL OF BASE PLATE MKD. BP3
(FOR COLUMN MKD. 3220/B'-3220/5' & 3220/B"-3220/5')
(SCALE 1:20)
(FOR EXACT ORIENTATION OF COLUMN REFER PLAN AT RL. 913.650)



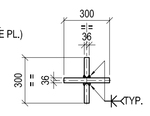
SECTION 9-9
(SCALE 1:20)



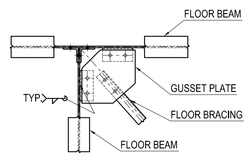
SECTION 10-10
(SCALE 1:20)



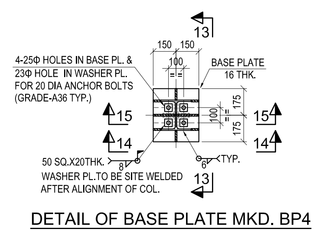
SECTION 11-11
(SCALE 1:20)



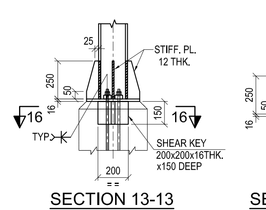
SECTION 12-12
(SCALE 1:20)



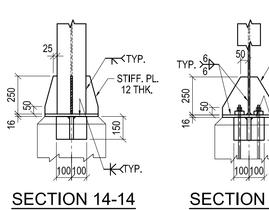
TYP. FLOOR BRCGS. CONNECTION
(SCALE 1:20)



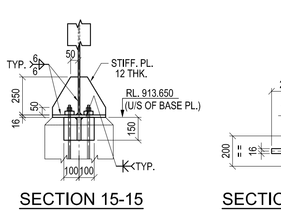
DETAIL OF BASE PLATE MKD. BP4
(FOR COLUMN MKD. 3220/A'-3220/3' & 3220/B'-3220/3')
(SCALE 1:20)
(FOR EXACT ORIENTATION OF COLUMN REFER PLAN AT RL. 913.650)



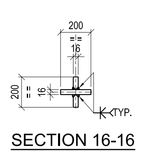
SECTION 13-13
(SCALE 1:20)



SECTION 14-14
(SCALE 1:20)



SECTION 15-15
(SCALE 1:20)



SECTION 16-16
(SCALE 1:20)

LEGENDS

T.O.S. - TOP OF STEEL
MC - MOMENT CONNECTION
U/S - UNDER SIDE
DN - DOWN
TYP. - TYPICAL
THK. - THICKNESS
PL. - PLATE
BRCG. - BRACING
DIA - DIAMETER
UNO - UNLESS NOTED OTHERWISE
MKD. - MARKED
DRG. - DRAWING
COL. - COLUMN
STIFF. - STIFFENER
T.O.G. - TOP OF GRATING
W.P. - WORK POINT
RL. - REDUCED LEVEL

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS ARE IN METERS UNO.
2. GRADE OF STRUCTURAL STEEL SHALL BE S275 JR AS PER EN 10025 UNO.
3. STRUCTURAL BOLT MATERIAL SHALL BE AS PER ASTM A325M
4. HANDRAIL PIPE/POST MATERIAL SHALL BE AS PER ASTM A53 GR.B
5. MAXIMUM SIZE OF WELD SHALL BE LESSER THICKNESS OF THE JOINED MEMBERS UNO.
6. ALL FLOOR GRATING SHALL BE MINIMUM WEBFORGE A255 AND STAIRCASE GRATING CONFORMS TO WEBFORGE T5/A255 MPG
7. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH THE DRAWING NO. MD-216-3230-EG-ST-GAD-2151

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-CV-DED-1003	DETAIL DRAWING ANCHOR BOLTS STANDARD DETAILS								
MD-216-3220-EG-ME-GAD-1075	BALL MILL SECTION F-F								
MD-216-3220-EG-ME-GAD-1074	BALL MILL SECTION E-E								
MD-216-3220-EG-ME-GAD-1073	BALL MILL SECTION D-D								
MD-216-3220-EG-ME-GAD-1072	BALL MILL SECTION C-C								
MD-216-3220-EG-ME-GAD-1071	BALL MILL SECTION B-B								
MD-216-3220-EG-ME-GAD-1070	BALL MILL SECTION A-A								
MD-216-3220-EG-ME-GAD-1068	BALL MILL PLAN AT EL +924.580								
MD-216-3220-EG-ME-GAD-1067	BALL MILL PLAN AT EL +922.180								
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL DESIGN CRITERIA	A01	26/02/2020	ISSUED FOR REFERENCE	BAB	PRI	BAB	AG	

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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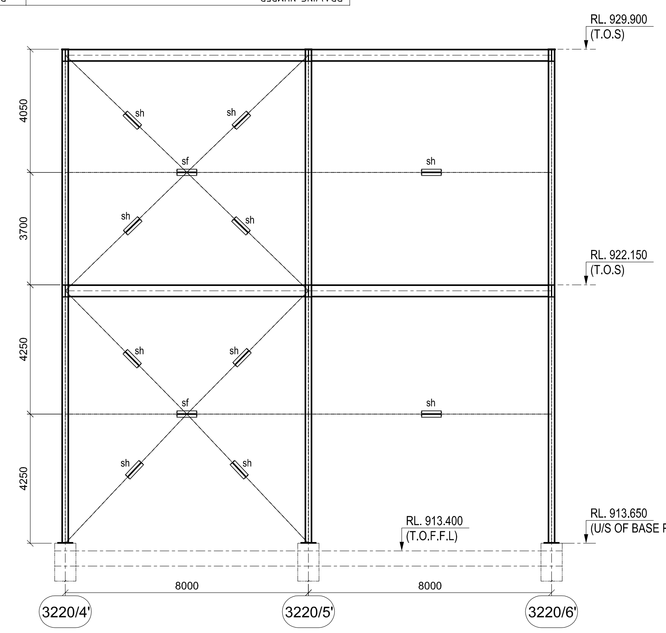
LARSEN & TOUBRO DRAWING NUMBER: O19045_450_S_G_26027_001 | REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ ZA01 _ MTL01 _ 32301 | DOCUMENT ID: OU100664463 | REVISION: 0

DRAWN	BAB	22/02/2020
CHECKED	PRI	25/02/2020
ENGINEER	BAB	24/02/2020
APPROVED	AG	26/02/2020
EMPLOYER		
SCALE	1:100	A0

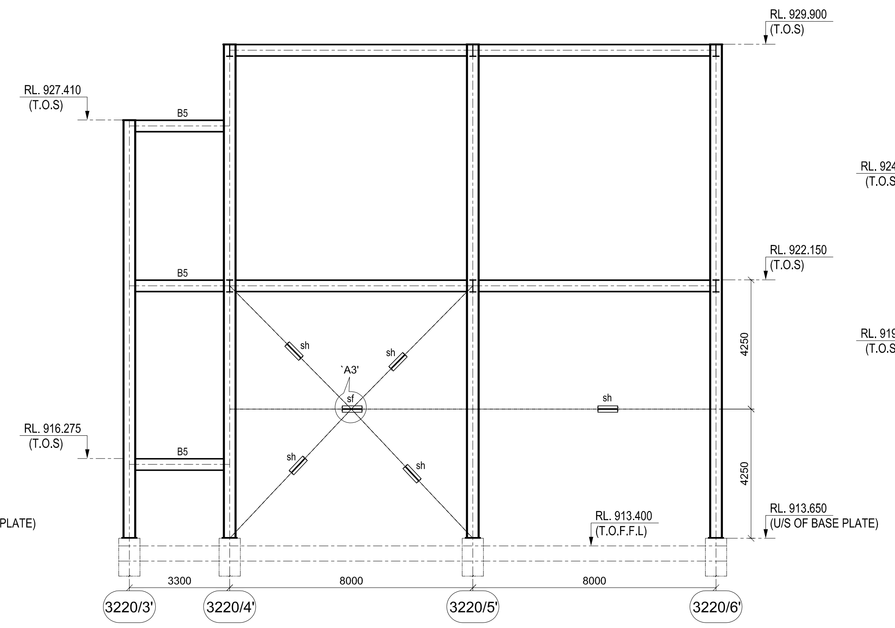
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRAVITY SEPARATION
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER: MD - 216 - 3230 - EG - ST - GAD - 2150 | REV: A01

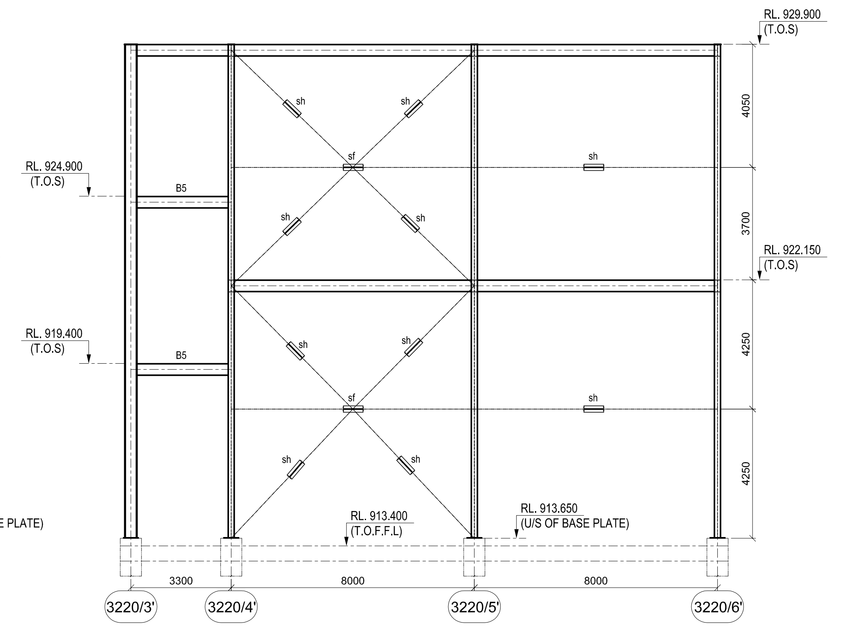
DRAWING NUMBER: MD-216-3230-EG-ST-GAD-2151 | REV: A01



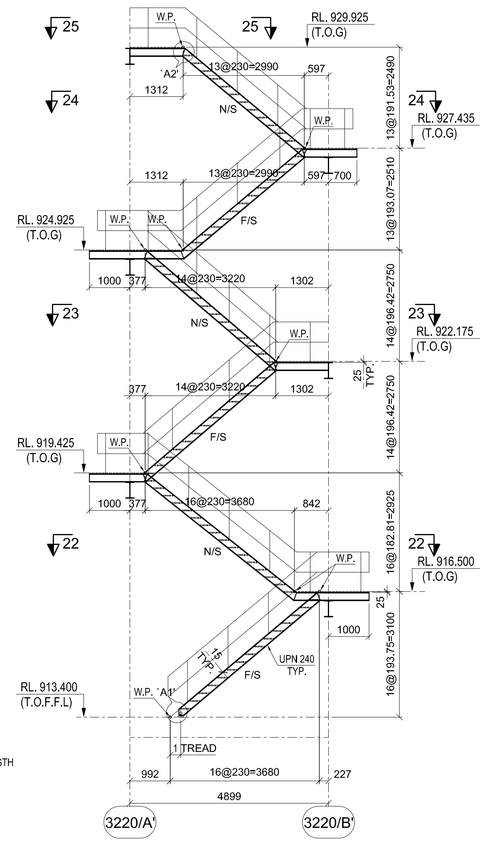
ELEVATION ALONG ROW 3220/B"
(SCALE 1:100)



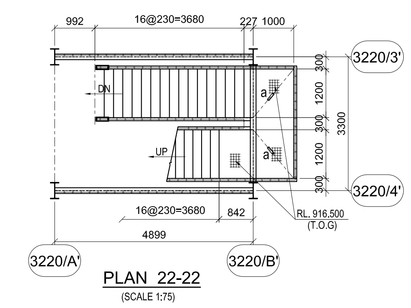
ELEVATION ALONG ROW 3220/B'
(SCALE 1:100)



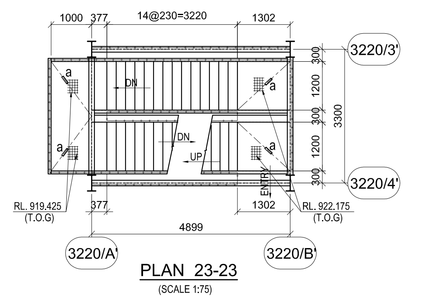
ELEVATION ALONG ROW 3220/A'
(SCALE 1:100)



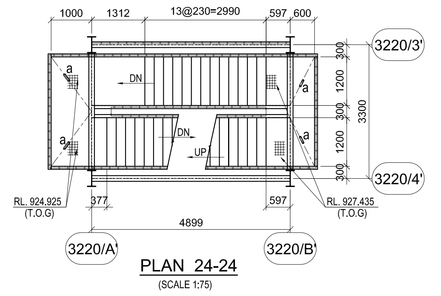
ELEVATION OF STAIRCASE BETWEEN LINE A' & B'
(SCALE 1:75)



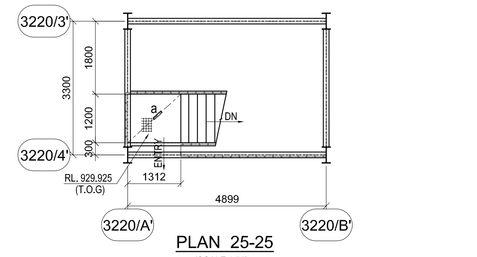
PLAN 22-22
(SCALE 1:75)



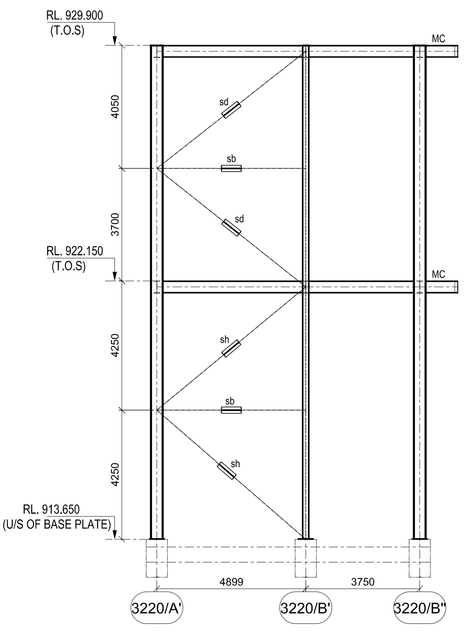
PLAN 23-23
(SCALE 1:75)



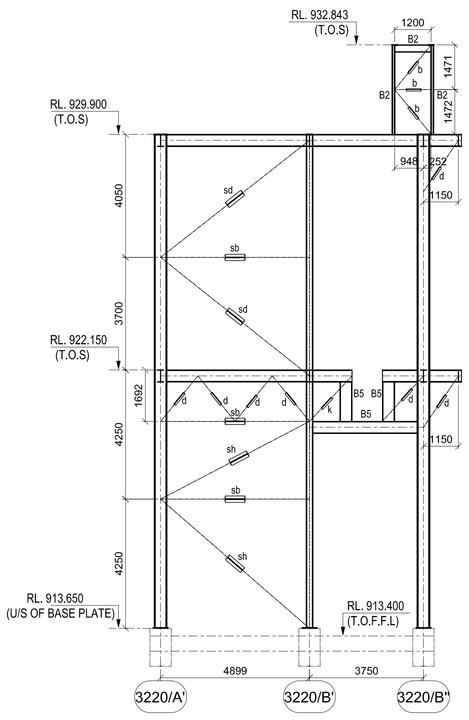
PLAN 24-24
(SCALE 1:75)



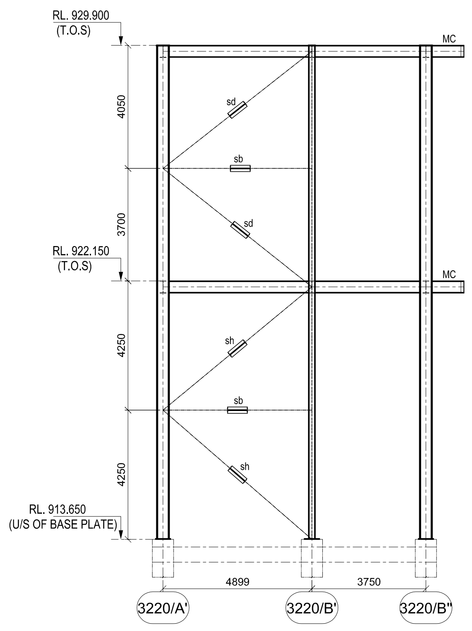
PLAN 25-25
(SCALE 1:75)



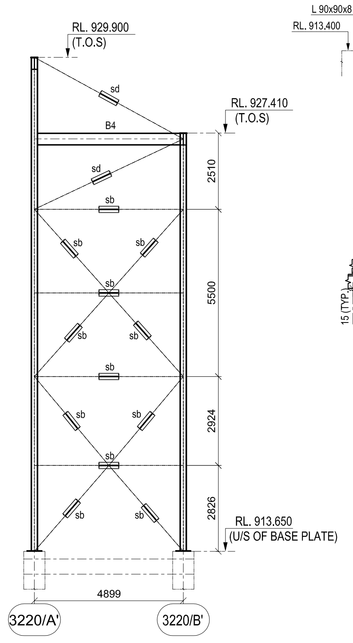
ELEVATION ALONG LINE 3220/6'
(SCALE 1:100)



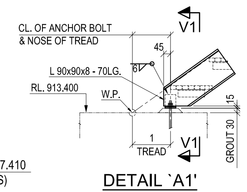
ELEVATION ALONG LINE 3220/5'
(SCALE 1:100)



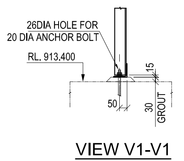
ELEVATION ALONG LINE 3220/4'
(SCALE 1:100)



ELEVATION ALONG LINE 3220/3'
(SCALE 1:100)



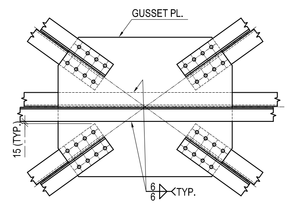
DETAIL `A1'
(SCALE 1:20)



VIEW V1-V1
(SCALE 1:20)



DETAIL `A2'
(SCALE 1:20)



DETAIL `A3'
(SCALE 1:20)

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS ARE IN METERS.
2. FOR OTHER NOTES REFER DRG. NO. MD-216-3230-EG-ST-GAD-2150
3. FOR MEMBER SCHEDULE & LEGENDS REFER DRAWING NO., MD-216-3230-EG-ST-GAD-2150
4. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH THE DRAWING NO., MD-216-3230-EG-ST-GAD-2150

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-CV-DED-1003	DETAIL DRAWING ANCHOR BOLTS STANDARD DETAILS								
MD-216-3220-EG-ME-GAD-1075	BALL MILL SECTION F-F								
MD-216-3220-EG-ME-GAD-1074	BALL MILL SECTION E-E								
MD-216-3220-EG-ME-GAD-1073	BALL MILL SECTION D-D								
MD-216-3220-EG-ME-GAD-1072	BALL MILL SECTION C-C								
MD-216-3220-EG-ME-GAD-1071	BALL MILL SECTION B-B								
MD-216-3220-EG-ME-GAD-1070	BALL MILL SECTION A-A								
MD-216-3220-EG-ME-GAD-1068	BALL MILL PLAN AT EL +924.580								
MD-216-3220-EG-ME-GAD-1067	BALL MILL PLAN AT EL +922.180								
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL DESIGN CRITERIA	A01	26/02/2020	ISSUED FOR REFERENCE	BAB	PRI	BAB	AG	

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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_450_S_G_26027_002			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
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DRAWN	BAB	22/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	PRI	25/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BAB	24/02/2020	GRAVITY SEPARATION
APPROVED	AG	26/02/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE 1:100	A0		DRAWING NUMBER: MD - 216 - 3230 - EG - ST - GAD - 2151
			REV: A01

ATTACHMENT - 2

Isometric View of STAAD Model

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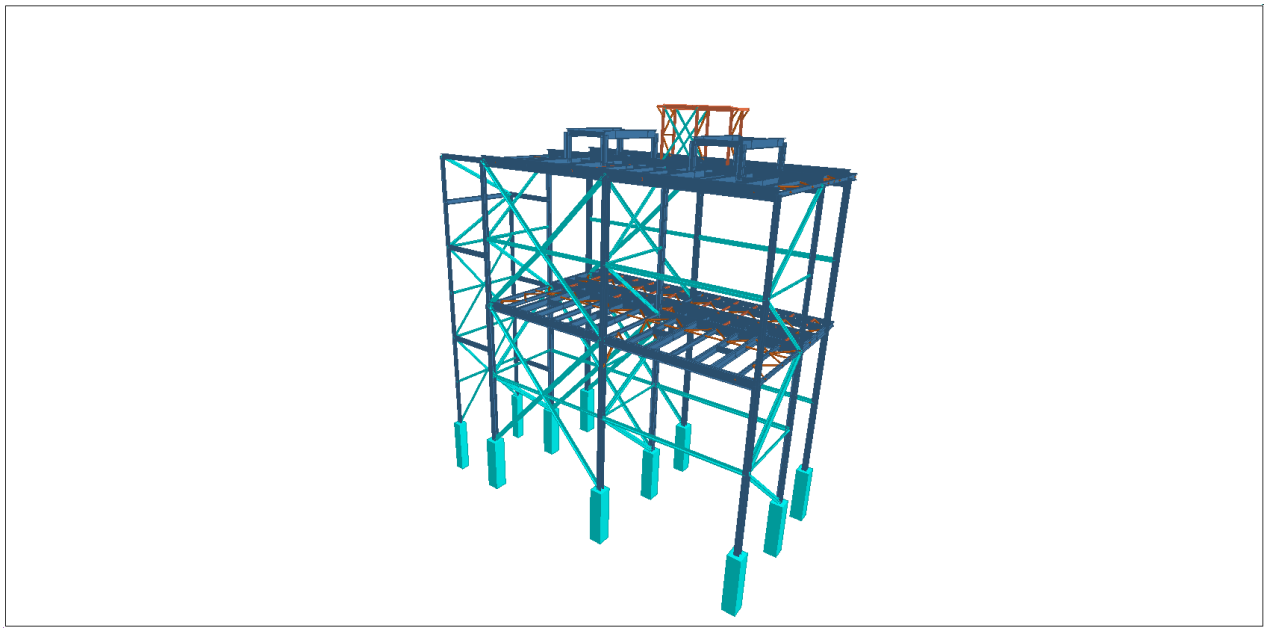
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	Part		
Job Title	Ref		
	By	Date 06-Feb-20	Chd
Client	File Gravity Seperation_R0.st	Date/Time	29-Feb-2020 18:10



3D Rendered View

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	Part		
Job Title	Ref		
	By	Date 06-Feb-20	Chd
Client	File Gravity Seperation_R0.st	Date/Time	29-Feb-2020 18:10



3D Rendered View

ATTACHMENT - 3

STAAD Input

PROPRIETARY & CONFIDENTIAL

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Job Information

	Engineer	Checked	Approved
Name:			
Date:	06-Feb-20		

Project ID	
Project Name	

Structure Type	SPACE FRAME

Number of Nodes	408	Highest Node	415
Number of Elements	779	Highest Beam	3069

Number of Basic Load Cases	17
Number of Combination Load Cases	90

Included in this printout are data for:

All	The Whole Structure

Included in this printout are results for load cases:

Type	L/C	Name
Primary	1	DEAD LOAD (DL)
Primary	2	LIVE LOAD (LL)
Primary	3	OPERATING FLUID LOAD (FL1)
Primary	4	OPERATING FLUID LOAD (FL2)
Primary	5	EQP DL (EQDL)
Primary	6	EQP LL (EQLL)
Primary	7	WIND LOAD (+X) WL(+X)
Primary	8	WIND LOAD (-X) WL(-X)
Primary	9	WIND LOAD (+Z) WL(+Z)
Primary	10	WIND LOAD (-Z) WL(-Z)
Primary	11	SEISMIC LOAD SL(+X)
Primary	12	SEISMIC LOAD SL(-X)
Primary	13	SEISMIC LOAD SL(+Z)
Primary	14	SEISMIC LOAD SL(-Z)
Primary	15	TEMPERATURE LOAD (TL)
Primary	16	PIPE LATERAL LOAD (PL)
Primary	17	EQP TEST LOAD (EQTL)
Combination	101	DL+EQDL+FL1
Combination	102	DL+EQDL+FL2
Combination	103	DL+EQDL+FL1+LL+EQLL+TL
Combination	104	DL+EQDL+FL2+LL+EQLL+TL
Combination	105	DL+EQDL+FL1+0.75(LL+EQLL)+0.75TL
Combination	106	DL+EQDL+FL2+0.75(LL+EQLL)+0.75TL
Combination	107	DL+EQDL+FL1+PL
Combination	108	DL+EQDL+FL2+PL
Combination	109	DL+EQDL+FL1+PL+LL+EQLL+TL
Combination	110	DL+EQDL+FL2+PL+LL+EQLL+TL
Combination	111	DL+EQDL+FL1+PL+0.75(LL+EQLL)+0.75
Combination	112	DL+EQDL+FL2+PL+0.75(LL+EQLL)+0.75

Job Information Cont...

Type	L/C	Name
Combination	116	DL+EQDL+FL1+WL(-Z)
Combination	117	DL+EQDL+FL2+WL(+X)
Combination	118	DL+EQDL+FL2+WL(-X)
Combination	119	DL+EQDL+FL2+WL(+Z)
Combination	120	DL+EQDL+FL2+WL(-Z)
Combination	121	DL+EQDL+FL1+0.7SL(+X)
Combination	122	DL+EQDL+FL1+0.7SL(-X)
Combination	123	DL+EQDL+FL1+0.7SL(+Z)
Combination	124	DL+EQDL+FL1+0.7SL(-Z)
Combination	125	DL+EQDL+FL2+0.7SL(+X)
Combination	126	DL+EQDL+FL2+0.7SL(-X)
Combination	127	DL+EQDL+FL2+0.7SL(+Z)
Combination	128	DL+EQDL+FL2+0.7SL(-Z)
Combination	129	DL+EQDL+FL1+0.75WL(+X)+0.75(LL+EQ
Combination	130	DL+EQDL+FL1+0.75WL(-X)+0.75(LL+EQI
Combination	131	DL+EQDL+FL1+0.75WL(+Z)+0.75(LL+EQ
Combination	132	DL+EQDL+FL1+0.75WL(-Z)+0.75(LL+EQI
Combination	133	DL+EQDL+FL2+0.75WL(+X)+0.75(LL+EQ
Combination	134	DL+EQDL+FL2+0.75WL(-X)+0.75(LL+EQI
Combination	135	DL+EQDL+FL2+0.75WL(+Z)+0.75(LL+EQ
Combination	136	DL+EQDL+FL2+0.75WL(-Z)+0.75(LL+EQI
Combination	137	DL+EQDL+FL1+0.525SL(+X)+0.75(LL+EC
Combination	138	DL+EQDL+FL1+0.525SL(-X)+0.75(LL+EQ
Combination	139	DL+EQDL+FL1+0.525SL(+Z)+0.75(LL+EC
Combination	140	DL+EQDL+FL1+0.525SL(-Z)+0.75(LL+EQ
Combination	141	DL+EQDL+FL2+0.525SL(+X)+0.75(LL+EC
Combination	142	DL+EQDL+FL2+0.525SL(-X)+0.75(LL+EQ
Combination	143	DL+EQDL+FL2+0.525SL(+Z)+0.75(LL+EC
Combination	144	DL+EQDL+FL2+0.525SL(-Z)+0.75(LL+EQ
Combination	145	0.6(DL+EQDL)+WL(+X)
Combination	146	0.6(DL+EQDL)+WL(-X)
Combination	147	0.6(DL+EQDL)+WL(+Z)
Combination	148	0.6(DL+EQDL)+WL(-Z)
Combination	149	0.6(DL+EQDL)+0.7SL(+X)
Combination	150	0.6(DL+EQDL)+0.7SL(-X)
Combination	151	0.6(DL+EQDL)+0.7SL(+Z)
Combination	152	0.6(DL+EQDL)+0.7SL(-Z)
Combination	153	DL+EQTL
Combination	201	1.4(DL+EQDL+FL1)
Combination	202	1.4(DL+EQDL+FL2)
Combination	203	1.4(DL+EQDL+FL1)+1.7(LL+EQLL)
Combination	204	1.4(DL+EQDL+FL2)+1.7(LL+EQLL)
Combination	205	1.4(DL+EQDL+FL1+TL)+1.7(LL+EQLL)
Combination	206	1.4(DL+EQDL+FL2+TL)+1.7(LL+EQLL)
Combination	207	1.4(DL+EQDL+FL1+PL)
Combination	208	1.4(DL+EQDL+FL2+PL)
Combination	209	1.4(DL+EQDL+FL1+PL)+1.7(LL+EQLL)
Combination	210	1.4(DL+EQDL+FL2+PL)+1.7(LL+EQLL)
Combination	211	1.4(DL+EQDL+FL1+PL+TL)+1.7(LL+EQLL

Job Information Cont...

Type	L/C	Name
Combination	212	1.4(DL+EQDL+FL2+PL+TL)+1.7(LL+EQLL
Combination	213	1.2(DL+EQDL)+0.8WL(+X)
Combination	214	1.2(DL+EQDL)+0.8WL(-X)
Combination	215	1.2(DL+EQDL)+0.8WL(+Z)
Combination	216	1.2(DL+EQDL)+0.8WL(-Z)
Combination	217	1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
Combination	218	1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
Combination	219	1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
Combination	220	1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
Combination	221	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5
Combination	222	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5
Combination	223	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5
Combination	224	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5
Combination	225	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5
Combination	226	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5
Combination	227	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5
Combination	228	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5
Combination	229	0.9(DL+EQDL)+1.6WL(+X)
Combination	230	0.9(DL+EQDL)+1.6WL(-X)
Combination	231	0.9(DL+EQDL)+1.6WL(+Z)
Combination	232	0.9(DL+EQDL)+1.6WL(-Z)
Combination	233	0.9(DL+EQDL)+1.6SL(+X)
Combination	234	0.9(DL+EQDL)+1.6SL(-X)
Combination	235	0.9(DL+EQDL)+1.6SL(+Z)
Combination	236	0.9(DL+EQDL)+1.6SL(-Z)
Combination	237	1.2DL+EQTL

Nodes

Node	X (m)	Y (m)	Z (m)
1	432.175	913.650	-371.650
2	432.175	913.650	-366.751
3	435.375	913.650	-375.400
4	443.375	913.650	-375.400
5	451.375	913.650	-375.400
6	435.375	913.650	-371.650
7	443.375	913.650	-371.650
8	451.375	913.650	-371.650
9	435.375	913.650	-366.751
10	443.375	913.650	-366.751
11	451.375	913.650	-366.751
12	432.175	916.475	-371.650
13	435.375	916.475	-371.650
14	432.175	916.475	-369.186
15	432.175	916.475	-366.751
16	435.375	917.900	-375.400
17	439.375	917.900	-375.400

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
18	443.375	917.900	-375.400
19	451.375	917.900	-375.400
20	435.375	917.900	-371.650
21	439.375	917.900	-371.650
22	443.375	917.900	-371.650
23	451.375	917.900	-371.650
24	435.375	917.900	-366.751
25	439.375	917.900	-366.751
26	443.375	917.900	-366.751
27	451.375	917.900	-366.751
29	432.175	919.399	-371.650
30	432.175	919.399	-366.751
31	435.375	919.399	-366.751
32	443.375	920.653	-375.400
33	443.375	920.653	-374.050
34	443.375	920.653	-373.050
35	443.375	920.653	-371.650
36	443.375	920.653	-369.201
37	443.375	920.653	-366.751
38	435.375	922.150	-376.550
39	436.375	922.150	-376.550
40	437.375	922.150	-376.550
41	438.575	922.150	-376.550
42	439.775	922.150	-376.550
43	440.975	922.150	-376.550
44	442.175	922.150	-376.550
45	443.375	922.150	-376.550
46	444.575	922.150	-376.550
47	445.775	922.150	-376.550
48	446.975	922.150	-376.550
49	448.175	922.150	-376.550
50	449.175	922.150	-376.550
51	450.375	922.150	-376.550
52	451.375	922.150	-376.550
53	435.375	922.150	-375.400
54	436.375	922.150	-375.400
55	437.375	922.150	-375.400
56	438.575	922.150	-375.400
57	439.375	922.150	-375.400
58	439.775	922.150	-375.400
59	440.975	922.150	-375.400
60	442.175	922.150	-375.400
61	443.375	922.150	-375.400
62	444.575	922.150	-375.400
63	445.775	922.150	-375.400
64	446.975	922.150	-375.400
65	448.175	922.150	-375.400
66	449.175	922.150	-375.400
67	450.375	922.150	-375.400

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
68	451.375	922.150	-375.400
69	435.375	922.150	-374.050
70	436.375	922.150	-374.050
71	437.375	922.150	-374.050
72	438.575	922.150	-374.050
73	439.775	922.150	-374.050
74	440.975	922.150	-374.050
75	442.175	922.150	-374.050
76	443.375	922.150	-374.050
77	444.575	922.150	-374.050
78	445.775	922.150	-374.050
79	446.975	922.150	-374.050
80	448.175	922.150	-374.050
81	449.175	922.150	-374.050
82	450.080	922.150	-374.050
83	450.375	922.150	-374.050
84	451.375	922.150	-374.050
85	435.375	922.150	-373.050
86	436.375	922.150	-373.050
87	437.375	922.150	-373.050
88	438.575	922.150	-373.050
89	439.775	922.150	-373.050
90	440.975	922.150	-373.050
91	442.175	922.150	-373.050
92	443.375	922.150	-373.050
93	444.575	922.150	-373.050
94	445.775	922.150	-373.050
95	446.975	922.150	-373.050
96	448.175	922.150	-373.050
97	449.175	922.150	-373.050
98	450.080	922.150	-373.050
99	450.375	922.150	-373.050
100	451.375	922.150	-373.050
101	437.375	922.150	-372.700
102	437.875	922.150	-372.700
103	438.575	922.150	-372.700
104	440.975	922.150	-372.700
105	441.275	922.150	-372.700
106	442.175	922.150	-372.700
107	444.575	922.150	-372.700
108	445.375	922.150	-372.700
109	445.775	922.150	-372.700
110	448.175	922.150	-372.700
111	448.775	922.150	-372.700
112	449.175	922.150	-372.700
113	432.175	922.150	-371.650
114	435.375	922.150	-371.650
115	436.375	922.150	-371.650
116	437.375	922.150	-371.650

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
117	438.575	922.150	-371.650
118	439.375	922.150	-371.650
119	439.775	922.150	-371.650
120	440.975	922.150	-371.650
121	442.175	922.150	-371.650
122	443.375	922.150	-371.650
123	444.575	922.150	-371.650
124	445.775	922.150	-371.650
125	446.975	922.150	-371.650
126	448.175	922.150	-371.650
127	449.175	922.150	-371.650
128	450.375	922.150	-371.650
129	451.375	922.150	-371.650
131	435.375	922.150	-370.425
132	436.375	922.150	-370.425
133	437.375	922.150	-370.425
134	438.575	922.150	-370.425
135	439.775	922.150	-370.425
136	440.975	922.150	-370.425
137	442.175	922.150	-370.425
138	443.375	922.150	-370.425
139	444.575	922.150	-370.425
140	445.775	922.150	-370.425
141	446.975	922.150	-370.425
142	448.175	922.150	-370.425
143	449.175	922.150	-370.425
144	450.375	922.150	-370.425
145	451.375	922.150	-370.425
146	435.375	922.150	-369.507
147	437.375	922.150	-369.300
148	437.875	922.150	-369.300
149	438.575	922.150	-369.300
150	440.975	922.150	-369.300
151	441.275	922.150	-369.300
152	442.175	922.150	-369.300
153	444.575	922.150	-369.300
154	445.375	922.150	-369.300
155	445.775	922.150	-369.300
156	448.175	922.150	-369.300
157	448.775	922.150	-369.300
158	449.175	922.150	-369.300
159	436.375	922.150	-368.588
160	443.375	922.150	-367.976
161	435.375	922.150	-367.669
162	435.375	922.150	-366.751
163	436.375	922.150	-366.751
164	437.375	922.150	-366.751
165	438.575	922.150	-366.751
166	439.375	922.150	-366.751

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
167	439.775	922.150	-366.751
168	440.975	922.150	-366.751
169	442.175	922.150	-366.751
170	443.375	922.150	-366.751
171	444.575	922.150	-366.751
172	445.775	922.150	-366.751
173	446.975	922.150	-366.751
174	448.175	922.150	-366.751
175	449.175	922.150	-366.751
176	450.375	922.150	-366.751
177	451.375	922.150	-366.751
178	432.175	924.899	-371.650
179	432.175	924.899	-366.751
180	435.375	924.899	-366.751
181	435.375	925.850	-375.400
182	439.376	925.850	-375.400
183	443.375	925.850	-375.400
184	451.375	925.850	-375.400
185	435.375	925.850	-371.650
186	443.375	925.850	-371.650
187	451.375	925.850	-371.650
188	435.375	925.850	-366.751
189	439.377	925.850	-366.751
190	443.375	925.850	-366.751
191	451.375	925.850	-366.751
192	432.175	927.409	-371.650
193	435.375	927.409	-371.650
194	432.175	927.409	-366.751
195	435.375	929.900	-376.550
196	436.375	929.900	-376.550
197	437.375	929.900	-376.550
198	438.487	929.900	-376.550
199	439.575	929.900	-376.550
200	440.663	929.900	-376.550
201	441.275	929.900	-376.550
202	442.175	929.900	-376.550
203	443.375	929.900	-376.550
204	444.575	929.900	-376.550
205	445.375	929.900	-376.550
206	446.025	929.900	-376.550
207	447.075	929.900	-376.550
208	448.125	929.900	-376.550
209	449.375	929.900	-376.550
210	450.375	929.900	-376.550
211	451.375	929.900	-376.550
212	438.487	929.900	-375.660
213	440.663	929.900	-375.660
214	446.025	929.900	-375.660
215	448.125	929.900	-375.660

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
216	441.275	929.900	-375.652
217	443.375	929.900	-375.652
218	445.375	929.900	-375.652
219	435.375	929.900	-375.400
220	436.375	929.900	-375.400
221	437.375	929.900	-375.400
222	438.487	929.900	-375.400
223	439.575	929.900	-375.400
224	440.663	929.900	-375.400
225	441.275	929.900	-375.400
226	442.175	929.900	-375.400
227	443.375	929.900	-375.400
228	444.575	929.900	-375.400
229	445.375	929.900	-375.400
230	446.025	929.900	-375.400
231	447.075	929.900	-375.400
232	448.125	929.900	-375.400
233	449.375	929.900	-375.400
234	450.375	929.900	-375.400
235	451.375	929.900	-375.400
236	438.487	929.900	-374.902
237	440.663	929.900	-374.902
238	446.025	929.900	-374.902
239	448.125	929.900	-374.902
240	437.375	929.900	-374.452
241	438.375	929.900	-374.452
242	438.487	929.900	-374.452
243	440.663	929.900	-374.452
244	440.775	929.900	-374.452
245	441.275	929.900	-374.452
246	442.175	929.900	-374.452
247	443.375	929.900	-374.452
248	444.575	929.900	-374.452
249	445.375	929.900	-374.452
250	445.875	929.900	-374.452
251	446.025	929.900	-374.452
252	448.125	929.900	-374.452
253	448.275	929.900	-374.452
254	449.375	929.900	-374.452
255	435.375	929.900	-374.050
256	436.375	929.900	-374.050
257	437.375	929.900	-374.050
258	438.487	929.900	-374.050
259	439.575	929.900	-374.050
260	440.663	929.900	-374.050
261	442.175	929.900	-374.050
262	443.375	929.900	-374.050
263	444.575	929.900	-374.050
264	446.025	929.900	-374.050

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
265	447.075	929.900	-374.050
266	448.125	929.900	-374.050
267	449.375	929.900	-374.050
268	450.375	929.900	-374.050
269	451.375	929.900	-374.050
270	435.375	929.900	-373.250
271	436.375	929.900	-373.250
272	437.375	929.900	-373.250
273	438.487	929.900	-373.250
274	439.575	929.900	-373.250
275	440.663	929.900	-373.250
276	442.175	929.900	-373.250
277	443.375	929.900	-373.250
278	444.575	929.900	-373.250
279	446.025	929.900	-373.250
280	447.075	929.900	-373.250
281	448.125	929.900	-373.250
282	449.375	929.900	-373.250
283	450.375	929.900	-373.250
284	451.375	929.900	-373.250
285	435.375	929.900	-372.450
286	435.375	929.900	-371.650
287	436.375	929.900	-371.650
288	437.375	929.900	-371.650
289	438.487	929.900	-371.650
290	439.575	929.900	-371.650
291	440.663	929.900	-371.650
292	442.175	929.900	-371.650
293	443.375	929.900	-371.650
294	444.575	929.900	-371.650
295	446.025	929.900	-371.650
296	447.075	929.900	-371.650
297	448.125	929.900	-371.650
298	449.375	929.900	-371.650
299	450.375	929.900	-371.650
300	451.375	929.900	-371.650
301	438.487	929.900	-370.849
302	439.575	929.900	-370.849
303	440.663	929.900	-370.849
304	446.025	929.900	-370.849
305	447.075	929.900	-370.849
306	448.125	929.900	-370.849
307	435.375	929.900	-370.401
308	442.175	929.900	-370.540
309	443.375	929.900	-370.540
310	444.575	929.900	-370.540
311	437.375	929.900	-369.901
312	438.375	929.900	-369.901
313	438.487	929.900	-369.901

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
314	440.663	929.900	-369.901
315	440.775	929.900	-369.901
316	442.175	929.900	-369.901
317	444.575	929.900	-369.901
318	445.875	929.900	-369.901
319	446.025	929.900	-369.901
320	448.125	929.900	-369.901
321	448.275	929.900	-369.901
322	449.375	929.900	-369.901
323	436.375	929.900	-369.201
324	435.375	929.900	-368.000
325	432.175	929.900	-366.751
326	435.375	929.900	-366.751
327	436.375	929.900	-366.751
328	437.375	929.900	-366.751
329	438.487	929.900	-366.751
330	439.575	929.900	-366.751
331	440.663	929.900	-366.751
332	442.175	929.900	-366.751
333	443.375	929.900	-366.751
334	444.575	929.900	-366.751
335	446.025	929.900	-366.751
336	447.075	929.900	-366.751
337	448.125	929.900	-366.751
338	449.375	929.900	-366.751
339	450.375	929.900	-366.751
340	451.375	929.900	-366.751
341	441.275	931.372	-375.652
342	443.375	931.372	-375.652
343	445.375	931.372	-375.652
344	441.275	931.372	-374.452
345	443.375	931.372	-374.452
346	445.375	931.372	-374.452
347	438.375	931.400	-374.452
348	440.775	931.400	-374.452
349	445.875	931.400	-374.452
350	448.275	931.400	-374.452
351	438.375	931.400	-373.752
352	440.775	931.400	-373.752
353	445.875	931.400	-373.752
354	448.275	931.400	-373.752
355	438.375	931.400	-370.802
356	440.775	931.400	-370.802
357	445.875	931.400	-370.802
358	448.275	931.400	-370.802
359	438.375	931.400	-369.901
360	440.775	931.400	-369.901
361	445.875	931.400	-369.901
362	448.275	931.400	-369.901

Nodes Cont...

Node	X (m)	Y (m)	Z (m)
363	440.862	932.843	-375.652
364	441.275	932.843	-375.652
365	442.325	932.843	-375.652
366	443.375	932.843	-375.652
367	445.375	932.843	-375.652
368	445.712	932.843	-375.652
369	440.862	932.843	-374.452
370	441.275	932.843	-374.452
371	443.375	932.843	-374.452
372	445.375	932.843	-374.452
373	445.712	932.843	-374.452
374	443.092	929.900	-370.540
375	443.892	929.900	-370.540
376	443.092	932.843	-374.452
377	443.892	932.843	-374.452
378	450.375	922.150	-368.588
380	451.375	922.150	-369.507
381	451.375	922.150	-367.669
382	450.375	929.900	-369.201
384	451.375	929.900	-370.425
385	451.375	929.900	-367.976
386	443.375	928.750	-375.400
387	451.375	929.900	-372.450
388	432.175	922.150	-366.751
390	432.175	922.150	-369.200
391	441.419	929.900	-373.250
392	445.300	929.900	-373.250
393	441.419	929.900	-371.650
394	441.419	929.900	-369.901
395	441.419	929.900	-366.751
396	445.300	929.900	-371.650
397	445.300	929.900	-369.901
398	445.300	929.900	-366.751
399	441.275	932.343	-374.452
400	441.275	932.343	-375.652
403	445.375	932.343	-374.452
404	445.375	932.343	-375.652
405	432.175	910.500	-371.650
406	432.175	910.500	-366.751
407	435.375	910.500	-375.400
408	443.375	910.500	-375.400
409	451.375	910.500	-375.400
410	435.375	910.500	-371.650
411	443.375	910.500	-371.650
412	451.375	910.500	-371.650
413	435.375	910.500	-366.751
414	443.375	910.500	-366.751
415	451.375	910.500	-366.751

Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
74	309	375	0.517	12	0
79	308	374	0.917	12	0
192	179	390	3.682	5	0
199	219	255	1.350	12	0
206	234	235	1.000	20	0
212	339	340	1.000	19	0
218	235	269	1.350	12	0
224	211	235	1.150	15	0
230	195	219	1.150	15	0
236	227	247	0.948	15	0
242	299	300	1.000	20	0
249	327	324	1.600	1	0
255	324	323	1.563	1	0
261	323	307	1.562	1	0
267	307	287	1.600	1	0
272	287	285	1.281	1	0
277	285	271	1.281	1	0
282	271	255	1.281	1	0
288	268	269	1.000	1	0
293	267	268	1.000	1	0
298	266	267	1.250	1	0
303	265	266	1.050	1	0
313	263	264	1.450	9	0
318	262	263	1.200	1	0
323	261	262	1.200	1	0
328	260	261	1.512	9	0
333	259	260	1.088	1	0
343	257	258	1.112	1	0
348	256	257	1.000	1	0
352	255	256	1.000	1	0
358	234	269	1.680	1	0
363	267	234	1.680	1	0
418	220	257	1.680	1	0
422	255	220	1.680	1	0
567	22	23	8.000	8	0
579	21	22	4.000	7	0
584	20	21	4.000	7	0
591	21	6	5.836	8	0
603	7	21	5.836	8	0
615	61	182	5.448	8	0
622	182	219	5.693	8	0
636	227	182	5.692	8	0
648	182	53	5.450	8	0
660	183	184	8.000	8	0
671	183	182	3.999	7	0
676	181	182	4.001	7	0
707	18	19	8.000	8	0
718	18	17	4.000	7	0
723	16	17	4.000	7	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
730	17	3	5.836	8	0
742	4	17	5.836	8	0
754	170	189	5.447	8	0
766	189	326	5.694	8	0
780	333	189	5.691	8	0
792	189	162	5.450	8	0
804	190	191	8.000	8	0
815	190	189	3.998	7	0
820	188	189	4.002	7	0
851	26	27	8.000	8	0
862	25	26	4.000	7	0
867	24	25	4.000	7	0
874	25	9	5.836	8	0
886	10	25	5.836	8	0
898	300	191	6.356	6	0
914	191	129	6.139	6	0
921	187	191	4.899	5	0
933	129	27	6.486	8	0
944	8	27	6.486	8	0
961	27	23	4.899	5	0
968	293	190	6.356	6	0
984	190	122	6.139	6	0
991	186	190	4.899	5	0
1015	7	26	6.486	8	0
1034	26	22	4.899	5	0
1041	286	188	6.356	6	0
1052	114	188	6.139	6	0
1065	185	188	4.899	5	0
1089	6	24	6.486	8	0
1101	20	24	4.899	5	0
1113	192	325	5.496	6	0
1125	192	179	5.505	6	0
1142	14	29	3.824	5	0
1149	14	30	3.805	5	0
1161	1	14	3.749	5	0
1173	14	2	3.730	5	0
1185	12	14	2.464	5	0
1190	14	15	2.435	5	0
1197	1	12	2.825	12	0
1203	128	144	1.225	12	0
1209	233	254	0.948	15	0
1215	234	268	1.350	12	0
1221	232	239	0.498	17	0
1227	231	265	1.350	12	0
1233	230	238	0.498	17	0
1239	228	248	0.948	15	0
1245	293	309	1.110	15	0
1251	226	246	0.948	15	0
1257	224	237	0.498	17	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1263	223	259	1.350	12	0
1269	222	236	0.498	17	0
1275	221	240	0.948	15	0
1287	61	76	1.350	12	0
1293	62	77	1.350	9	0
1299	63	78	1.350	9	0
1305	64	79	1.350	9	0
1311	65	80	1.350	9	0
1317	66	81	1.350	9	0
1323	67	83	1.350	9	0
1329	60	75	1.350	9	0
1335	59	74	1.350	9	0
1341	58	73	1.350	9	0
1347	56	72	1.350	9	0
1353	55	71	1.350	9	0
1359	54	70	1.350	9	0
1366	115	132	1.225	12	0
1372	116	133	1.225	16	0
1378	117	134	1.225	15	0
1384	119	135	1.225	13	0
1390	120	136	1.225	16	0
1396	121	137	1.225	15	0
1402	122	138	1.225	12	0
1408	123	139	1.225	15	0
1414	124	140	1.225	16	0
1420	125	141	1.225	13	0
1426	126	142	1.225	15	0
1432	127	143	1.225	16	0
1437	287	323	2.449	12	0
1442	288	311	1.749	15	0
1447	289	301	0.801	18	0
1452	291	303	0.801	18	0
1457	292	308	1.110	15	0
1462	295	304	0.801	18	0
1467	297	306	0.801	18	0
1472	298	322	1.749	15	0
1477	299	382	2.449	12	0
1482	294	310	1.110	15	0
1489	302	303	1.088	12	0
1495	305	306	1.050	12	0
1501	302	330	4.098	12	0
1507	305	336	4.098	12	0
1512	86	115	1.400	9	0
1517	88	103	0.350	13	0
1522	87	101	0.350	13	0
1527	95	125	1.400	9	0
1532	96	110	0.350	13	0
1537	97	112	0.350	13	0
1542	99	128	1.400	9	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1547	89	119	1.400	9	0
1552	90	104	0.350	13	0
1557	92	122	1.400	9	0
1562	91	106	0.350	13	0
1567	93	107	0.350	13	0
1572	94	109	0.350	13	0
1579	99	100	1.000	17	0
1585	83	84	1.000	13	0
1591	67	68	1.000	16	0
1597	226	227	1.200	20	0
1604	325	326	3.200	14	0
1610	179	180	3.200	13	0
1616	30	31	3.200	13	0
1622	192	193	3.200	13	0
1628	12	13	3.200	13	0
1634	113	114	3.200	13	0
1641	192	194	4.899	13	0
1647	178	179	4.899	5	0
1653	29	30	4.899	5	0
1659	2	15	2.825	12	0
1664	210	234	1.150	9	0
1669	196	220	1.150	9	0
1674	209	233	1.150	9	0
1679	208	215	0.890	12	0
1684	207	231	1.150	9	0
1689	206	214	0.890	12	0
1694	204	228	1.150	9	0
1699	197	221	1.150	9	0
1704	198	212	0.890	12	0
1709	199	223	1.150	9	0
1714	200	213	0.890	12	0
1719	202	226	1.150	9	0
1724	51	67	1.150	9	0
1729	39	54	1.150	9	0
1734	50	66	1.150	9	0
1739	49	65	1.150	9	0
1744	48	64	1.150	9	0
1749	47	63	1.150	9	0
1754	46	62	1.150	9	0
1759	40	55	1.150	9	0
1764	41	56	1.150	9	0
1769	42	58	1.150	9	0
1774	43	59	1.150	9	0
1779	44	60	1.150	9	0
1785	176	177	1.000	18	0
1791	128	129	1.000	19	0
1797	38	53	1.150	13	0
1809	45	61	1.150	13	0
1815	203	217	0.898	15	0

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Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1821	52	68	1.150	13	0
1827	300	384	1.225	12	0
1833	332	333	1.200	19	0
1839	292	293	1.200	20	0
1845	51	52	1.000	13	0
1851	210	211	1.000	15	0
1857	3	16	4.250	22	90
1863	6	13	2.825	22	0
1869	5	19	4.250	22	90
1875	4	18	4.250	22	90
1881	7	22	4.250	22	0
1892	10	26	4.250	22	90
1898	8	23	4.250	22	0
1904	11	27	4.250	22	90
1936	232	233	1.250	20	0
1939	334	398	0.725	19	0
1942	336	337	1.050	19	0
1945	337	338	1.250	19	0
1948	338	339	1.000	19	0
1954	297	298	1.250	20	0
1957	298	299	1.000	20	0
1999	309	333	3.789	15	0
2002	301	313	0.948	18	0
2005	303	314	0.948	18	0
2008	308	316	0.639	15	0
2011	304	319	0.948	18	0
2014	306	320	0.948	18	0
2017	310	317	0.639	15	0
2020	301	302	1.088	12	0
2023	304	305	1.050	12	0
2026	97	98	0.905	17	0
2029	98	99	0.295	17	0
2032	81	82	0.905	13	0
2035	82	83	0.295	13	0
2038	60	61	1.200	16	0
2041	61	62	1.200	16	0
2044	62	63	1.200	16	0
2047	63	64	1.200	16	0
2050	64	65	1.200	16	0
2053	65	66	1.000	16	0
2056	66	67	1.200	16	0
2059	225	226	0.900	20	0
2068	15	30	2.924	12	0
2071	175	176	1.200	18	0
2074	127	128	1.200	19	0
2077	85	114	1.400	16	0
2080	69	85	1.000	16	0
2083	53	69	1.350	16	0
2086	217	227	0.252	15	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
2089	100	129	1.400	16	0
2092	84	100	1.000	16	0
2095	68	84	1.350	16	0
2101	327	328	1.000	19	0
2104	328	329	1.112	19	0
2107	330	331	1.088	19	0
2110	331	395	0.756	19	0
2113	291	393	0.756	20	0
2116	50	51	1.200	13	0
2119	209	210	1.000	15	0
2125	16	53	4.250	22	90
2131	20	114	4.250	22	0
2134	13	20	1.425	22	0
2137	184	235	4.050	22	90
2140	19	68	4.250	22	90
2158	24	31	1.499	22	90
2167	26	37	2.753	22	90
2170	129	187	3.700	21	0
2176	27	177	4.250	22	90
2179	231	232	1.050	20	0
2182	333	334	1.200	19	0
2185	335	336	1.050	19	0
2188	296	297	1.050	20	0
2194	113	178	2.749	12	0
2197	29	113	2.751	12	0
2200	85	86	1.000	17	0
2203	87	88	1.200	17	0
2206	94	95	1.200	17	0
2209	95	96	1.200	17	0
2212	96	97	1.000	17	0
2215	75	76	1.200	13	0
2218	76	77	1.200	13	0
2221	77	78	1.200	13	0
2224	78	79	1.200	13	0
2227	79	80	1.200	13	0
2230	80	81	1.000	13	0
2233	59	60	1.200	16	0
2236	223	224	1.088	20	0
2242	179	194	2.510	12	0
2245	30	388	2.751	12	0
2248	162	163	1.000	18	0
2251	163	164	1.000	18	0
2254	164	165	1.200	18	0
2257	166	167	0.400	18	0
2260	167	168	1.200	18	0
2263	168	169	1.200	18	0
2266	169	170	1.200	18	0
2269	170	171	1.200	18	0
2272	171	172	1.200	18	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
2275	172	173	1.200	18	0
2278	173	174	1.200	18	0
2281	174	175	1.000	18	0
2284	114	115	1.000	19	0
2287	115	116	1.000	19	0
2290	116	117	1.200	19	0
2293	118	119	0.400	19	0
2296	119	120	1.200	19	0
2299	120	121	1.200	19	0
2302	121	122	1.200	19	0
2305	122	123	1.200	19	0
2308	123	124	1.200	19	0
2311	124	125	1.200	19	0
2314	125	126	1.200	19	0
2317	126	127	1.000	19	0
2320	114	131	1.225	12	0
2323	129	145	1.225	12	0
2326	329	330	1.088	19	0
2329	290	291	1.088	20	0
2332	38	39	1.000	13	0
2335	49	50	1.000	13	0
2338	195	196	1.000	15	0
2341	208	209	1.250	15	0
2344	181	219	4.050	22	90
2347	53	181	3.700	22	90
2353	185	193	1.559	21	0
2356	114	185	3.700	21	0
2365	68	184	3.700	22	90
2371	183	386	2.900	23	90
2374	61	183	3.700	22	90
2377	122	186	3.700	22	0
2383	180	188	0.951	21	90
2386	162	180	2.749	21	90
2389	31	162	2.751	22	90
2392	190	333	4.050	22	90
2395	170	190	3.700	22	90
2398	191	340	4.050	21	90
2401	177	191	3.700	21	90
2404	230	231	1.050	20	0
2407	295	296	1.050	20	0
2410	178	192	2.510	12	0
2413	86	87	1.000	17	0
2416	88	89	1.200	17	0
2419	89	90	1.200	17	0
2422	91	92	1.200	17	0
2425	92	93	1.200	17	0
2428	93	94	1.200	17	0
2431	74	75	1.200	13	0
2434	58	59	1.200	16	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
2437	222	223	1.088	20	0
2440	289	290	1.088	20	0
2443	48	49	1.200	13	0
2446	207	208	1.050	15	0
2449	193	286	2.491	21	0
2452	188	326	4.050	21	90
2455	229	230	0.650	20	0
2458	294	396	0.725	20	0
2461	90	91	1.200	17	0
2464	73	74	1.200	13	0
2467	57	58	0.400	16	0
2470	221	222	1.112	20	0
2473	288	289	1.112	20	0
2476	47	48	1.200	13	0
2479	206	207	1.050	15	0
2482	227	228	1.200	20	0
2485	293	294	1.200	20	0
2488	72	73	1.200	13	0
2491	55	56	1.200	16	0
2494	220	221	1.000	20	0
2497	287	288	1.000	20	0
2500	46	47	1.200	13	0
2503	205	206	0.650	15	0
2506	71	72	1.200	13	0
2509	54	55	1.000	16	0
2512	219	220	1.000	20	0
2515	286	287	1.000	20	0
2518	45	46	1.200	13	0
2521	203	204	1.200	15	0
2524	70	71	1.000	13	0
2527	53	54	1.000	16	0
2530	39	40	1.000	13	0
2533	40	41	1.200	13	0
2536	41	42	1.200	13	0
2539	42	43	1.200	13	0
2542	43	44	1.200	13	0
2545	44	45	1.200	13	0
2548	196	197	1.000	15	0
2551	197	198	1.112	15	0
2554	198	199	1.088	15	0
2557	199	200	1.088	15	0
2560	201	202	0.900	15	0
2563	202	203	1.200	15	0
2566	69	70	1.000	13	0
2600	24	114	6.486	8	0
2607	9	24	4.250	22	90
2629	212	222	0.260	12	0
2630	213	224	0.260	12	0
2631	236	242	0.450	17	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
2632	237	243	0.450	17	0
2633	242	258	0.402	17	0
2634	243	260	0.402	17	0
2635	313	329	3.150	18	0
2636	314	331	3.150	18	0
2637	312	359	1.500	22	0
2638	315	360	1.500	22	0
2639	241	347	1.500	22	0
2640	244	348	1.500	22	0
2641	348	352	0.700	17	0
2642	347	351	0.700	17	0
2643	347	348	2.400	14	0
2644	359	360	2.400	14	0
2645	240	257	0.402	15	0
2646	242	241	0.112	15	0
2647	246	261	0.402	15	0
2648	243	244	0.112	15	0
2649	311	328	3.150	15	0
2650	313	312	0.112	15	0
2651	316	332	3.150	15	0
2652	314	315	0.112	15	0
2653	311	312	1.000	15	0
2654	241	240	1.000	15	0
2655	315	394	0.644	15	0
2656	244	245	0.500	15	0
2661	214	230	0.260	12	0
2662	215	232	0.260	12	0
2663	238	251	0.450	17	0
2665	239	252	0.450	17	0
2666	248	263	0.402	15	0
2668	251	264	0.402	17	0
2669	248	249	0.800	15	0
2670	252	266	0.402	17	0
2672	254	267	0.402	15	0
2673	252	253	0.150	15	0
2674	317	334	3.150	15	0
2676	319	335	3.150	18	0
2677	317	397	0.725	15	0
2678	320	337	3.150	18	0
2680	322	338	3.150	15	0
2681	320	321	0.150	15	0
2682	253	254	1.100	15	0
2683	318	319	0.150	15	0
2684	318	361	1.500	22	0
2685	321	322	1.100	15	0
2686	321	362	1.500	22	0
2687	250	251	0.150	15	0
2688	250	349	1.500	22	0
2689	253	350	1.500	22	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
2690	349	350	2.400	14	0
2691	349	353	0.700	17	0
2692	361	362	2.400	14	0
2693	350	354	0.700	17	0
2694	351	355	2.950	17	0
2695	352	356	2.950	17	0
2696	353	357	2.950	17	0
2697	354	358	2.950	17	0
2698	355	359	0.901	17	0
2699	356	360	0.901	17	0
2700	357	361	0.901	17	0
2702	358	362	0.901	17	0
2703	245	246	0.900	15	0
2704	245	344	1.472	10	0
2705	216	341	1.472	10	0
2706	224	225	0.612	20	0
2708	225	216	0.252	12	0
2709	200	201	0.612	15	0
2710	216	201	0.898	12	0
2711	364	370	1.200	10	0
2714	217	342	1.472	10	0
2715	364	365	1.050	10	0
2716	247	262	0.402	15	0
2717	371	366	1.200	10	0
2718	247	345	1.472	10	0
2720	369	363	1.200	10	0
2721	369	370	0.413	10	0
2722	370	376	1.817	11	0
2723	373	368	1.200	10	0
2724	371	377	0.517	11	0
2725	372	373	0.337	10	0
2726	249	250	0.500	15	0
2727	249	346	1.472	10	0
2728	218	343	1.472	10	0
2729	228	229	0.800	20	0
2730	218	229	0.252	12	0
2731	204	205	0.800	15	0
2732	218	205	0.898	12	0
2733	372	367	1.200	10	0
2734	366	367	2.000	10	0
2735	363	364	0.413	10	0
2736	367	368	0.337	10	0
2737	255	270	0.800	12	0
2738	268	283	0.800	12	0
2739	269	284	0.800	12	0
2740	267	282	0.800	15	0
2741	265	280	0.800	12	0
2742	263	278	0.800	15	0
2743	262	277	0.800	15	0

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Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
2744	261	276	0.800	15	0
2745	257	272	0.800	15	0
2746	256	271	0.800	12	0
2747	233	234	1.000	20	0
2748	258	259	1.088	1	0
2749	258	273	0.800	17	0
2750	259	274	0.800	12	0
2752	260	275	0.800	17	0
2753	264	265	1.050	1	0
2754	264	279	0.800	17	0
2756	266	281	0.800	17	0
2757	263	227	1.806	2	0
2758	227	261	1.806	2	0
2759	326	327	1.000	19	0
2760	324	326	1.249	12	0
2761	307	324	2.401	12	0
2763	285	286	0.800	12	0
2764	271	287	1.600	12	0
2765	220	256	1.350	12	0
2766	286	307	1.249	12	0
2767	323	327	2.450	12	0
2768	187	300	4.050	21	0
2770	23	129	4.250	22	0
2772	186	293	4.050	22	0
2774	165	166	0.800	18	0
2776	56	57	0.800	16	0
2777	161	162	0.918	12	0
2778	146	161	1.838	12	0
2779	272	288	1.600	15	0
2780	271	272	1.000	1	0
2781	273	289	1.600	17	0
2782	272	273	1.112	1	0
2783	274	290	1.600	12	0
2784	273	274	1.088	1	0
2785	275	291	1.600	17	0
2786	274	275	1.088	1	0
2787	276	292	1.600	15	0
2788	275	391	0.756	9	0
2789	277	293	1.600	15	0
2790	276	277	1.200	1	0
2791	278	294	1.600	15	0
2792	277	278	1.200	1	0
2793	279	295	1.600	17	0
2794	278	392	0.725	9	0
2795	280	296	1.600	12	0
2796	279	280	1.050	1	0
2797	281	297	1.600	17	0
2798	280	281	1.050	1	0
2799	282	298	1.600	15	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
2800	281	282	1.250	1	0
2801	283	299	1.600	12	0
2802	282	283	1.000	1	0
2803	284	387	0.800	12	0
2804	283	284	1.000	1	0
2805	271	257	1.281	1	0
2806	257	273	1.370	1	0
2807	273	259	1.350	1	0
2808	259	275	1.350	2	0
2809	275	261	1.711	2	0
2810	261	277	1.442	2	0
2811	277	263	1.442	2	0
2812	263	279	1.656	2	0
2813	279	265	1.320	2	0
2814	265	281	1.320	1	0
2815	281	267	1.484	1	0
2816	267	283	1.281	1	0
2817	283	269	1.281	1	0
2818	247	370	3.615	4	0
2819	245	371	3.615	4	0
2820	216	366	3.615	4	0
2821	217	364	3.615	4	0
2822	365	366	1.050	10	0
2823	370	365	1.595	1	0
2824	365	371	1.595	1	0
2825	344	399	0.971	10	0
2826	341	400	0.971	10	0
2827	342	366	1.471	10	0
2828	345	371	1.471	10	0
2829	346	403	0.971	10	0
2830	343	404	0.971	10	0
2831	216	344	1.899	1	0
2832	344	364	1.898	1	0
2833	217	345	1.899	1	0
2834	345	366	1.898	1	0
2835	218	346	1.899	1	0
2836	346	367	1.898	1	0
2837	342	345	1.200	1	0
2838	343	346	1.200	1	0
2839	341	344	1.200	1	0
2840	18	32	2.753	22	90
2841	22	35	2.753	22	0
2842	32	61	1.497	22	90
2843	35	122	1.497	22	0
2844	32	33	1.350	14	0
2845	34	35	1.400	14	0
2846	92	34	1.497	14	0
2847	33	34	1.000	14	0
2848	76	33	1.497	14	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
2849	92	35	2.050	3	0
2851	37	170	1.497	22	90
2852	35	36	2.449	5	0
2854	36	37	2.450	5	0
2856	138	160	2.449	12	0
2858	160	170	1.225	12	0
2860	35	138	1.934	2	0
2861	138	36	1.934	2	0
2862	36	160	1.934	2	0
2863	160	37	1.934	2	0
2864	35	26	5.620	8	0
2873	102	103	0.700	13	0
2874	148	149	0.700	13	0
2877	108	109	0.400	13	0
2878	154	155	0.400	13	0
2881	101	116	1.050	13	0
2882	101	102	0.500	13	0
2883	147	164	2.549	16	0
2884	147	148	0.500	13	0
2885	106	121	1.050	13	0
2886	105	106	0.900	13	0
2887	152	169	2.549	15	0
2888	151	152	0.900	13	0
2889	103	117	1.050	13	0
2890	104	120	1.050	13	0
2891	104	105	0.300	13	0
2892	149	165	2.549	15	0
2893	150	168	2.549	16	0
2894	150	151	0.300	13	0
2895	107	123	1.050	13	0
2896	107	108	0.800	13	0
2897	153	171	2.549	15	0
2898	153	154	0.800	13	0
2899	112	127	1.050	13	0
2900	111	112	0.400	13	0
2901	158	175	2.549	16	0
2902	157	158	0.400	13	0
2903	155	172	2.549	16	0
2904	156	174	2.549	15	0
2905	110	126	1.050	13	0
2906	109	124	1.050	13	0
2907	110	111	0.600	13	0
2909	156	157	0.600	13	0
2910	92	123	1.844	1	0
2911	92	121	1.844	1	0
2912	76	60	1.806	1	0
2913	76	62	1.806	1	0
2914	62	78	1.806	1	0
2915	78	64	1.806	1	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
2916	64	80	1.806	1	0
2917	80	66	1.680	1	0
2918	66	83	1.806	1	0
2919	60	74	1.806	1	0
2920	74	58	1.806	1	0
2921	58	72	1.806	1	0
2922	72	55	1.806	1	0
2923	55	70	1.680	1	0
2924	70	53	1.680	1	0
2925	83	68	1.680	1	0
2926	137	152	1.125	15	0
2927	137	138	1.200	1	0
2928	136	150	1.125	16	0
2929	136	137	1.200	1	0
2930	135	167	3.674	13	0
2931	135	136	1.200	1	0
2932	134	149	1.125	15	0
2933	134	135	1.200	1	0
2934	133	147	1.125	16	0
2935	133	134	1.200	1	0
2936	132	159	1.837	12	0
2937	132	133	1.000	1	0
2939	131	146	0.918	12	0
2940	131	132	1.000	1	0
2941	139	153	1.125	15	0
2942	139	138	1.200	1	0
2943	140	155	1.125	16	0
2944	140	139	1.200	1	0
2945	141	173	3.674	13	0
2946	141	140	1.200	1	0
2947	142	156	1.125	15	0
2948	142	141	1.200	1	0
2949	143	158	1.125	16	0
2950	143	142	1.000	1	0
2951	144	378	1.837	12	0
2952	144	143	1.200	1	0
2953	145	380	0.918	12	0
2954	145	144	1.000	1	0
2955	121	138	1.715	1	0
2956	138	123	1.715	1	0
2957	123	140	1.715	1	0
2958	140	125	1.715	1	0
2959	125	142	1.715	1	0
2960	142	127	1.581	1	0
2961	127	144	1.715	1	0
2962	144	129	1.581	1	0
2963	121	136	1.715	1	0
2964	136	119	1.715	1	0
2965	119	134	1.715	1	0

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Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
2966	134	116	1.715	1	0
2967	116	132	1.581	1	0
2968	132	114	1.581	1	0
2969	132	146	1.357	1	0
2970	161	163	1.357	1	0
2972	159	163	1.837	12	0
2973	146	159	1.358	1	0
2974	159	161	1.358	1	0
2975	194	325	2.491	12	0
2976	12	29	2.924	12	0
2978	117	118	0.800	19	0
2980	270	285	0.800	12	0
2981	270	271	1.000	1	0
2982	374	309	0.283	12	0
2983	375	310	0.683	12	0
2984	376	371	0.283	11	0
2985	377	372	1.483	11	0
2986	70	86	1.000	1	0
2987	70	85	1.414	1	0
2988	85	115	1.720	1	0
2991	83	99	1.000	1	0
2992	83	100	1.414	1	0
2993	100	128	1.720	1	0
2994	53	39	1.524	1	0
2995	68	51	1.524	1	0
2996	378	176	1.837	12	0
2997	380	381	1.838	12	0
2999	144	380	1.357	1	0
3000	380	378	1.358	1	0
3001	381	177	0.918	12	0
3002	378	381	1.358	1	0
3003	381	176	1.357	1	0
3004	195	220	1.524	1	0
3005	211	234	1.524	1	0
3006	382	339	2.450	12	0
3007	384	385	2.449	12	0
3009	299	384	1.581	1	0
3010	384	382	1.581	1	0
3011	385	340	1.225	12	0
3012	382	385	1.581	1	0
3013	385	339	1.581	1	0
3014	17	61	5.836	8	0
3015	17	53	5.836	8	0
3016	114	21	5.836	8	0
3017	122	21	5.836	8	0
3018	162	25	5.836	8	0
3019	25	170	5.836	8	0
3020	386	227	1.150	23	90
3021	386	203	1.626	2	0

Beams Cont...

Beam	Node A	Node B	Length (m)	Property	β (degrees)
3022	32	45	1.888	2	0
3023	387	300	0.800	12	0
3024	283	387	1.281	1	0
3025	387	299	1.281	1	0
3026	33	61	2.016	2	0
3027	388	179	2.749	12	0
3028	113	390	2.450	5	0
3029	178	390	3.682	5	0
3030	390	29	3.684	5	0
3032	390	388	2.449	5	0
3034	390	30	3.683	5	0
3035	391	276	0.756	9	0
3036	392	279	0.725	9	0
3037	393	292	0.756	20	0
3038	391	393	1.600	9	0
3039	394	316	0.756	15	0
3040	393	394	1.749	9	0
3041	395	332	0.756	19	0
3042	394	395	3.150	10	0
3043	396	295	0.725	20	0
3044	392	396	1.600	9	0
3045	397	318	0.575	15	0
3046	396	397	1.749	9	0
3047	398	335	0.725	19	0
3048	397	398	3.150	10	0
3049	399	370	0.500	10	0
3050	400	364	0.500	10	0
3053	403	372	0.500	10	0
3054	404	367	0.500	10	0
3055	400	363	0.649	1	0
3056	399	369	0.649	1	0
3057	404	368	0.603	1	0
3058	403	373	0.603	1	0
3059	405	1	3.150	25	0
3060	406	2	3.150	25	0
3061	407	3	3.150	24	0
3062	408	4	3.150	24	0
3063	409	5	3.150	24	0
3064	410	6	3.150	24	0
3065	411	7	3.150	24	0
3066	412	8	3.150	24	0
3067	413	9	3.150	24	0
3068	414	10	3.150	24	0
3069	415	11	3.150	24	0

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Section Properties

Prop	Section	Area (cm^2)	I_{yy} (cm^4)	I_{zz} (cm^4)	J (cm^4)	Material
1	L65X65X6	7.530	47.478	12.502	0.914	STEEL
2	L90X90X8	13.900	169.093	44.227	3.004	STEEL
3	L150X150X13	37.600	1.28E+3	331.877	21.494	STEEL
4	L50X50X6SA	11.380	64.548	40.800	1.354	STEEL
5	L65X65X6SA	15.060	131.290	92.800	1.786	STEEL
6	L90X90X8SA	27.800	441.798	332.000	5.871	STEEL
7	L100X100X8SA	31.000	595.648	460.000	6.554	STEEL
8	L130X130X10SA	50.400	1.59E+3	1.28E+3	16.667	STEEL
9	UPN100	13.500	29.300	206.000	2.800	STEEL
10	UPN120	17.000	43.200	364.000	4.000	STEEL
11	UPN240	42.300	248.000	3.6E+3	19.500	STEEL
12	IPE200	28.500	142.000	1.94E+3	7.000	STEEL
13	IPE240	39.100	284.000	3.89E+3	12.900	STEEL
14	IPE240	39.100	284.000	3.89E+3	12.900	STEEL
15	IPE300	53.800	604.000	8.36E+3	20.100	STEEL
16	IPE360	72.700	1.04E+3	16.3E+3	37.300	STEEL
17	IPE400	84.500	1.32E+3	23.1E+3	51.100	STEEL
18	IPE450	98.800	1.68E+3	33.7E+3	66.900	STEEL
19	IPE500	116.000	2.14E+3	48.2E+3	89.300	STEEL
20	W800X10X250X20	176.000	5.21E+3	189E+3	158.667	STEEL
21	HE200A	53.800	1.34E+3	3.69E+3	21.000	STEEL
22	HE260A	86.800	3.67E+3	10.5E+3	52.400	STEEL
23	HE260A TB TB	134.800	5.27E+3	18.8E+3	75.440	STEEL
24	Rect 0.70x0.70	4.9E+3	2E+6	2E+6	3.38E+6	CONCRETE
25	Rect 0.50x0.50	2.5E+3	521E+3	521E+3	879E+3	CONCRETE

Materials

Mat	Name	E (kN/mm^2)	ν	Density (kg/m^3)	α (/°C)
1	CONCRETE	21.718	0.170	2.4E+3	10E -6
2	ALUMINUM	68.948	0.330	2.71E+3	23E -6
3	STEEL_50_KSI	199.948	0.300	7.83E+3	11.7E -6
4	STAINLESSSTEEL	197.930	0.300	7.83E+3	18E -6
5	STEEL_36_KSI	199.948	0.300	7.83E+3	11.7E -6
6	STEEL_275_NMM2	205.000	0.300	7.85E+3	12E -6
7	STEEL	196.133	0.300	7.75E+3	12E -6
8	STEEL_355_NMM2	205.000	0.300	7.85E+3	12E -6

Supports

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
405	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
406	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
407	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
408	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
409	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
410	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
411	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
412	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
413	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
414	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
415	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

Releases

Beam ends not shown in this table are fixed in all directions.

Beam	Node	x	y	z	rx	ry	rz
74	309	Fixed	Fixed	Fixed	Fixed	Pin	Pin
79	308	Fixed	Fixed	Fixed	Fixed	Pin	Pin
192	179	Fixed	Fixed	Fixed	Fixed	Pin	Pin
192	390	Fixed	Fixed	Fixed	Fixed	Pin	Pin
199	219	Fixed	Fixed	Fixed	Fixed	Pin	Pin
206	235	Fixed	Fixed	Fixed	Fixed	Pin	Pin
212	340	Fixed	Fixed	Fixed	Fixed	Fixed	Pin
218	235	Fixed	Fixed	Fixed	Fixed	Pin	Pin
236	227	Fixed	Fixed	Fixed	Fixed	Pin	Pin
242	300	Fixed	Fixed	Fixed	Fixed	Pin	Pin
249	327	Fixed	Fixed	Fixed	Fixed	Pin	Pin
249	324	Fixed	Fixed	Fixed	Fixed	Pin	Pin
255	324	Fixed	Fixed	Fixed	Fixed	Pin	Pin
255	323	Fixed	Fixed	Fixed	Fixed	Pin	Pin
261	323	Fixed	Fixed	Fixed	Fixed	Pin	Pin
261	307	Fixed	Fixed	Fixed	Fixed	Pin	Pin
267	307	Fixed	Fixed	Fixed	Fixed	Pin	Pin
267	287	Fixed	Fixed	Fixed	Fixed	Pin	Pin
272	287	Fixed	Fixed	Fixed	Fixed	Pin	Pin
272	285	Fixed	Fixed	Fixed	Fixed	Pin	Pin
277	285	Fixed	Fixed	Fixed	Fixed	Pin	Pin
277	271	Fixed	Fixed	Fixed	Fixed	Pin	Pin
282	271	Fixed	Fixed	Fixed	Fixed	Pin	Pin
282	255	Fixed	Fixed	Fixed	Fixed	Pin	Pin
288	268	Fixed	Fixed	Fixed	Fixed	Pin	Pin
288	269	Fixed	Fixed	Fixed	Fixed	Pin	Pin
293	267	Fixed	Fixed	Fixed	Fixed	Pin	Pin
293	268	Fixed	Fixed	Fixed	Fixed	Pin	Pin
298	266	Fixed	Fixed	Fixed	Fixed	Pin	Pin
298	267	Fixed	Fixed	Fixed	Fixed	Pin	Pin
303	265	Fixed	Fixed	Fixed	Fixed	Pin	Pin
303	266	Fixed	Fixed	Fixed	Fixed	Pin	Pin

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CONNECTED User: Suman Podder

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
313	263	Fixed	Fixed	Fixed	Fixed	Pin	Pin
313	264	Fixed	Fixed	Fixed	Fixed	Pin	Pin
318	262	Fixed	Fixed	Fixed	Fixed	Pin	Pin
318	263	Fixed	Fixed	Fixed	Fixed	Pin	Pin
323	261	Fixed	Fixed	Fixed	Fixed	Pin	Pin
323	262	Fixed	Fixed	Fixed	Fixed	Pin	Pin
328	260	Fixed	Fixed	Fixed	Fixed	Pin	Pin
328	261	Fixed	Fixed	Fixed	Fixed	Pin	Pin
333	259	Fixed	Fixed	Fixed	Fixed	Pin	Pin
333	260	Fixed	Fixed	Fixed	Fixed	Pin	Pin
343	257	Fixed	Fixed	Fixed	Fixed	Pin	Pin
343	258	Fixed	Fixed	Fixed	Fixed	Pin	Pin
348	256	Fixed	Fixed	Fixed	Fixed	Pin	Pin
348	257	Fixed	Fixed	Fixed	Fixed	Pin	Pin
352	255	Fixed	Fixed	Fixed	Fixed	Pin	Pin
352	256	Fixed	Fixed	Fixed	Fixed	Pin	Pin
358	234	Fixed	Fixed	Fixed	Fixed	Pin	Pin
358	269	Fixed	Fixed	Fixed	Fixed	Pin	Pin
363	267	Fixed	Fixed	Fixed	Fixed	Pin	Pin
363	234	Fixed	Fixed	Fixed	Fixed	Pin	Pin
418	220	Fixed	Fixed	Fixed	Fixed	Pin	Pin
418	257	Fixed	Fixed	Fixed	Fixed	Pin	Pin
422	255	Fixed	Fixed	Fixed	Fixed	Pin	Pin
422	220	Fixed	Fixed	Fixed	Fixed	Pin	Pin
567	22	Fixed	Fixed	Fixed	Fixed	Pin	Pin
567	23	Fixed	Fixed	Fixed	Fixed	Pin	Pin
579	22	Fixed	Fixed	Fixed	Fixed	Pin	Pin
584	20	Fixed	Fixed	Fixed	Fixed	Pin	Pin
591	21	Fixed	Fixed	Fixed	Fixed	Pin	Pin
591	6	Fixed	Fixed	Fixed	Fixed	Pin	Pin
603	7	Fixed	Fixed	Fixed	Fixed	Pin	Pin
603	21	Fixed	Fixed	Fixed	Fixed	Pin	Pin
615	182	Fixed	Fixed	Fixed	Fixed	Pin	Pin
622	182	Fixed	Fixed	Fixed	Fixed	Pin	Pin
622	219	Fixed	Fixed	Fixed	Fixed	Pin	Pin
636	227	Fixed	Fixed	Fixed	Fixed	Pin	Pin
636	182	Fixed	Fixed	Fixed	Fixed	Pin	Pin
648	182	Fixed	Fixed	Fixed	Fixed	Pin	Pin
648	53	Fixed	Fixed	Fixed	Fixed	Pin	Pin
660	183	Fixed	Fixed	Fixed	Fixed	Pin	Pin
660	184	Fixed	Fixed	Fixed	Fixed	Pin	Pin
671	183	Fixed	Fixed	Fixed	Fixed	Pin	Pin
676	181	Fixed	Fixed	Fixed	Fixed	Pin	Pin
707	18	Fixed	Fixed	Fixed	Fixed	Pin	Pin
707	19	Fixed	Fixed	Fixed	Fixed	Pin	Pin
718	18	Fixed	Fixed	Fixed	Fixed	Pin	Pin
723	16	Fixed	Fixed	Fixed	Fixed	Pin	Pin
730	17	Fixed	Fixed	Fixed	Fixed	Pin	Pin
730	3	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
742	4	Fixed	Fixed	Fixed	Fixed	Pin	Pin
742	17	Fixed	Fixed	Fixed	Fixed	Pin	Pin
754	189	Fixed	Fixed	Fixed	Fixed	Pin	Pin
766	189	Fixed	Fixed	Fixed	Fixed	Pin	Pin
766	326	Fixed	Fixed	Fixed	Fixed	Pin	Pin
780	333	Fixed	Fixed	Fixed	Fixed	Pin	Pin
780	189	Fixed	Fixed	Fixed	Fixed	Pin	Pin
792	189	Fixed	Fixed	Fixed	Fixed	Pin	Pin
792	162	Fixed	Fixed	Fixed	Fixed	Pin	Pin
804	190	Fixed	Fixed	Fixed	Fixed	Pin	Pin
804	191	Fixed	Fixed	Fixed	Fixed	Pin	Pin
815	190	Fixed	Fixed	Fixed	Fixed	Pin	Pin
820	188	Fixed	Fixed	Fixed	Fixed	Pin	Pin
851	26	Fixed	Fixed	Fixed	Fixed	Pin	Pin
851	27	Fixed	Fixed	Fixed	Fixed	Pin	Pin
862	26	Fixed	Fixed	Fixed	Fixed	Pin	Pin
867	24	Fixed	Fixed	Fixed	Fixed	Pin	Pin
874	25	Fixed	Fixed	Fixed	Fixed	Pin	Pin
874	9	Fixed	Fixed	Fixed	Fixed	Pin	Pin
886	10	Fixed	Fixed	Fixed	Fixed	Pin	Pin
886	25	Fixed	Fixed	Fixed	Fixed	Pin	Pin
898	300	Fixed	Fixed	Fixed	Fixed	Pin	Pin
898	191	Fixed	Fixed	Fixed	Fixed	Pin	Pin
914	191	Fixed	Fixed	Fixed	Fixed	Pin	Pin
914	129	Fixed	Fixed	Fixed	Fixed	Pin	Pin
921	187	Fixed	Fixed	Fixed	Fixed	Pin	Pin
921	191	Fixed	Fixed	Fixed	Fixed	Pin	Pin
933	129	Fixed	Fixed	Fixed	Fixed	Pin	Pin
933	27	Fixed	Fixed	Fixed	Fixed	Pin	Pin
944	8	Fixed	Fixed	Fixed	Fixed	Pin	Pin
944	27	Fixed	Fixed	Fixed	Fixed	Pin	Pin
961	27	Fixed	Fixed	Fixed	Fixed	Pin	Pin
961	23	Fixed	Fixed	Fixed	Fixed	Pin	Pin
968	293	Fixed	Fixed	Fixed	Fixed	Pin	Pin
968	190	Fixed	Fixed	Fixed	Fixed	Pin	Pin
984	190	Fixed	Fixed	Fixed	Fixed	Pin	Pin
984	122	Fixed	Fixed	Fixed	Fixed	Pin	Pin
991	186	Fixed	Fixed	Fixed	Fixed	Pin	Pin
991	190	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1015	7	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1015	26	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1034	26	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1034	22	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1041	286	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1041	188	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1052	114	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1052	188	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1065	185	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1065	188	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
1089	6	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1089	24	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1101	20	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1101	24	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1113	192	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1113	325	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1125	192	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1125	179	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1142	14	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1142	29	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1149	14	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1149	30	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1161	1	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1161	14	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1173	14	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1173	2	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1185	12	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1190	15	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1203	128	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1209	233	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1215	234	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1221	232	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1227	231	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1233	230	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1239	228	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1245	293	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1251	226	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1257	224	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1263	223	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1269	222	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1275	221	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1287	61	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1287	76	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1293	62	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1293	77	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1299	63	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1299	78	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1305	64	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1305	79	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1311	65	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1311	80	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1317	66	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1317	81	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1323	67	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1323	83	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1329	60	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1329	75	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1335	59	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1335	74	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
1341	58	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1341	73	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1347	56	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1347	72	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1353	55	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1353	71	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1359	54	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1359	70	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1366	115	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1372	116	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1378	117	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1384	119	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1390	120	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1396	121	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1402	122	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1408	123	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1414	124	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1420	125	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1426	126	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1432	127	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1437	287	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1442	288	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1447	289	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1452	291	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1457	292	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1462	295	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1467	297	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1472	298	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1477	299	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1482	294	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1489	303	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1495	306	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1501	302	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1501	330	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1507	305	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1507	336	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1512	86	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1512	115	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1517	88	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1522	87	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1527	95	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1527	125	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1532	96	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1537	97	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1542	99	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1542	128	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1547	89	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1547	119	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1552	90	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
1557	92	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1557	122	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1562	91	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1567	93	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1572	94	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1579	100	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1585	84	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1591	68	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1597	227	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1604	325	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1604	326	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1610	179	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1610	180	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1616	30	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1616	31	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1622	192	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1622	193	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1628	12	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1628	13	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1634	113	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1634	114	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1641	192	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1641	194	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1647	178	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1647	179	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1653	29	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1653	30	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1664	210	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1664	234	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1669	196	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1669	220	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1674	209	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1674	233	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1679	208	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1684	207	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1684	231	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1689	206	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1694	204	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1694	228	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1699	197	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1699	221	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1704	198	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1709	199	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1709	223	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1714	200	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1719	202	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1719	226	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1724	51	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1724	67	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
1729	39	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1729	54	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1734	50	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1734	66	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1739	49	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1739	65	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1744	48	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1744	64	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1749	47	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1749	63	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1754	46	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1754	62	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1759	40	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1759	55	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1764	41	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1764	56	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1769	42	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1769	58	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1774	43	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1774	59	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1779	44	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1779	60	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1785	177	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1791	129	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1809	45	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1809	61	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1815	203	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1827	300	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1833	333	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1839	293	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1845	52	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1851	211	Fixed	Fixed	Fixed	Fixed	Pin	Pin
1999	333	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2020	301	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2023	304	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2038	61	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2041	61	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2077	114	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2083	53	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2086	227	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2089	129	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2095	68	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2182	333	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2200	85	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2248	162	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2266	170	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2269	170	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2284	114	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2302	122	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
2305	122	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2320	114	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2323	129	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2332	38	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2338	195	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2482	227	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2485	293	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2512	219	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2515	286	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2527	53	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2566	69	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2600	24	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2600	114	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2629	222	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2630	224	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2635	329	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2636	331	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2637	312	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2638	315	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2639	241	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2640	244	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2646	242	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2648	243	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2649	328	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2650	313	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2651	332	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2652	314	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2653	311	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2654	240	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2661	230	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2662	232	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2669	248	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2673	252	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2674	334	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2676	335	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2677	317	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2678	337	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2680	338	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2681	320	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2682	254	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2683	319	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2684	318	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2685	322	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2686	321	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2687	251	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2688	250	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2689	253	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2703	246	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2704	245	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
2705	216	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2708	225	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2710	201	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2711	364	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2711	370	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2714	217	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2715	364	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2717	371	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2717	366	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2718	247	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2720	369	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2720	363	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2722	370	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2723	373	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2723	368	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2724	371	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2727	249	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2728	218	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2730	229	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2732	205	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2733	372	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2733	367	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2734	366	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2734	367	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2748	258	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2748	259	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2753	264	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2753	265	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2757	263	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2757	227	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2758	227	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2758	261	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2759	326	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2760	326	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2763	286	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2764	287	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2765	220	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2766	286	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2767	327	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2777	162	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2779	288	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2780	271	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2780	272	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2781	289	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2782	272	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2782	273	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2783	290	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2784	273	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2784	274	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
2785	291	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2786	274	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2786	275	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2787	292	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2788	275	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2789	293	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2790	276	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2790	277	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2791	294	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2792	277	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2792	278	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2793	295	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2794	278	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2795	296	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2796	279	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2796	280	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2797	297	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2798	280	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2798	281	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2799	298	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2800	281	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2800	282	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2801	299	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2802	282	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2802	283	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2804	283	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2804	284	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2805	271	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2805	257	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2806	257	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2806	273	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2807	273	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2807	259	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2808	259	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2808	275	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2809	275	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2809	261	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2810	261	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2810	277	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2811	277	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2811	263	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2812	263	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2812	279	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2813	279	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2813	265	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2814	265	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2814	281	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2815	281	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2815	267	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
2816	267	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2816	283	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2817	283	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2817	269	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2818	247	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2818	370	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2819	245	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2819	371	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2820	216	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2820	366	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2821	217	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2821	364	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2822	366	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2823	370	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2823	365	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2824	365	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2824	371	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2831	216	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2831	344	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2832	344	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2832	364	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2833	217	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2833	345	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2834	345	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2834	366	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2835	218	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2835	346	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2836	346	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2836	367	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2837	342	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2837	345	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2838	343	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2838	346	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2839	341	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2839	344	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2844	32	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2845	35	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2846	92	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2846	34	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2848	76	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2848	33	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2849	92	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2849	35	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2852	35	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2854	37	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2858	170	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2860	35	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2860	138	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2861	138	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
2861	36	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2862	36	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2862	160	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2863	160	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2863	37	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2864	35	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2864	26	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2873	103	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2874	149	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2877	109	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2878	155	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2881	116	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2882	101	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2883	164	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2884	147	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2885	121	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2886	106	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2887	169	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2888	152	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2889	117	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2890	120	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2891	104	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2892	165	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2893	168	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2894	150	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2895	123	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2896	107	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2897	171	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2898	153	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2899	127	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2900	112	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2901	175	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2902	158	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2903	172	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2904	174	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2905	126	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2906	124	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2907	110	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2909	156	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2910	92	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2910	123	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2911	92	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2911	121	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2912	76	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2912	60	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2913	76	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2913	62	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2914	62	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2914	78	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
2915	78	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2915	64	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2916	64	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2916	80	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2917	80	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2917	66	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2918	66	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2918	83	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2919	60	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2919	74	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2920	74	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2920	58	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2921	58	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2921	72	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2922	72	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2922	55	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2923	55	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2923	70	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2924	70	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2924	53	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2925	83	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2925	68	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2927	137	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2927	138	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2929	136	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2929	137	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2930	167	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2931	135	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2931	136	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2933	134	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2933	135	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2935	133	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2935	134	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2937	132	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2937	133	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2940	131	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2940	132	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2942	139	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2942	138	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2944	140	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2944	139	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2945	173	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2946	141	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2946	140	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2948	142	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2948	141	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2950	143	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2950	142	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2952	144	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
2952	143	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2954	145	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2954	144	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2955	121	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2955	138	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2956	138	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2956	123	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2957	123	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2957	140	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2958	140	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2958	125	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2959	125	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2959	142	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2960	142	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2960	127	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2961	127	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2961	144	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2962	144	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2962	129	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2963	121	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2963	136	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2964	136	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2964	119	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2965	119	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2965	134	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2966	134	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2966	116	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2967	116	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2967	132	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2968	132	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2968	114	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2969	132	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2969	146	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2970	161	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2970	163	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2972	163	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2973	146	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2973	159	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2974	159	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2974	161	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2981	270	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2981	271	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2982	309	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2983	310	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2984	371	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2985	372	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2986	70	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2986	86	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2987	70	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
2987	85	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2988	85	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2988	115	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2991	83	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2991	99	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2992	83	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2992	100	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2993	100	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2993	128	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2994	53	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2994	39	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2995	68	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2995	51	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2996	176	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2999	144	Fixed	Fixed	Fixed	Fixed	Pin	Pin
2999	380	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3000	380	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3000	378	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3001	177	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3002	378	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3002	381	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3003	381	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3003	176	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3004	195	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3004	220	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3005	211	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3005	234	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3006	339	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3009	299	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3009	384	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3010	384	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3010	382	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3011	340	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3012	382	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3012	385	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3013	385	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3013	339	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3014	17	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3014	61	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3015	17	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3015	53	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3016	114	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3016	21	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3017	122	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3017	21	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3018	162	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3018	25	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3019	25	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3019	170	Fixed	Fixed	Fixed	Fixed	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
3021	386	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3021	203	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3022	32	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3022	45	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3023	300	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3024	283	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3024	387	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3025	387	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3025	299	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3026	33	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3026	61	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3028	113	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3029	178	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3029	390	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3030	390	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3030	29	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3032	388	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3034	390	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3034	30	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3035	276	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3036	279	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3038	391	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3038	393	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3039	316	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3040	393	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3040	394	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3042	394	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3042	395	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3044	392	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3044	396	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3046	396	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3046	397	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3048	397	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3048	398	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3055	400	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3055	363	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3056	399	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3056	369	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3057	404	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3057	368	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3058	403	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3058	373	Fixed	Fixed	Fixed	Fixed	Pin	Pin
3059	1	Fixed	Fixed	Fixed	Pin	Pin	Pin
3060	2	Fixed	Fixed	Fixed	Pin	Pin	Pin
3061	3	Fixed	Fixed	Fixed	Pin	Pin	Pin
3062	4	Fixed	Fixed	Fixed	Pin	Pin	Pin
3063	5	Fixed	Fixed	Fixed	Pin	Pin	Pin
3064	6	Fixed	Fixed	Fixed	Pin	Pin	Pin
3065	7	Fixed	Fixed	Fixed	Pin	Pin	Pin

Releases Cont...

Beam	Node	x	y	z	rx	ry	rz
3066	8	Fixed	Fixed	Fixed	Pin	Pin	Pin
3067	9	Fixed	Fixed	Fixed	Pin	Pin	Pin
3068	10	Fixed	Fixed	Fixed	Pin	Pin	Pin
3069	11	Fixed	Fixed	Fixed	Pin	Pin	Pin

Primary Load Cases

Number	Name	Type
1	DEAD LOAD (DL)	None
2	LIVE LOAD (LL)	None
3	OPERATING FLUID LOAD (FL1)	None
4	OPERATING FLUID LOAD (FL2)	None
5	EQP DL (EQDL)	None
6	EQP LL (EQLL)	None
7	WIND LOAD (+X) WL(+X)	None
8	WIND LOAD (-X) WL(-X)	None
9	WIND LOAD (+Z) WL(+Z)	None
10	WIND LOAD (-Z) WL(-Z)	None
11	SEISMIC LOAD SL(+X)	None
12	SEISMIC LOAD SL(-X)	None
13	SEISMIC LOAD SL(+Z)	None
14	SEISMIC LOAD SL(-Z)	None
15	TEMPERATURE LOAD (TL)	None
16	PIPE LATERAL LOAD (PL)	None
17	EQP TEST LOAD (EQTL)	None

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
101	DL+EQDL+FL1	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
102	DL+EQDL+FL2	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
103	DL+EQDL+FL1+LL+EQLL+TL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		2	LIVE LOAD (LL)	1.00
		6	EQP LL (EQLL)	1.00
		15	TEMPERATURE LOAD (TL)	1.00
104	DL+EQDL+FL2+LL+EQLL+TL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		2	LIVE LOAD (LL)	1.00
		6	EQP LL (EQLL)	1.00

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		15	TEMPERATURE LOAD (TL)	1.00
105	DL+EQDL+FL1+0.75(LL+EQLL)+0.75TL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
		15	TEMPERATURE LOAD (TL)	0.75
106	DL+EQDL+FL2+0.75(LL+EQLL)+0.75TL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
		15	TEMPERATURE LOAD (TL)	0.75
107	DL+EQDL+FL1+PL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		16	PIPE LATERAL LOAD (PL)	1.00
108	DL+EQDL+FL2+PL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		16	PIPE LATERAL LOAD (PL)	1.00
109	DL+EQDL+FL1+PL+LL+EQLL+TL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		16	PIPE LATERAL LOAD (PL)	1.00
		2	LIVE LOAD (LL)	1.00
		6	EQP LL (EQLL)	1.00
		15	TEMPERATURE LOAD (TL)	1.00
110	DL+EQDL+FL2+PL+LL+EQLL+TL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		16	PIPE LATERAL LOAD (PL)	1.00
		2	LIVE LOAD (LL)	1.00
		6	EQP LL (EQLL)	1.00
		15	TEMPERATURE LOAD (TL)	1.00
111	DL+EQDL+FL1+PL+0.75(LL+EQLL)+0.75	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		16	PIPE LATERAL LOAD (PL)	1.00
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
		15	TEMPERATURE LOAD (TL)	0.75
112	DL+EQDL+FL2+PL+0.75(LL+EQLL)+0.75	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		16	PIPE LATERAL LOAD (PL)	1.00
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
		15	TEMPERATURE LOAD (TL)	0.75

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
113	DL+EQDL+FL1+WL(+X)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		7	WIND LOAD (+X) WL(+X)	1.00
114	DL+EQDL+FL1+WL(-X)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		8	WIND LOAD (-X) WL(-X)	1.00
115	DL+EQDL+FL1+WL(+Z)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		9	WIND LOAD (+Z) WL(+Z)	1.00
116	DL+EQDL+FL1+WL(-Z)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		10	WIND LOAD (-Z) WL(-Z)	1.00
117	DL+EQDL+FL2+WL(+X)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		7	WIND LOAD (+X) WL(+X)	1.00
118	DL+EQDL+FL2+WL(-X)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		8	WIND LOAD (-X) WL(-X)	1.00
119	DL+EQDL+FL2+WL(+Z)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		9	WIND LOAD (+Z) WL(+Z)	1.00
120	DL+EQDL+FL2+WL(-Z)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		10	WIND LOAD (-Z) WL(-Z)	1.00
121	DL+EQDL+FL1+0.7SL(+X)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		11	SEISMIC LOAD SL(+X)	0.70
122	DL+EQDL+FL1+0.7SL(-X)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		12	SEISMIC LOAD SL(-X)	0.70
123	DL+EQDL+FL1+0.7SL(+Z)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		13	SEISMIC LOAD SL(+Z)	0.70
124	DL+EQDL+FL1+0.7SL(-Z)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		14	SEISMIC LOAD SL(-Z)	0.70
125	DL+EQDL+FL2+0.7SL(+X)	1	DEAD LOAD (DL)	1.00

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		11	SEISMIC LOAD SL(+X)	0.70
126	DL+EQDL+FL2+0.7SL(-X)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		12	SEISMIC LOAD SL(-X)	0.70
127	DL+EQDL+FL2+0.7SL(+Z)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		13	SEISMIC LOAD SL(+Z)	0.70
128	DL+EQDL+FL2+0.7SL(-Z)	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		14	SEISMIC LOAD SL(-Z)	0.70
129	DL+EQDL+FL1+0.75WL(+X)+0.75(LL+EQ	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		7	WIND LOAD (+X) WL(+X)	0.75
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
130	DL+EQDL+FL1+0.75WL(-X)+0.75(LL+EQL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		8	WIND LOAD (-X) WL(-X)	0.75
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
131	DL+EQDL+FL1+0.75WL(+Z)+0.75(LL+EQ	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		9	WIND LOAD (+Z) WL(+Z)	0.75
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
132	DL+EQDL+FL1+0.75WL(-Z)+0.75(LL+EQL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		10	WIND LOAD (-Z) WL(-Z)	0.75
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
133	DL+EQDL+FL2+0.75WL(+X)+0.75(LL+EQ	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		7	WIND LOAD (+X) WL(+X)	0.75
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
134	DL+EQDL+FL2+0.75WL(-X)+0.75(LL+EQL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		8	WIND LOAD (-X) WL(-X)	0.75

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
135	DL+EQDL+FL2+0.75WL(+Z)+0.75(LL+EQ	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		9	WIND LOAD (+Z) WL(+Z)	0.75
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
136	DL+EQDL+FL2+0.75WL(-Z)+0.75(LL+EQL	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		10	WIND LOAD (-Z) WL(-Z)	0.75
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
137	DL+EQDL+FL1+0.525SL(+X)+0.75(LL+EC	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		11	SEISMIC LOAD SL(+X)	0.52
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
138	DL+EQDL+FL1+0.525SL(-X)+0.75(LL+EQ	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		12	SEISMIC LOAD SL(-X)	0.52
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
139	DL+EQDL+FL1+0.525SL(+Z)+0.75(LL+EC	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		13	SEISMIC LOAD SL(+Z)	0.52
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
140	DL+EQDL+FL1+0.525SL(-Z)+0.75(LL+EQ	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		3	OPERATING FLUID LOAD (FL1)	1.00
		14	SEISMIC LOAD SL(-Z)	0.52
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
141	DL+EQDL+FL2+0.525SL(+X)+0.75(LL+EC	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		11	SEISMIC LOAD SL(+X)	0.52
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
142	DL+EQDL+FL2+0.525SL(-X)+0.75(LL+EQ	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		12	SEISMIC LOAD SL(-X)	0.52
		2	LIVE LOAD (LL)	0.75

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		6	EQP LL (EQLL)	0.75
143	DL+EQDL+FL2+0.525SL(+Z)+0.75(LL+EC	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		13	SEISMIC LOAD SL(+Z)	0.52
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
144	DL+EQDL+FL2+0.525SL(-Z)+0.75(LL+EQ	1	DEAD LOAD (DL)	1.00
		5	EQP DL (EQDL)	1.00
		4	OPERATING FLUID LOAD (FL2)	1.00
		14	SEISMIC LOAD SL(-Z)	0.52
		2	LIVE LOAD (LL)	0.75
		6	EQP LL (EQLL)	0.75
145	0.6(DL+EQDL)+WL(+X)	1	DEAD LOAD (DL)	0.60
		5	EQP DL (EQDL)	0.60
		7	WIND LOAD (+X) WL(+X)	1.00
146	0.6(DL+EQDL)+WL(-X)	1	DEAD LOAD (DL)	0.60
		5	EQP DL (EQDL)	0.60
		8	WIND LOAD (-X) WL(-X)	1.00
147	0.6(DL+EQDL)+WL(+Z)	1	DEAD LOAD (DL)	0.60
		5	EQP DL (EQDL)	0.60
		9	WIND LOAD (+Z) WL(+Z)	1.00
148	0.6(DL+EQDL)+WL(-Z)	1	DEAD LOAD (DL)	0.60
		5	EQP DL (EQDL)	0.60
		10	WIND LOAD (-Z) WL(-Z)	1.00
149	0.6(DL+EQDL)+0.7SL(+X)	1	DEAD LOAD (DL)	0.60
		5	EQP DL (EQDL)	0.60
		11	SEISMIC LOAD SL(+X)	0.70
150	0.6(DL+EQDL)+0.7SL(-X)	1	DEAD LOAD (DL)	0.60
		5	EQP DL (EQDL)	0.60
		12	SEISMIC LOAD SL(-X)	0.70
151	0.6(DL+EQDL)+0.7SL(+Z)	1	DEAD LOAD (DL)	0.60
		5	EQP DL (EQDL)	0.60
		13	SEISMIC LOAD SL(+Z)	0.70
152	0.6(DL+EQDL)+0.7SL(-Z)	1	DEAD LOAD (DL)	0.60
		5	EQP DL (EQDL)	0.60
		14	SEISMIC LOAD SL(-Z)	0.70
153	DL+EQTL	1	DEAD LOAD (DL)	1.00
		17	EQP TEST LOAD (EQTL)	1.00
201	1.4(DL+EQDL+FL1)	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		3	OPERATING FLUID LOAD (FL1)	1.40
202	1.4(DL+EQDL+FL2)	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		4	OPERATING FLUID LOAD (FL2)	1.40
203	1.4(DL+EQDL+FL1)+1.7(LL+EQLL)	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		3	OPERATING FLUID LOAD (FL1)	1.40
		2	LIVE LOAD (LL)	1.70

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		6	EQP LL (EQLL)	1.70
204	1.4(DL+EQDL+FL2)+1.7(LL+EQLL)	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		4	OPERATING FLUID LOAD (FL2)	1.40
		2	LIVE LOAD (LL)	1.70
		6	EQP LL (EQLL)	1.70
205	1.4(DL+EQDL+FL1+TL)+1.7(LL+EQLL)	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		3	OPERATING FLUID LOAD (FL1)	1.40
		15	TEMPERATURE LOAD (TL)	1.40
		2	LIVE LOAD (LL)	1.70
		6	EQP LL (EQLL)	1.70
206	1.4(DL+EQDL+FL2+TL)+1.7(LL+EQLL)	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		4	OPERATING FLUID LOAD (FL2)	1.40
		15	TEMPERATURE LOAD (TL)	1.40
		2	LIVE LOAD (LL)	1.70
		6	EQP LL (EQLL)	1.70
207	1.4(DL+EQDL+FL1+PL)	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		3	OPERATING FLUID LOAD (FL1)	1.40
		16	PIPE LATERAL LOAD (PL)	1.40
208	1.4(DL+EQDL+FL2+PL)	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		4	OPERATING FLUID LOAD (FL2)	1.40
		16	PIPE LATERAL LOAD (PL)	1.40
209	1.4(DL+EQDL+FL1+PL)+1.7(LL+EQLL)	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		3	OPERATING FLUID LOAD (FL1)	1.40
		16	PIPE LATERAL LOAD (PL)	1.40
		2	LIVE LOAD (LL)	1.70
		6	EQP LL (EQLL)	1.70
210	1.4(DL+EQDL+FL2+PL)+1.7(LL+EQLL)	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		4	OPERATING FLUID LOAD (FL2)	1.40
		16	PIPE LATERAL LOAD (PL)	1.40
		2	LIVE LOAD (LL)	1.70
		6	EQP LL (EQLL)	1.70
211	1.4(DL+EQDL+FL1+PL+TL)+1.7(LL+EQLL	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		3	OPERATING FLUID LOAD (FL1)	1.40
		16	PIPE LATERAL LOAD (PL)	1.40
		15	TEMPERATURE LOAD (TL)	1.40
		2	LIVE LOAD (LL)	1.70
		6	EQP LL (EQLL)	1.70
212	1.4(DL+EQDL+FL2+PL+TL)+1.7(LL+EQLL	1	DEAD LOAD (DL)	1.40
		5	EQP DL (EQDL)	1.40
		4	OPERATING FLUID LOAD (FL2)	1.40
		16	PIPE LATERAL LOAD (PL)	1.40

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		15	TEMPERATURE LOAD (TL)	1.40
		2	LIVE LOAD (LL)	1.70
		6	EQP LL (EQLL)	1.70
213	1.2(DL+EQDL)+0.8WL(+X)	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		7	WIND LOAD (+X) WL(+X)	0.80
214	1.2(DL+EQDL)+0.8WL(-X)	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		8	WIND LOAD (-X) WL(-X)	0.80
215	1.2(DL+EQDL)+0.8WL(+Z)	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		9	WIND LOAD (+Z) WL(+Z)	0.80
216	1.2(DL+EQDL)+0.8WL(-Z)	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		10	WIND LOAD (-Z) WL(-Z)	0.80
217	1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		7	WIND LOAD (+X) WL(+X)	1.60
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
218	1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		8	WIND LOAD (-X) WL(-X)	1.60
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
219	1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		9	WIND LOAD (+Z) WL(+Z)	1.60
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
220	1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		10	WIND LOAD (-Z) WL(-Z)	1.60
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
221	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		11	SEISMIC LOAD SL(+X)	1.00
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
		3	OPERATING FLUID LOAD (FL1)	0.50
222	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		12	SEISMIC LOAD SL(-X)	1.00
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
		3	OPERATING FLUID LOAD (FL1)	0.50
223	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
		13	SEISMIC LOAD SL(+Z)	1.00
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
		3	OPERATING FLUID LOAD (FL1)	0.50
224	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		14	SEISMIC LOAD SL(-Z)	1.00
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
		3	OPERATING FLUID LOAD (FL1)	0.50
225	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		11	SEISMIC LOAD SL(+X)	1.00
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
		4	OPERATING FLUID LOAD (FL2)	0.50
226	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		12	SEISMIC LOAD SL(-X)	1.00
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
		4	OPERATING FLUID LOAD (FL2)	0.50
227	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		13	SEISMIC LOAD SL(+Z)	1.00
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
		4	OPERATING FLUID LOAD (FL2)	0.50
228	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5	1	DEAD LOAD (DL)	1.20
		5	EQP DL (EQDL)	1.20
		14	SEISMIC LOAD SL(-Z)	1.00
		2	LIVE LOAD (LL)	0.50
		6	EQP LL (EQLL)	0.50
		4	OPERATING FLUID LOAD (FL2)	0.50
229	0.9(DL+EQDL)+1.6WL(+X)	1	DEAD LOAD (DL)	0.90
		5	EQP DL (EQDL)	0.90
		7	WIND LOAD (+X) WL(+X)	1.60
230	0.9(DL+EQDL)+1.6WL(-X)	1	DEAD LOAD (DL)	0.90
		5	EQP DL (EQDL)	0.90
		8	WIND LOAD (-X) WL(-X)	1.60
231	0.9(DL+EQDL)+1.6WL(+Z)	1	DEAD LOAD (DL)	0.90
		5	EQP DL (EQDL)	0.90
		9	WIND LOAD (+Z) WL(+Z)	1.60
232	0.9(DL+EQDL)+1.6WL(-Z)	1	DEAD LOAD (DL)	0.90
		5	EQP DL (EQDL)	0.90
		10	WIND LOAD (-Z) WL(-Z)	1.60
233	0.9(DL+EQDL)+1.6SL(+X)	1	DEAD LOAD (DL)	0.90
		5	EQP DL (EQDL)	0.90
		11	SEISMIC LOAD SL(+X)	1.60

Combination Load Cases Cont...

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
234	0.9(DL+EQDL)+1.6SL(-X)	1	DEAD LOAD (DL)	0.90
		5	EQP DL (EQDL)	0.90
		12	SEISMIC LOAD SL(-X)	1.60
235	0.9(DL+EQDL)+1.6SL(+Z)	1	DEAD LOAD (DL)	0.90
		5	EQP DL (EQDL)	0.90
		13	SEISMIC LOAD SL(+Z)	1.60
236	0.9(DL+EQDL)+1.6SL(-Z)	1	DEAD LOAD (DL)	0.90
		5	EQP DL (EQDL)	0.90
		14	SEISMIC LOAD SL(-Z)	1.60
237	1.2DL+EQTL	1	DEAD LOAD (DL)	1.20
		17	EQP TEST LOAD (EQTL)	1.00

Load Generators

There is no data of this type.

1 DEAD LOAD (DL) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
12	-	-2.550	-	-	-	-
13	-	-2.550	-	-	-	-
30	-	-2.550	-	-	-	-
31	-	-2.550	-	-	-	-
113	-	-2.550	-	-	-	-
114	-	-2.550	-	-	-	-
179	-	-2.550	-	-	-	-
180	-	-2.550	-	-	-	-
192	-	-2.550	-	-	-	-
193	-	-2.550	-	-	-	-
325	-	-2.550	-	-	-	-
326	-	-2.550	-	-	-	-
374	-	-1.422	-	-	-	-
375	-	-1.422	-	-	-	-
376	-	-1.422	-	-	-	-
377	-	-1.422	-	-	-	-

1 DEAD LOAD (DL) : Selfweight

Direction	Factor	Assigned Geometry
Y	-1.050	ALL

2 LIVE LOAD (LL) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
12	-	-8.238	-	-	-	-
13	-	-8.238	-	-	-	-
30	-	-8.238	-	-	-	-
31	-	-8.238	-	-	-	-
113	-	-8.238	-	-	-	-
114	-	-8.238	-	-	-	-
179	-	-8.238	-	-	-	-
180	-	-8.238	-	-	-	-
192	-	-8.238	-	-	-	-
193	-	-8.238	-	-	-	-
325	-	-8.238	-	-	-	-
326	-	-8.238	-	-	-	-
374	-	-3.069	-	-	-	-
375	-	-3.069	-	-	-	-
376	-	-3.069	-	-	-	-
377	-	-3.069	-	-	-	-

3 OPERATING FLUID LOAD (FL1) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
181	-	-19.613	-	-16.671	-	-
183	-	-19.613	-	-16.671	-	-
184	-	-19.613	-	-16.671	-	-
195	-	-19.613	-	-	-	-
199	-	-19.613	-	-	-	-
203	-	-19.613	-	-	-	-
207	-	-19.613	-	-	-	-
211	-	-19.613	-	-	-	-
351	-	-	2.942	-	-	-
	-	-58.840	-	-	-	-
	2.942	-	-	-	-	-
352	-	-	2.942	-	-	-
	-	-58.840	-	-	-	-
	2.942	-	-	-	-	-
353	-	-	2.942	-	-	-
	-	-58.840	-	-	-	-
	2.942	-	-	-	-	-
354	-	-	2.942	-	-	-
	-	-58.840	-	-	-	-
	2.942	-	-	-	-	-
355	-	-	2.942	-	-	-
	-	-49.033	-	-	-	-
	2.942	-	-	-	-	-
356	-	-	2.942	-	-	-
	-	-49.033	-	-	-	-
	2.942	-	-	-	-	-
357	-	-	2.942	-	-	-

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3 OPERATING FLUID LOAD (FL1) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
357	-	-49.033	-	-	-	-
	2.942	-	-	-	-	-
358	-	-	2.942	-	-	-
	-	-49.033	-	-	-	-
	2.942	-	-	-	-	-

3 OPERATING FLUID LOAD (FL1) : Beam Loads

Beam	Type	Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
1579	UNI kN/m	GY	-1.961	-	-	-	-
1585	UNI kN/m	GY	-1.961	-	-	-	-
2026	UNI kN/m	GY	-1.961	-	-	-	-
2029	UNI kN/m	GY	-1.961	-	-	-	-
2032	UNI kN/m	GY	-1.961	-	-	-	-
2035	UNI kN/m	GY	-1.961	-	-	-	-
2200	UNI kN/m	GY	-1.961	-	-	-	-
2203	UNI kN/m	GY	-1.961	-	-	-	-
2206	UNI kN/m	GY	-1.961	-	-	-	-
2209	UNI kN/m	GY	-1.961	-	-	-	-
2212	UNI kN/m	GY	-1.961	-	-	-	-
2215	UNI kN/m	GY	-1.961	-	-	-	-
2218	UNI kN/m	GY	-1.961	-	-	-	-
2221	UNI kN/m	GY	-1.961	-	-	-	-
2224	UNI kN/m	GY	-1.961	-	-	-	-
2227	UNI kN/m	GY	-1.961	-	-	-	-
2230	UNI kN/m	GY	-1.961	-	-	-	-
2413	UNI kN/m	GY	-1.961	-	-	-	-
2416	UNI kN/m	GY	-1.961	-	-	-	-
2419	UNI kN/m	GY	-1.961	-	-	-	-
2422	UNI kN/m	GY	-1.961	-	-	-	-
2425	UNI kN/m	GY	-1.961	-	-	-	-
2428	UNI kN/m	GY	-1.961	-	-	-	-
2431	UNI kN/m	GY	-1.961	-	-	-	-
2461	UNI kN/m	GY	-1.961	-	-	-	-
2464	UNI kN/m	GY	-1.961	-	-	-	-
2488	UNI kN/m	GY	-1.961	-	-	-	-
2506	UNI kN/m	GY	-1.961	-	-	-	-
2524	UNI kN/m	GY	-1.961	-	-	-	-
2566	UNI kN/m	GY	-1.961	-	-	-	-

4 OPERATING FLUID LOAD (FL2) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
181	-	-19.613	-	-16.671	-	-
183	-	-19.613	-	-16.671	-	-
184	-	-19.613	-	-16.671	-	-
195	-	-19.613	-	-	-	-
199	-	-19.613	-	-	-	-
203	-	-19.613	-	-	-	-
207	-	-19.613	-	-	-	-
211	-	-19.613	-	-	-	-
351	-	-	-2.942	-	-	-
	-	-58.840	-	-	-	-
	-2.942	-	-	-	-	-
352	-	-	-2.942	-	-	-
	-	-58.840	-	-	-	-
	-2.942	-	-	-	-	-
353	-	-	-2.942	-	-	-
	-	-58.840	-	-	-	-
	-2.942	-	-	-	-	-
354	-	-	-2.942	-	-	-
	-	-58.840	-	-	-	-
	-2.942	-	-	-	-	-
355	-	-	-2.942	-	-	-
	-	-49.033	-	-	-	-
	-2.942	-	-	-	-	-
356	-	-	-2.942	-	-	-
	-	-49.033	-	-	-	-
	-2.942	-	-	-	-	-
357	-	-	-2.942	-	-	-
	-	-49.033	-	-	-	-
	-2.942	-	-	-	-	-
358	-	-	-2.942	-	-	-
	-	-49.033	-	-	-	-
	-2.942	-	-	-	-	-

4 OPERATING FLUID LOAD (FL2) : Beam Loads

Beam	Type	Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
1579	UNI kN/m	GY	-1.961	-	-	-	-
1585	UNI kN/m	GY	-1.961	-	-	-	-
2026	UNI kN/m	GY	-1.961	-	-	-	-
2029	UNI kN/m	GY	-1.961	-	-	-	-
2032	UNI kN/m	GY	-1.961	-	-	-	-
2035	UNI kN/m	GY	-1.961	-	-	-	-
2200	UNI kN/m	GY	-1.961	-	-	-	-
2203	UNI kN/m	GY	-1.961	-	-	-	-
2206	UNI kN/m	GY	-1.961	-	-	-	-
2209	UNI kN/m	GY	-1.961	-	-	-	-
2212	UNI kN/m	GY	-1.961	-	-	-	-
2215	UNI kN/m	GY	-1.961	-	-	-	-
2218	UNI kN/m	GY	-1.961	-	-	-	-
2221	UNI kN/m	GY	-1.961	-	-	-	-
2224	UNI kN/m	GY	-1.961	-	-	-	-
2227	UNI kN/m	GY	-1.961	-	-	-	-
2230	UNI kN/m	GY	-1.961	-	-	-	-
2413	UNI kN/m	GY	-1.961	-	-	-	-
2416	UNI kN/m	GY	-1.961	-	-	-	-
2419	UNI kN/m	GY	-1.961	-	-	-	-
2422	UNI kN/m	GY	-1.961	-	-	-	-
2425	UNI kN/m	GY	-1.961	-	-	-	-
2428	UNI kN/m	GY	-1.961	-	-	-	-
2431	UNI kN/m	GY	-1.961	-	-	-	-
2461	UNI kN/m	GY	-1.961	-	-	-	-
2464	UNI kN/m	GY	-1.961	-	-	-	-
2488	UNI kN/m	GY	-1.961	-	-	-	-
2506	UNI kN/m	GY	-1.961	-	-	-	-
2524	UNI kN/m	GY	-1.961	-	-	-	-
2566	UNI kN/m	GY	-1.961	-	-	-	-

5 EQP DL (EQDL) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
102	-	-58.840	-	-	-	-
105	-	-58.840	-	-	-	-
108	-	-58.840	-	-	-	-
111	-	-58.840	-	-	-	-
148	-	-58.840	-	-	-	-
151	-	-58.840	-	-	-	-
154	-	-58.840	-	-	-	-
157	-	-58.840	-	-	-	-
212	-	-11.768	-	-	-	-
213	-	-11.768	-	-	-	-
214	-	-11.768	-	-	-	-
215	-	-11.768	-	-	-	-
236	-	-11.768	-	-	-	-

5 EQP DL (EQDL) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
237	-	-11.768	-	-	-	-
238	-	-11.768	-	-	-	-
239	-	-11.768	-	-	-	-
351	-	-14.710	-	-	-	-
352	-	-14.710	-	-	-	-
353	-	-14.710	-	-	-	-
354	-	-14.710	-	-	-	-
355	-	-14.710	-	-	-	-
356	-	-14.710	-	-	-	-
357	-	-14.710	-	-	-	-
358	-	-14.710	-	-	-	-

6 EQP LL (EQLL) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
351	-	-122.583	-	-	-	-
352	-	-122.583	-	-	-	-
353	-	-122.583	-	-	-	-
354	-	-122.583	-	-	-	-
355	-	-93.163	-	-	-	-
356	-	-93.163	-	-	-	-
357	-	-93.163	-	-	-	-
358	-	-93.163	-	-	-	-

7 WIND LOAD (+X) WL(+X) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
102	11.768	-	-	-	-	-
105	11.768	-	-	-	-	-
108	11.768	-	-	-	-	-
111	11.768	-	-	-	-	-
148	11.768	-	-	-	-	-
151	11.768	-	-	-	-	-
154	11.768	-	-	-	-	-
157	11.768	-	-	-	-	-
212	0.490	-	-	-	-	-
213	0.490	-	-	-	-	-
214	0.981	-	-	-	-	-
215	0.981	-	-	-	-	-
236	0.490	-	-	-	-	-
237	0.490	-	-	-	-	-
238	0.981	-	-	-	-	-
239	0.981	-	-	-	-	-
351	1.961	-	-	-	-	-
352	1.961	-	-	-	-	-
353	1.961	-	-	-	-	-

7 WIND LOAD (+X) WL(+X) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
354	1.961	-	-	-	-	-
355	1.961	-	-	-	-	-
356	1.961	-	-	-	-	-
357	1.961	-	-	-	-	-
358	1.961	-	-	-	-	-

7 WIND LOAD (+X) WL(+X) : Beam Loads

Beam	Type		Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
1857	UNI	kN/m	GX	0.490	-	-	-	-
1863	UNI	kN/m	GX	1.079	-	-	-	-
1869	UNI	kN/m	GX	0.490	-	-	-	-
1875	UNI	kN/m	GX	0.490	-	-	-	-
1881	UNI	kN/m	GX	1.275	-	-	-	-
1892	UNI	kN/m	GX	0.686	-	-	-	-
1898	UNI	kN/m	GX	1.079	-	-	-	-
1904	UNI	kN/m	GX	0.588	-	-	-	-
2125	UNI	kN/m	GX	0.490	-	-	-	-
2131	UNI	kN/m	GX	1.079	-	-	-	-
2134	UNI	kN/m	GX	1.079	-	-	-	-
2137	UNI	kN/m	GX	0.490	-	-	-	-
2140	UNI	kN/m	GX	0.490	-	-	-	-
2158	UNI	kN/m	GX	0.588	-	-	-	-
2167	UNI	kN/m	GX	0.686	-	-	-	-
2170	UNI	kN/m	GX	1.177	-	-	-	-
2176	UNI	kN/m	GX	0.588	-	-	-	-
2344	UNI	kN/m	GX	0.490	-	-	-	-
2347	UNI	kN/m	GX	0.490	-	-	-	-
2353	UNI	kN/m	GX	1.177	-	-	-	-
2356	UNI	kN/m	GX	1.177	-	-	-	-
2365	UNI	kN/m	GX	0.490	-	-	-	-
2371	UNI	kN/m	GX	0.588	-	-	-	-
2374	UNI	kN/m	GX	0.588	-	-	-	-
2377	UNI	kN/m	GX	1.373	-	-	-	-
2383	UNI	kN/m	GX	0.686	-	-	-	-
2386	UNI	kN/m	GX	0.686	-	-	-	-
2389	UNI	kN/m	GX	0.588	-	-	-	-
2392	UNI	kN/m	GX	0.785	-	-	-	-
2395	UNI	kN/m	GX	0.785	-	-	-	-
2398	UNI	kN/m	GX	0.686	-	-	-	-
2401	UNI	kN/m	GX	0.686	-	-	-	-
2449	UNI	kN/m	GX	1.177	-	-	-	-
2452	UNI	kN/m	GX	0.686	-	-	-	-
2607	UNI	kN/m	GX	0.588	-	-	-	-
2637	UNI	kN/m	GX	1.275	-	-	-	-
2638	UNI	kN/m	GX	1.275	-	-	-	-
2639	UNI	kN/m	GX	1.275	-	-	-	-

7 WIND LOAD (+X) WL(+X) : Beam Loads Cont...

Beam	Type	Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
2640	UNI kN/m	GX	1.275	-	-	-	-
2684	UNI kN/m	GX	1.275	-	-	-	-
2686	UNI kN/m	GX	1.275	-	-	-	-
2688	UNI kN/m	GX	1.275	-	-	-	-
2689	UNI kN/m	GX	1.275	-	-	-	-
2704	UNI kN/m	GX	0.196	-	-	-	-
2705	UNI kN/m	GX	0.196	-	-	-	-
2714	UNI kN/m	GX	0.294	-	-	-	-
2718	UNI kN/m	GX	0.294	-	-	-	-
2727	UNI kN/m	GX	0.196	-	-	-	-
2728	UNI kN/m	GX	0.196	-	-	-	-
2768	UNI kN/m	GX	1.177	-	-	-	-
2770	UNI kN/m	GX	1.079	-	-	-	-
2772	UNI kN/m	GX	1.373	-	-	-	-
2825	UNI kN/m	GX	0.196	-	-	-	-
2826	UNI kN/m	GX	0.196	-	-	-	-
2827	UNI kN/m	GX	0.294	-	-	-	-
2828	UNI kN/m	GX	0.294	-	-	-	-
2829	UNI kN/m	GX	0.196	-	-	-	-
2830	UNI kN/m	GX	0.196	-	-	-	-
2840	UNI kN/m	GX	0.490	-	-	-	-
2841	UNI kN/m	GX	1.275	-	-	-	-
2842	UNI kN/m	GX	0.490	-	-	-	-
2843	UNI kN/m	GX	1.275	-	-	-	-
2851	UNI kN/m	GX	0.686	-	-	-	-
3020	UNI kN/m	GX	0.588	-	-	-	-
3049	UNI kN/m	GX	0.196	-	-	-	-
3050	UNI kN/m	GX	0.196	-	-	-	-
3053	UNI kN/m	GX	0.196	-	-	-	-
3054	UNI kN/m	GX	0.196	-	-	-	-

8 WIND LOAD (-X) WL(-X) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
102	-11.768	-	-	-	-	-
105	-11.768	-	-	-	-	-
108	-11.768	-	-	-	-	-
111	-11.768	-	-	-	-	-
148	-11.768	-	-	-	-	-
151	-11.768	-	-	-	-	-
154	-11.768	-	-	-	-	-
157	-11.768	-	-	-	-	-
212	-0.490	-	-	-	-	-
213	-0.490	-	-	-	-	-
214	-0.981	-	-	-	-	-
215	-0.981	-	-	-	-	-
236	-0.490	-	-	-	-	-

8 WIND LOAD (-X) WL(-X) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
237	-0.490	-	-	-	-	-
238	-0.981	-	-	-	-	-
239	-0.981	-	-	-	-	-
351	-1.961	-	-	-	-	-
352	-1.961	-	-	-	-	-
353	-1.961	-	-	-	-	-
354	-1.961	-	-	-	-	-
355	-1.961	-	-	-	-	-
356	-1.961	-	-	-	-	-
357	-1.961	-	-	-	-	-
358	-1.961	-	-	-	-	-

8 WIND LOAD (-X) WL(-X) : Beam Loads

Beam	Type	Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
1857	UNI kN/m	GX	-0.490	-	-	-	-
1863	UNI kN/m	GX	-1.079	-	-	-	-
1869	UNI kN/m	GX	-0.490	-	-	-	-
1875	UNI kN/m	GX	-0.490	-	-	-	-
1881	UNI kN/m	GX	-1.275	-	-	-	-
1892	UNI kN/m	GX	-0.686	-	-	-	-
1898	UNI kN/m	GX	-1.079	-	-	-	-
1904	UNI kN/m	GX	-0.588	-	-	-	-
2125	UNI kN/m	GX	-0.490	-	-	-	-
2131	UNI kN/m	GX	-1.079	-	-	-	-
2134	UNI kN/m	GX	-1.079	-	-	-	-
2137	UNI kN/m	GX	-0.490	-	-	-	-
2140	UNI kN/m	GX	-0.490	-	-	-	-
2158	UNI kN/m	GX	-0.588	-	-	-	-
2167	UNI kN/m	GX	-0.686	-	-	-	-
2170	UNI kN/m	GX	-1.177	-	-	-	-
2176	UNI kN/m	GX	-0.588	-	-	-	-
2344	UNI kN/m	GX	-0.490	-	-	-	-
2347	UNI kN/m	GX	-0.490	-	-	-	-
2353	UNI kN/m	GX	-1.177	-	-	-	-
2356	UNI kN/m	GX	-1.177	-	-	-	-
2365	UNI kN/m	GX	-0.490	-	-	-	-
2371	UNI kN/m	GX	-0.588	-	-	-	-
2374	UNI kN/m	GX	-0.588	-	-	-	-
2377	UNI kN/m	GX	-1.373	-	-	-	-
2383	UNI kN/m	GX	-0.686	-	-	-	-
2386	UNI kN/m	GX	-0.686	-	-	-	-
2389	UNI kN/m	GX	-0.588	-	-	-	-
2392	UNI kN/m	GX	-0.785	-	-	-	-
2395	UNI kN/m	GX	-0.785	-	-	-	-
2398	UNI kN/m	GX	-0.686	-	-	-	-
2401	UNI kN/m	GX	-0.686	-	-	-	-

8 WIND LOAD (-X) WL(-X) : Beam Loads Cont...

Beam	Type	Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
2449	UNI kN/m	GX	-1.177	-	-	-	-
2452	UNI kN/m	GX	-0.686	-	-	-	-
2607	UNI kN/m	GX	-0.588	-	-	-	-
2637	UNI kN/m	GX	-1.275	-	-	-	-
2638	UNI kN/m	GX	-1.275	-	-	-	-
2639	UNI kN/m	GX	-1.275	-	-	-	-
2640	UNI kN/m	GX	-1.275	-	-	-	-
2684	UNI kN/m	GX	-1.275	-	-	-	-
2686	UNI kN/m	GX	-1.275	-	-	-	-
2688	UNI kN/m	GX	-1.275	-	-	-	-
2689	UNI kN/m	GX	-1.275	-	-	-	-
2704	UNI kN/m	GX	-0.196	-	-	-	-
2705	UNI kN/m	GX	-0.196	-	-	-	-
2714	UNI kN/m	GX	-0.294	-	-	-	-
2718	UNI kN/m	GX	-0.294	-	-	-	-
2727	UNI kN/m	GX	-0.196	-	-	-	-
2728	UNI kN/m	GX	-0.196	-	-	-	-
2768	UNI kN/m	GX	-1.177	-	-	-	-
2770	UNI kN/m	GX	-1.079	-	-	-	-
2772	UNI kN/m	GX	-1.373	-	-	-	-
2825	UNI kN/m	GX	-0.196	-	-	-	-
2826	UNI kN/m	GX	-0.196	-	-	-	-
2827	UNI kN/m	GX	-0.294	-	-	-	-
2828	UNI kN/m	GX	-0.294	-	-	-	-
2829	UNI kN/m	GX	-0.196	-	-	-	-
2830	UNI kN/m	GX	-0.196	-	-	-	-
2840	UNI kN/m	GX	-0.490	-	-	-	-
2841	UNI kN/m	GX	-1.275	-	-	-	-
2842	UNI kN/m	GX	-0.490	-	-	-	-
2843	UNI kN/m	GX	-1.275	-	-	-	-
2851	UNI kN/m	GX	-0.686	-	-	-	-
3020	UNI kN/m	GX	-0.588	-	-	-	-
3049	UNI kN/m	GX	-0.196	-	-	-	-
3050	UNI kN/m	GX	-0.196	-	-	-	-
3053	UNI kN/m	GX	-0.196	-	-	-	-
3054	UNI kN/m	GX	-0.196	-	-	-	-

9 WIND LOAD (+Z) WL(+Z) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
102	-	-	11.768	-	-	-
105	-	-	11.768	-	-	-
108	-	-	11.768	-	-	-
111	-	-	11.768	-	-	-
148	-	-	11.768	-	-	-
151	-	-	11.768	-	-	-
154	-	-	11.768	-	-	-
157	-	-	11.768	-	-	-
212	-	-	0.490	-	-	-
213	-	-	0.490	-	-	-
214	-	-	0.981	-	-	-
215	-	-	0.981	-	-	-
236	-	-	0.490	-	-	-
237	-	-	0.490	-	-	-
238	-	-	0.981	-	-	-
239	-	-	0.981	-	-	-
351	-	-	0.981	-	-	-
352	-	-	0.981	-	-	-
353	-	-	0.981	-	-	-
354	-	-	0.981	-	-	-
355	-	-	0.981	-	-	-
356	-	-	0.981	-	-	-
357	-	-	0.981	-	-	-
358	-	-	0.981	-	-	-

9 WIND LOAD (+Z) WL(+Z) : Beam Loads

Beam	Type	Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
1197	UNI kN/m	GZ	0.294	-	-	-	-
1659	UNI kN/m	GZ	0.294	-	-	-	-
1857	UNI kN/m	GZ	1.177	-	-	-	-
1863	UNI kN/m	GZ	1.275	-	-	-	-
1869	UNI kN/m	GZ	1.177	-	-	-	-
1875	UNI kN/m	GZ	2.256	-	-	-	-
1881	UNI kN/m	GZ	1.961	-	-	-	-
1892	UNI kN/m	GZ	2.059	-	-	-	-
1898	UNI kN/m	GZ	0.981	-	-	-	-
1904	UNI kN/m	GZ	1.079	-	-	-	-
2068	UNI kN/m	GZ	0.294	-	-	-	-
2125	UNI kN/m	GZ	1.177	-	-	-	-
2131	UNI kN/m	GZ	1.275	-	-	-	-
2134	UNI kN/m	GZ	1.275	-	-	-	-
2137	UNI kN/m	GZ	1.275	-	-	-	-
2140	UNI kN/m	GZ	1.177	-	-	-	-
2158	UNI kN/m	GZ	1.373	-	-	-	-
2167	UNI kN/m	GZ	2.059	-	-	-	-
2170	UNI kN/m	GZ	1.079	-	-	-	-

9 WIND LOAD (+Z) WL(+Z) : Beam Loads Cont...

Beam	Type	Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
2176	UNI kN/m	GZ	1.079	-	-	-	-
2194	UNI kN/m	GZ	0.294	-	-	-	-
2197	UNI kN/m	GZ	0.294	-	-	-	-
2242	UNI kN/m	GZ	0.294	-	-	-	-
2245	UNI kN/m	GZ	0.294	-	-	-	-
2344	UNI kN/m	GZ	1.275	-	-	-	-
2347	UNI kN/m	GZ	1.275	-	-	-	-
2353	UNI kN/m	GZ	1.373	-	-	-	-
2356	UNI kN/m	GZ	1.373	-	-	-	-
2365	UNI kN/m	GZ	1.275	-	-	-	-
2371	UNI kN/m	GZ	2.452	-	-	-	-
2374	UNI kN/m	GZ	2.452	-	-	-	-
2377	UNI kN/m	GZ	2.059	-	-	-	-
2383	UNI kN/m	GZ	1.471	-	-	-	-
2386	UNI kN/m	GZ	1.471	-	-	-	-
2389	UNI kN/m	GZ	1.373	-	-	-	-
2392	UNI kN/m	GZ	2.157	-	-	-	-
2395	UNI kN/m	GZ	2.157	-	-	-	-
2398	UNI kN/m	GZ	1.079	-	-	-	-
2401	UNI kN/m	GZ	1.079	-	-	-	-
2410	UNI kN/m	GZ	0.294	-	-	-	-
2449	UNI kN/m	GZ	1.373	-	-	-	-
2452	UNI kN/m	GZ	1.471	-	-	-	-
2607	UNI kN/m	GZ	1.373	-	-	-	-
2637	UNI kN/m	GZ	0.686	-	-	-	-
2638	UNI kN/m	GZ	0.686	-	-	-	-
2639	UNI kN/m	GZ	0.686	-	-	-	-
2640	UNI kN/m	GZ	0.686	-	-	-	-
2684	UNI kN/m	GZ	0.686	-	-	-	-
2686	UNI kN/m	GZ	0.686	-	-	-	-
2688	UNI kN/m	GZ	0.686	-	-	-	-
2689	UNI kN/m	GZ	0.686	-	-	-	-
2704	UNI kN/m	GZ	0.392	-	-	-	-
2705	UNI kN/m	GZ	0.392	-	-	-	-
2714	UNI kN/m	GZ	0.785	-	-	-	-
2718	UNI kN/m	GZ	0.785	-	-	-	-
2727	UNI kN/m	GZ	0.392	-	-	-	-
2728	UNI kN/m	GZ	0.392	-	-	-	-
2768	UNI kN/m	GZ	1.079	-	-	-	-
2770	UNI kN/m	GZ	0.981	-	-	-	-
2772	UNI kN/m	GZ	2.059	-	-	-	-
2825	UNI kN/m	GZ	0.392	-	-	-	-
2826	UNI kN/m	GZ	0.392	-	-	-	-
2827	UNI kN/m	GZ	0.785	-	-	-	-
2828	UNI kN/m	GZ	0.785	-	-	-	-
2829	UNI kN/m	GZ	0.392	-	-	-	-
2830	UNI kN/m	GZ	0.392	-	-	-	-
2840	UNI kN/m	GZ	2.256	-	-	-	-
2841	UNI kN/m	GZ	1.961	-	-	-	-

9 WIND LOAD (+Z) WL(+Z) : Beam Loads Cont...

Beam	Type	Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
2842	UNI kN/m	GZ	2.256	-	-	-	-
2843	UNI kN/m	GZ	1.961	-	-	-	-
2851	UNI kN/m	GZ	2.059	-	-	-	-
2975	UNI kN/m	GZ	0.294	-	-	-	-
2976	UNI kN/m	GZ	0.294	-	-	-	-
3020	UNI kN/m	GZ	2.452	-	-	-	-
3027	UNI kN/m	GZ	0.294	-	-	-	-
3049	UNI kN/m	GZ	0.392	-	-	-	-
3050	UNI kN/m	GZ	0.392	-	-	-	-
3053	UNI kN/m	GZ	0.392	-	-	-	-
3054	UNI kN/m	GZ	0.392	-	-	-	-

10 WIND LOAD (-Z) WL(-Z) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
102	-	-	-11.768	-	-	-
105	-	-	-11.768	-	-	-
108	-	-	-11.768	-	-	-
111	-	-	-11.768	-	-	-
148	-	-	-11.768	-	-	-
151	-	-	-11.768	-	-	-
154	-	-	-11.768	-	-	-
157	-	-	-11.768	-	-	-
212	-	-	-0.490	-	-	-
213	-	-	-0.490	-	-	-
214	-	-	-0.981	-	-	-
215	-	-	-0.981	-	-	-
236	-	-	-0.490	-	-	-
237	-	-	-0.490	-	-	-
238	-	-	-0.981	-	-	-
239	-	-	-0.981	-	-	-
351	-	-	-0.981	-	-	-
352	-	-	-0.981	-	-	-
353	-	-	-0.981	-	-	-
354	-	-	-0.981	-	-	-
355	-	-	-0.981	-	-	-
356	-	-	-0.981	-	-	-
357	-	-	-0.981	-	-	-
358	-	-	-0.981	-	-	-

10 WIND LOAD (-Z) WL(-Z) : Beam Loads

Beam	Type	Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
1197	UNI kN/m	GZ	-0.294	-	-	-	-
1659	UNI kN/m	GZ	-0.294	-	-	-	-
1857	UNI kN/m	GZ	-1.177	-	-	-	-
1863	UNI kN/m	GZ	-1.275	-	-	-	-
1869	UNI kN/m	GZ	-1.177	-	-	-	-
1875	UNI kN/m	GZ	-2.256	-	-	-	-
1881	UNI kN/m	GZ	-1.961	-	-	-	-
1892	UNI kN/m	GZ	-2.059	-	-	-	-
1898	UNI kN/m	GZ	-0.981	-	-	-	-
1904	UNI kN/m	GZ	-1.079	-	-	-	-
2068	UNI kN/m	GZ	-0.294	-	-	-	-
2125	UNI kN/m	GZ	-1.177	-	-	-	-
2131	UNI kN/m	GZ	-1.275	-	-	-	-
2134	UNI kN/m	GZ	-1.275	-	-	-	-
2137	UNI kN/m	GZ	-1.275	-	-	-	-
2140	UNI kN/m	GZ	-1.177	-	-	-	-
2158	UNI kN/m	GZ	-1.373	-	-	-	-
2167	UNI kN/m	GZ	-2.059	-	-	-	-
2170	UNI kN/m	GZ	-1.079	-	-	-	-
2176	UNI kN/m	GZ	-1.079	-	-	-	-
2194	UNI kN/m	GZ	-0.294	-	-	-	-
2197	UNI kN/m	GZ	-0.294	-	-	-	-
2242	UNI kN/m	GZ	-0.294	-	-	-	-
2245	UNI kN/m	GZ	-0.294	-	-	-	-
2344	UNI kN/m	GZ	-1.275	-	-	-	-
2347	UNI kN/m	GZ	-1.275	-	-	-	-
2353	UNI kN/m	GZ	-1.373	-	-	-	-
2356	UNI kN/m	GZ	-1.373	-	-	-	-
2365	UNI kN/m	GZ	-1.275	-	-	-	-
2371	UNI kN/m	GZ	-2.452	-	-	-	-
2374	UNI kN/m	GZ	-2.452	-	-	-	-
2377	UNI kN/m	GZ	-2.059	-	-	-	-
2383	UNI kN/m	GZ	-1.471	-	-	-	-
2386	UNI kN/m	GZ	-1.471	-	-	-	-
2389	UNI kN/m	GZ	-1.373	-	-	-	-
2392	UNI kN/m	GZ	-2.157	-	-	-	-
2395	UNI kN/m	GZ	-2.157	-	-	-	-
2398	UNI kN/m	GZ	-1.079	-	-	-	-
2401	UNI kN/m	GZ	-1.079	-	-	-	-
2410	UNI kN/m	GZ	-0.294	-	-	-	-
2449	UNI kN/m	GZ	-1.373	-	-	-	-
2452	UNI kN/m	GZ	-1.471	-	-	-	-
2607	UNI kN/m	GZ	-1.373	-	-	-	-
2637	UNI kN/m	GZ	-0.686	-	-	-	-
2638	UNI kN/m	GZ	-0.686	-	-	-	-
2639	UNI kN/m	GZ	-0.686	-	-	-	-
2640	UNI kN/m	GZ	-0.686	-	-	-	-
2684	UNI kN/m	GZ	-0.686	-	-	-	-
2686	UNI kN/m	GZ	-0.686	-	-	-	-

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10 WIND LOAD (-Z) WL(-Z) : Beam Loads Cont...

Beam	Type	Direction	Fa	Da (m)	Fb	Db	Ecc. (m)
2688	UNI kN/m	GZ	-0.686	-	-	-	-
2689	UNI kN/m	GZ	-0.686	-	-	-	-
2704	UNI kN/m	GZ	-0.392	-	-	-	-
2705	UNI kN/m	GZ	-0.392	-	-	-	-
2714	UNI kN/m	GZ	-0.785	-	-	-	-
2718	UNI kN/m	GZ	-0.785	-	-	-	-
2727	UNI kN/m	GZ	-0.392	-	-	-	-
2728	UNI kN/m	GZ	-0.392	-	-	-	-
2768	UNI kN/m	GZ	-1.079	-	-	-	-
2770	UNI kN/m	GZ	-0.981	-	-	-	-
2772	UNI kN/m	GZ	-2.059	-	-	-	-
2825	UNI kN/m	GZ	-0.392	-	-	-	-
2826	UNI kN/m	GZ	-0.392	-	-	-	-
2827	UNI kN/m	GZ	-0.785	-	-	-	-
2828	UNI kN/m	GZ	-0.785	-	-	-	-
2829	UNI kN/m	GZ	-0.392	-	-	-	-
2830	UNI kN/m	GZ	-0.392	-	-	-	-
2840	UNI kN/m	GZ	-2.256	-	-	-	-
2841	UNI kN/m	GZ	-1.961	-	-	-	-
2842	UNI kN/m	GZ	-2.256	-	-	-	-
2843	UNI kN/m	GZ	-1.961	-	-	-	-
2851	UNI kN/m	GZ	-2.059	-	-	-	-
2975	UNI kN/m	GZ	-0.294	-	-	-	-
2976	UNI kN/m	GZ	-0.294	-	-	-	-
3020	UNI kN/m	GZ	-2.452	-	-	-	-
3027	UNI kN/m	GZ	-0.294	-	-	-	-
3049	UNI kN/m	GZ	-0.392	-	-	-	-
3050	UNI kN/m	GZ	-0.392	-	-	-	-
3053	UNI kN/m	GZ	-0.392	-	-	-	-
3054	UNI kN/m	GZ	-0.392	-	-	-	-

11 SEISMIC LOAD SL(+X) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
1	0.373	-	-	-	-	-
2	0.226	-	-	-	-	-
3	0.245	-	-	-	-	-
4	0.618	-	-	-	-	-
5	0.137	-	-	-	-	-
6	0.471	-	-	-	-	-
7	1.216	-	-	-	-	-
8	0.118	-	-	-	-	-
9	0.490	-	-	-	-	-
10	0.147	-	-	-	-	-
11	0.265	-	-	-	-	-
53	2.971	-	-	-	-	-
61	5.747	-	-	-	-	-

11 SEISMIC LOAD SL(+X) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
68	2.834	-	-	-	-	-
102	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
105	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
108	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
111	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
113	1.167	-	-	-	-	-
114	4.746	-	-	-	-	-
122	9.463	-	-	-	-	-
129	3.991	-	-	-	-	-
148	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
151	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
154	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
157	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
162	2.932	-	-	-	-	-
170	3.707	-	-	-	-	-
177	2.226	-	-	-	-	-
212	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
213	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
214	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
215	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
219	4.256	-	-	-	-	-
227	9.375	-	-	-	-	-
235	4.227	-	-	-	-	-
236	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
237	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
238	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
239	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
286	4.688	-	-	-	-	-
293	8.787	-	-	-	-	-
300	4.384	-	-	-	-	-
325	0.647	-	-	-	-	-
326	2.687	-	-	-	-	-
333	4.550	-	-	-	-	-

11 SEISMIC LOAD SL(+X) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
340	2.050	-	-	-	-	-
351	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
352	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
353	-	-	6.865	-	-	-
	6.865	-	-	-	-	-
354	-	-	6.865	-	-	-
	6.865	-	-	-	-	-
355	-	-	4.903	-	-	-
	4.903	-	-	-	-	-
356	-	-	4.903	-	-	-
	4.903	-	-	-	-	-
357	-	-	4.903	-	-	-
	4.903	-	-	-	-	-
358	-	-	4.903	-	-	-
	4.903	-	-	-	-	-
388	0.686	-	-	-	-	-
405	0.373	-	-	-	-	-
406	0.226	-	-	-	-	-
407	0.245	-	-	-	-	-
408	0.618	-	-	-	-	-
409	0.137	-	-	-	-	-
410	0.471	-	-	-	-	-
411	1.216	-	-	-	-	-
412	0.118	-	-	-	-	-
413	0.490	-	-	-	-	-
414	0.147	-	-	-	-	-
415	0.265	-	-	-	-	-

12 SEISMIC LOAD SL(-X) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
1	-0.373	-	-	-	-	-
2	-0.226	-	-	-	-	-
3	-0.245	-	-	-	-	-
4	-0.618	-	-	-	-	-
5	-0.137	-	-	-	-	-
6	-0.471	-	-	-	-	-
7	-1.216	-	-	-	-	-
8	-0.118	-	-	-	-	-
9	-0.490	-	-	-	-	-
10	-0.147	-	-	-	-	-
11	-0.265	-	-	-	-	-
53	-2.971	-	-	-	-	-
61	-5.747	-	-	-	-	-
68	-2.834	-	-	-	-	-

12 SEISMIC LOAD SL(-X) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
102	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
105	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
108	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
111	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
113	-1.167	-	-	-	-	-
114	-4.746	-	-	-	-	-
122	-9.463	-	-	-	-	-
129	-3.991	-	-	-	-	-
148	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
151	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
154	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
157	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
162	-2.932	-	-	-	-	-
170	-3.707	-	-	-	-	-
177	-2.226	-	-	-	-	-
212	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
213	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
214	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
215	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
219	-4.256	-	-	-	-	-
227	-9.375	-	-	-	-	-
235	-4.227	-	-	-	-	-
236	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
237	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
238	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
239	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
286	-4.688	-	-	-	-	-
293	-8.787	-	-	-	-	-
300	-4.384	-	-	-	-	-
325	-0.647	-	-	-	-	-
326	-2.687	-	-	-	-	-
333	-4.550	-	-	-	-	-
340	-2.050	-	-	-	-	-

12 SEISMIC LOAD SL(-X) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
351	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
352	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
353	-	-	-6.865	-	-	-
	-6.865	-	-	-	-	-
354	-	-	-6.865	-	-	-
	-6.865	-	-	-	-	-
355	-	-	-4.903	-	-	-
	-4.903	-	-	-	-	-
356	-	-	-4.903	-	-	-
	-4.903	-	-	-	-	-
357	-	-	-4.903	-	-	-
	-4.903	-	-	-	-	-
358	-	-	-4.903	-	-	-
	-4.903	-	-	-	-	-
388	-0.686	-	-	-	-	-
405	-0.373	-	-	-	-	-
406	-0.226	-	-	-	-	-
407	-0.245	-	-	-	-	-
408	-0.618	-	-	-	-	-
409	-0.137	-	-	-	-	-
410	-0.471	-	-	-	-	-
411	-1.216	-	-	-	-	-
412	-0.118	-	-	-	-	-
413	-0.490	-	-	-	-	-
414	-0.147	-	-	-	-	-
415	-0.265	-	-	-	-	-

13 SEISMIC LOAD SL(+Z) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
1	-	-	0.373	-	-	-
2	-	-	0.226	-	-	-
3	-	-	0.245	-	-	-
4	-	-	0.618	-	-	-
5	-	-	0.137	-	-	-
6	-	-	0.471	-	-	-
7	-	-	1.216	-	-	-
8	-	-	0.118	-	-	-
9	-	-	0.490	-	-	-
10	-	-	0.147	-	-	-
11	-	-	0.265	-	-	-
53	-	-	2.971	-	-	-
61	-	-	5.747	-	-	-
68	-	-	2.834	-	-	-
102	-	-	5.884	-	-	-

13 SEISMIC LOAD SL(+Z) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
102	5.884	-	-	-	-	-
105	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
108	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
111	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
113	-	-	1.167	-	-	-
114	-	-	4.746	-	-	-
122	-	-	9.463	-	-	-
129	-	-	3.991	-	-	-
148	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
151	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
154	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
157	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
162	-	-	2.932	-	-	-
170	-	-	3.707	-	-	-
177	-	-	2.226	-	-	-
212	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
213	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
214	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
215	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
219	-	-	4.256	-	-	-
227	-	-	9.375	-	-	-
235	-	-	4.227	-	-	-
236	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
237	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
238	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
239	-	-	2.942	-	-	-
	2.942	-	-	-	-	-
286	-	-	4.688	-	-	-
293	-	-	8.787	-	-	-
300	-	-	4.384	-	-	-
325	-	-	0.647	-	-	-
326	-	-	2.687	-	-	-
333	-	-	4.550	-	-	-
340	-	-	2.050	-	-	-
351	-	-	5.884	-	-	-

13 SEISMIC LOAD SL(+Z) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
351	5.884	-	-	-	-	-
352	-	-	5.884	-	-	-
	5.884	-	-	-	-	-
353	-	-	6.865	-	-	-
	6.865	-	-	-	-	-
354	-	-	6.865	-	-	-
	6.865	-	-	-	-	-
355	-	-	4.903	-	-	-
	4.903	-	-	-	-	-
356	-	-	4.903	-	-	-
	4.903	-	-	-	-	-
357	-	-	4.903	-	-	-
	4.903	-	-	-	-	-
358	-	-	4.903	-	-	-
	4.903	-	-	-	-	-
388	-	-	0.686	-	-	-
405	-	-	0.373	-	-	-
406	-	-	0.226	-	-	-
407	-	-	0.245	-	-	-
408	-	-	0.618	-	-	-
409	-	-	0.137	-	-	-
410	-	-	0.471	-	-	-
411	-	-	1.216	-	-	-
412	-	-	0.118	-	-	-
413	-	-	0.490	-	-	-
414	-	-	0.147	-	-	-
415	-	-	0.265	-	-	-

14 SEISMIC LOAD SL(-Z) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
1	-	-	-0.373	-	-	-
2	-	-	-0.226	-	-	-
3	-	-	-0.245	-	-	-
4	-	-	-0.618	-	-	-
5	-	-	-0.137	-	-	-
6	-	-	-0.471	-	-	-
7	-	-	-1.216	-	-	-
8	-	-	-0.118	-	-	-
9	-	-	-0.490	-	-	-
10	-	-	-0.147	-	-	-
11	-	-	-0.265	-	-	-
53	-	-	-2.971	-	-	-
61	-	-	-5.747	-	-	-
68	-	-	-2.834	-	-	-
102	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-

14 SEISMIC LOAD SL(-Z) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
105	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
108	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
111	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
113	-	-	-1.167	-	-	-
114	-	-	-4.746	-	-	-
122	-	-	-9.463	-	-	-
129	-	-	-3.991	-	-	-
148	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
151	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
154	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
157	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
162	-	-	-2.932	-	-	-
170	-	-	-3.707	-	-	-
177	-	-	-2.226	-	-	-
212	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
213	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
214	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
215	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
219	-	-	-4.256	-	-	-
227	-	-	-9.375	-	-	-
235	-	-	-4.227	-	-	-
236	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
237	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
238	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
239	-	-	-2.942	-	-	-
	-2.942	-	-	-	-	-
286	-	-	-4.688	-	-	-
293	-	-	-8.787	-	-	-
300	-	-	-4.384	-	-	-
325	-	-	-0.647	-	-	-
326	-	-	-2.687	-	-	-
333	-	-	-4.550	-	-	-
340	-	-	-2.050	-	-	-
351	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-

14 SEISMIC LOAD SL(-Z) : Node Loads Cont...

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
352	-	-	-5.884	-	-	-
	-5.884	-	-	-	-	-
353	-	-	-6.865	-	-	-
	-6.865	-	-	-	-	-
354	-	-	-6.865	-	-	-
	-6.865	-	-	-	-	-
355	-	-	-4.903	-	-	-
	-4.903	-	-	-	-	-
356	-	-	-4.903	-	-	-
	-4.903	-	-	-	-	-
357	-	-	-4.903	-	-	-
	-4.903	-	-	-	-	-
358	-	-	-4.903	-	-	-
	-4.903	-	-	-	-	-
388	-	-	-0.686	-	-	-
405	-	-	-0.373	-	-	-
406	-	-	-0.226	-	-	-
407	-	-	-0.245	-	-	-
408	-	-	-0.618	-	-	-
409	-	-	-0.137	-	-	-
410	-	-	-0.471	-	-	-
411	-	-	-1.216	-	-	-
412	-	-	-0.118	-	-	-
413	-	-	-0.490	-	-	-
414	-	-	-0.147	-	-	-
415	-	-	-0.265	-	-	-

15 TEMPERATURE LOAD (TL) : Temperature Loads

Beam	Temperature Dif. Axial (°C)	Cross (°C)
74	25.000	-
79	25.000	-
192	25.000	-
199	25.000	-
206	25.000	-
212	25.000	-
218	25.000	-
224	25.000	-
230	25.000	-
236	25.000	-
242	25.000	-
249	25.000	-
255	25.000	-
261	25.000	-
267	25.000	-
272	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
277	25.000	-
282	25.000	-
288	25.000	-
293	25.000	-
298	25.000	-
303	25.000	-
313	25.000	-
318	25.000	-
323	25.000	-
328	25.000	-
333	25.000	-
343	25.000	-
348	25.000	-
352	25.000	-
358	25.000	-
363	25.000	-
418	25.000	-
422	25.000	-
567	25.000	-
579	25.000	-
584	25.000	-
591	25.000	-
603	25.000	-
615	25.000	-
622	25.000	-
636	25.000	-
648	25.000	-
660	25.000	-
671	25.000	-
676	25.000	-
707	25.000	-
718	25.000	-
723	25.000	-
730	25.000	-
742	25.000	-
754	25.000	-
766	25.000	-
780	25.000	-
792	25.000	-
804	25.000	-
815	25.000	-
820	25.000	-
851	25.000	-
862	25.000	-
867	25.000	-
874	25.000	-
886	25.000	-
898	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
914	25.000	-
921	25.000	-
933	25.000	-
944	25.000	-
961	25.000	-
968	25.000	-
984	25.000	-
991	25.000	-
1015	25.000	-
1034	25.000	-
1041	25.000	-
1052	25.000	-
1065	25.000	-
1089	25.000	-
1101	25.000	-
1113	25.000	-
1125	25.000	-
1142	25.000	-
1149	25.000	-
1161	25.000	-
1173	25.000	-
1185	25.000	-
1190	25.000	-
1197	25.000	-
1203	25.000	-
1209	25.000	-
1215	25.000	-
1221	25.000	-
1227	25.000	-
1233	25.000	-
1239	25.000	-
1245	25.000	-
1251	25.000	-
1257	25.000	-
1263	25.000	-
1269	25.000	-
1275	25.000	-
1287	25.000	-
1293	25.000	-
1299	25.000	-
1305	25.000	-
1311	25.000	-
1317	25.000	-
1323	25.000	-
1329	25.000	-
1335	25.000	-
1341	25.000	-
1347	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
1353	25.000	-
1359	25.000	-
1366	25.000	-
1372	25.000	-
1378	25.000	-
1384	25.000	-
1390	25.000	-
1396	25.000	-
1402	25.000	-
1408	25.000	-
1414	25.000	-
1420	25.000	-
1426	25.000	-
1432	25.000	-
1437	25.000	-
1442	25.000	-
1447	25.000	-
1452	25.000	-
1457	25.000	-
1462	25.000	-
1467	25.000	-
1472	25.000	-
1477	25.000	-
1482	25.000	-
1489	25.000	-
1495	25.000	-
1501	25.000	-
1507	25.000	-
1512	25.000	-
1517	25.000	-
1522	25.000	-
1527	25.000	-
1532	25.000	-
1537	25.000	-
1542	25.000	-
1547	25.000	-
1552	25.000	-
1557	25.000	-
1562	25.000	-
1567	25.000	-
1572	25.000	-
1579	25.000	-
1585	25.000	-
1591	25.000	-
1597	25.000	-
1604	25.000	-
1610	25.000	-
1616	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
1622	25.000	-
1628	25.000	-
1634	25.000	-
1641	25.000	-
1647	25.000	-
1653	25.000	-
1659	25.000	-
1664	25.000	-
1669	25.000	-
1674	25.000	-
1679	25.000	-
1684	25.000	-
1689	25.000	-
1694	25.000	-
1699	25.000	-
1704	25.000	-
1709	25.000	-
1714	25.000	-
1719	25.000	-
1724	25.000	-
1729	25.000	-
1734	25.000	-
1739	25.000	-
1744	25.000	-
1749	25.000	-
1754	25.000	-
1759	25.000	-
1764	25.000	-
1769	25.000	-
1774	25.000	-
1779	25.000	-
1785	25.000	-
1791	25.000	-
1797	25.000	-
1809	25.000	-
1815	25.000	-
1821	25.000	-
1827	25.000	-
1833	25.000	-
1839	25.000	-
1845	25.000	-
1851	25.000	-
1857	25.000	-
1863	25.000	-
1869	25.000	-
1875	25.000	-
1881	25.000	-
1892	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
1898	25.000	-
1904	25.000	-
1936	25.000	-
1939	25.000	-
1942	25.000	-
1945	25.000	-
1948	25.000	-
1954	25.000	-
1957	25.000	-
1999	25.000	-
2002	25.000	-
2005	25.000	-
2008	25.000	-
2011	25.000	-
2014	25.000	-
2017	25.000	-
2020	25.000	-
2023	25.000	-
2026	25.000	-
2029	25.000	-
2032	25.000	-
2035	25.000	-
2038	25.000	-
2041	25.000	-
2044	25.000	-
2047	25.000	-
2050	25.000	-
2053	25.000	-
2056	25.000	-
2059	25.000	-
2068	25.000	-
2071	25.000	-
2074	25.000	-
2077	25.000	-
2080	25.000	-
2083	25.000	-
2086	25.000	-
2089	25.000	-
2092	25.000	-
2095	25.000	-
2101	25.000	-
2104	25.000	-
2107	25.000	-
2110	25.000	-
2113	25.000	-
2116	25.000	-
2119	25.000	-
2125	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
2131	25.000	-
2134	25.000	-
2137	25.000	-
2140	25.000	-
2158	25.000	-
2167	25.000	-
2170	25.000	-
2176	25.000	-
2179	25.000	-
2182	25.000	-
2185	25.000	-
2188	25.000	-
2194	25.000	-
2197	25.000	-
2200	25.000	-
2203	25.000	-
2206	25.000	-
2209	25.000	-
2212	25.000	-
2215	25.000	-
2218	25.000	-
2221	25.000	-
2224	25.000	-
2227	25.000	-
2230	25.000	-
2233	25.000	-
2236	25.000	-
2242	25.000	-
2245	25.000	-
2248	25.000	-
2251	25.000	-
2254	25.000	-
2257	25.000	-
2260	25.000	-
2263	25.000	-
2266	25.000	-
2269	25.000	-
2272	25.000	-
2275	25.000	-
2278	25.000	-
2281	25.000	-
2284	25.000	-
2287	25.000	-
2290	25.000	-
2293	25.000	-
2296	25.000	-
2299	25.000	-
2302	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
2305	25.000	-
2308	25.000	-
2311	25.000	-
2314	25.000	-
2317	25.000	-
2320	25.000	-
2323	25.000	-
2326	25.000	-
2329	25.000	-
2332	25.000	-
2335	25.000	-
2338	25.000	-
2341	25.000	-
2344	25.000	-
2347	25.000	-
2353	25.000	-
2356	25.000	-
2365	25.000	-
2371	25.000	-
2374	25.000	-
2377	25.000	-
2383	25.000	-
2386	25.000	-
2389	25.000	-
2392	25.000	-
2395	25.000	-
2398	25.000	-
2401	25.000	-
2404	25.000	-
2407	25.000	-
2410	25.000	-
2413	25.000	-
2416	25.000	-
2419	25.000	-
2422	25.000	-
2425	25.000	-
2428	25.000	-
2431	25.000	-
2434	25.000	-
2437	25.000	-
2440	25.000	-
2443	25.000	-
2446	25.000	-
2449	25.000	-
2452	25.000	-
2455	25.000	-
2458	25.000	-
2461	25.000	-

Software licensed to STAAD.Pro Advanced
CONNECTED User: Suman Podder

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
2464	25.000	-
2467	25.000	-
2470	25.000	-
2473	25.000	-
2476	25.000	-
2479	25.000	-
2482	25.000	-
2485	25.000	-
2488	25.000	-
2491	25.000	-
2494	25.000	-
2497	25.000	-
2500	25.000	-
2503	25.000	-
2506	25.000	-
2509	25.000	-
2512	25.000	-
2515	25.000	-
2518	25.000	-
2521	25.000	-
2524	25.000	-
2527	25.000	-
2530	25.000	-
2533	25.000	-
2536	25.000	-
2539	25.000	-
2542	25.000	-
2545	25.000	-
2548	25.000	-
2551	25.000	-
2554	25.000	-
2557	25.000	-
2560	25.000	-
2563	25.000	-
2566	25.000	-
2600	25.000	-
2607	25.000	-
2629	25.000	-
2630	25.000	-
2631	25.000	-
2632	25.000	-
2633	25.000	-
2634	25.000	-
2635	25.000	-
2636	25.000	-
2637	25.000	-
2638	25.000	-
2639	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
2640	25.000	-
2641	25.000	-
2642	25.000	-
2643	25.000	-
2644	25.000	-
2645	25.000	-
2646	25.000	-
2647	25.000	-
2648	25.000	-
2649	25.000	-
2650	25.000	-
2651	25.000	-
2652	25.000	-
2653	25.000	-
2654	25.000	-
2655	25.000	-
2656	25.000	-
2661	25.000	-
2662	25.000	-
2663	25.000	-
2665	25.000	-
2666	25.000	-
2668	25.000	-
2669	25.000	-
2670	25.000	-
2672	25.000	-
2673	25.000	-
2674	25.000	-
2676	25.000	-
2677	25.000	-
2678	25.000	-
2680	25.000	-
2681	25.000	-
2682	25.000	-
2683	25.000	-
2684	25.000	-
2685	25.000	-
2686	25.000	-
2687	25.000	-
2688	25.000	-
2689	25.000	-
2690	25.000	-
2691	25.000	-
2692	25.000	-
2693	25.000	-
2694	25.000	-
2695	25.000	-
2696	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
2697	25.000	-
2698	25.000	-
2699	25.000	-
2700	25.000	-
2702	25.000	-
2703	25.000	-
2704	25.000	-
2705	25.000	-
2706	25.000	-
2708	25.000	-
2709	25.000	-
2710	25.000	-
2711	25.000	-
2714	25.000	-
2715	25.000	-
2716	25.000	-
2717	25.000	-
2718	25.000	-
2720	25.000	-
2721	25.000	-
2722	25.000	-
2723	25.000	-
2724	25.000	-
2725	25.000	-
2726	25.000	-
2727	25.000	-
2728	25.000	-
2729	25.000	-
2730	25.000	-
2731	25.000	-
2732	25.000	-
2733	25.000	-
2734	25.000	-
2735	25.000	-
2736	25.000	-
2737	25.000	-
2738	25.000	-
2739	25.000	-
2740	25.000	-
2741	25.000	-
2742	25.000	-
2743	25.000	-
2744	25.000	-
2745	25.000	-
2746	25.000	-
2747	25.000	-
2748	25.000	-
2749	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
2750	25.000	-
2752	25.000	-
2753	25.000	-
2754	25.000	-
2756	25.000	-
2757	25.000	-
2758	25.000	-
2759	25.000	-
2760	25.000	-
2761	25.000	-
2763	25.000	-
2764	25.000	-
2765	25.000	-
2766	25.000	-
2767	25.000	-
2768	25.000	-
2770	25.000	-
2772	25.000	-
2774	25.000	-
2776	25.000	-
2777	25.000	-
2778	25.000	-
2779	25.000	-
2780	25.000	-
2781	25.000	-
2782	25.000	-
2783	25.000	-
2784	25.000	-
2785	25.000	-
2786	25.000	-
2787	25.000	-
2788	25.000	-
2789	25.000	-
2790	25.000	-
2791	25.000	-
2792	25.000	-
2793	25.000	-
2794	25.000	-
2795	25.000	-
2796	25.000	-
2797	25.000	-
2798	25.000	-
2799	25.000	-
2800	25.000	-
2801	25.000	-
2802	25.000	-
2803	25.000	-
2804	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
2805	25.000	-
2806	25.000	-
2807	25.000	-
2808	25.000	-
2809	25.000	-
2810	25.000	-
2811	25.000	-
2812	25.000	-
2813	25.000	-
2814	25.000	-
2815	25.000	-
2816	25.000	-
2817	25.000	-
2818	25.000	-
2819	25.000	-
2820	25.000	-
2821	25.000	-
2822	25.000	-
2823	25.000	-
2824	25.000	-
2825	25.000	-
2826	25.000	-
2827	25.000	-
2828	25.000	-
2829	25.000	-
2830	25.000	-
2831	25.000	-
2832	25.000	-
2833	25.000	-
2834	25.000	-
2835	25.000	-
2836	25.000	-
2837	25.000	-
2838	25.000	-
2839	25.000	-
2840	25.000	-
2841	25.000	-
2842	25.000	-
2843	25.000	-
2844	25.000	-
2845	25.000	-
2846	25.000	-
2847	25.000	-
2848	25.000	-
2849	25.000	-
2851	25.000	-
2852	25.000	-
2854	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
2856	25.000	-
2858	25.000	-
2860	25.000	-
2861	25.000	-
2862	25.000	-
2863	25.000	-
2864	25.000	-
2873	25.000	-
2874	25.000	-
2877	25.000	-
2878	25.000	-
2881	25.000	-
2882	25.000	-
2883	25.000	-
2884	25.000	-
2885	25.000	-
2886	25.000	-
2887	25.000	-
2888	25.000	-
2889	25.000	-
2890	25.000	-
2891	25.000	-
2892	25.000	-
2893	25.000	-
2894	25.000	-
2895	25.000	-
2896	25.000	-
2897	25.000	-
2898	25.000	-
2899	25.000	-
2900	25.000	-
2901	25.000	-
2902	25.000	-
2903	25.000	-
2904	25.000	-
2905	25.000	-
2906	25.000	-
2907	25.000	-
2909	25.000	-
2910	25.000	-
2911	25.000	-
2912	25.000	-
2913	25.000	-
2914	25.000	-
2915	25.000	-
2916	25.000	-
2917	25.000	-
2918	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
2919	25.000	-
2920	25.000	-
2921	25.000	-
2922	25.000	-
2923	25.000	-
2924	25.000	-
2925	25.000	-
2926	25.000	-
2927	25.000	-
2928	25.000	-
2929	25.000	-
2930	25.000	-
2931	25.000	-
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2933	25.000	-
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2935	25.000	-
2936	25.000	-
2937	25.000	-
2939	25.000	-
2940	25.000	-
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2954	25.000	-
2955	25.000	-
2956	25.000	-
2957	25.000	-
2958	25.000	-
2959	25.000	-
2960	25.000	-
2961	25.000	-
2962	25.000	-
2963	25.000	-
2964	25.000	-
2965	25.000	-
2966	25.000	-
2967	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
2968	25.000	-
2969	25.000	-
2970	25.000	-
2972	25.000	-
2973	25.000	-
2974	25.000	-
2975	25.000	-
2976	25.000	-
2978	25.000	-
2980	25.000	-
2981	25.000	-
2982	25.000	-
2983	25.000	-
2984	25.000	-
2985	25.000	-
2986	25.000	-
2987	25.000	-
2988	25.000	-
2991	25.000	-
2992	25.000	-
2993	25.000	-
2994	25.000	-
2995	25.000	-
2996	25.000	-
2997	25.000	-
2999	25.000	-
3000	25.000	-
3001	25.000	-
3002	25.000	-
3003	25.000	-
3004	25.000	-
3005	25.000	-
3006	25.000	-
3007	25.000	-
3009	25.000	-
3010	25.000	-
3011	25.000	-
3012	25.000	-
3013	25.000	-
3014	25.000	-
3015	25.000	-
3016	25.000	-
3017	25.000	-
3018	25.000	-
3019	25.000	-
3020	25.000	-
3021	25.000	-
3022	25.000	-

15 TEMPERATURE LOAD (TL) : Temperature Loads Cont...

Beam	Temperature Dif. Axial (°C)	Cross (°C)
3023	25.000	-
3024	25.000	-
3025	25.000	-
3026	25.000	-
3027	25.000	-
3028	25.000	-
3029	25.000	-
3030	25.000	-
3032	25.000	-
3034	25.000	-
3035	25.000	-
3036	25.000	-
3037	25.000	-
3038	25.000	-
3039	25.000	-
3040	25.000	-
3041	25.000	-
3042	25.000	-
3043	25.000	-
3044	25.000	-
3045	25.000	-
3046	25.000	-
3047	25.000	-
3048	25.000	-
3049	25.000	-
3050	25.000	-
3053	25.000	-
3054	25.000	-
3055	25.000	-
3056	25.000	-
3057	25.000	-
3058	25.000	-

16 PIPE LATERAL LOAD (PL) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
181	4.903	-	4.903	-	-	-
183	4.903	-	4.903	-	-	-
184	4.903	-	4.903	-	-	-
195	5.884	-	5.884	-	-	-
199	5.884	-	5.884	-	-	-
203	5.884	-	5.884	-	-	-
207	5.884	-	5.884	-	-	-
211	5.884	-	5.884	-	-	-

17 EQP TEST LOAD (EQTL) : Node Loads

Node	FX (kN)	FY (kN)	FZ (kN)	MX (kN-m)	MY (kN-m)	MZ (kN-m)
102	-	-117.680	-	-	-	-
105	-	-117.680	-	-	-	-
108	-	-117.680	-	-	-	-
111	-	-117.680	-	-	-	-
148	-	-117.680	-	-	-	-
151	-	-117.680	-	-	-	-
154	-	-117.680	-	-	-	-
157	-	-117.680	-	-	-	-
212	-	-58.840	-	-	-	-
213	-	-58.840	-	-	-	-
214	-	-58.840	-	-	-	-
215	-	-58.840	-	-	-	-
236	-	-58.840	-	-	-	-
237	-	-58.840	-	-	-	-
238	-	-58.840	-	-	-	-
239	-	-58.840	-	-	-	-

ATTACHMENT - 4

STAAD Output

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     **************************************************
     *                                                *
     *          STAAD.Pro CONNECT Edition             *
     *          Version  22.00.00.15                  *
     *          Proprietary Program of                *
     *          Bentley Systems, Inc.                 *
     *          Date=    FEB 29, 2020                 *
     *          Time=    18:10:28                     *
     *                                                *
     *  Licensed to: Larsen & Toubro Ltd.             *
     **************************************************


    1. STAAD SPACE TRIMBLE SOLUTIONS CORPORATION
INPUT FILE: D:\MAADEN-GOLD-O19045\3230 GRAVITY SEPERATION\CIVIL\LATEST\Gravity Seperation_R0.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 06-FEB-20
    4. END JOB INFORMATION
    5. UNIT METER MTON
    6. JOINT COORDINATES
    7. 1 432.175 913.65 -371.65; 2 432.175 913.65 -366.751
    8. 3 435.375 913.65 -375.4; 4 443.375 913.65 -375.4
    9. 5 451.375 913.65 -375.4; 6 435.375 913.65 -371.65
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   13. 13 435.375 916.475 -371.65; 14 432.175 916.475 -369.186
   14. 15 432.175 916.475 -366.751; 16 435.375 917.9 -375.4
   15. 17 439.375 917.9 -375.4; 18 443.375 917.9 -375.4
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   17. 21 439.375 917.9 -371.65; 22 443.375 917.9 -371.65
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   19. 25 439.375 917.9 -366.751; 26 443.375 917.9 -366.751
   20. 27 451.375 917.9 -366.751; 29 432.175 919.399 -371.65
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   23. 34 443.375 920.653 -373.05; 35 443.375 920.653 -371.65
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   25. 38 435.375 922.15 -376.55; 39 436.375 922.15 -376.55
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   28. 44 442.175 922.15 -376.55; 45 443.375 922.15 -376.55
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   38. 64 446.975 922.15 -375.4; 65 448.175 922.15 -375.4
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357. 3016 114 21; 3017 122 21; 3018 162 25; 3019 25 170; 3020 386 227
358. 3021 386 203; 3022 32 45; 3023 387 300; 3024 283 387; 3025 387 299
359. 3026 33 61; 3027 388 179; 3028 113 390; 3029 178 390; 3030 390 29
360. 3032 390 388; 3034 390 30; 3035 391 276; 3036 392 279; 3037 393 292
361. 3038 391 393; 3039 394 316; 3040 393 394; 3041 395 332; 3042 394 395
362. 3043 396 295; 3044 392 396; 3045 397 318; 3046 396 397; 3047 398 335
363. 3048 397 398; 3049 399 370; 3050 400 364; 3053 403 372; 3054 404 367
364. 3055 400 363; 3056 399 369; 3057 404 368; 3058 403 373; 3059 405 1
365. 3060 406 2; 3061 407 3; 3062 408 4; 3063 409 5; 3064 410 6; 3065 411 7
366. 3066 412 8; 3067 413 9; 3068 414 10; 3069 415 11
367. *
368. *
369. START GROUP DEFINITION
370. FLOOR
371. _922.15_1 1203 1287 1293 1299 1305 1311 1317 1323 1329 1335 1341 1347 -
372. 1353 1359 1366 1372 1378 1384 1390 1396 1402 1408 1414 1420 1426 1432 -
373. 1512 1517 1522 1527 1532 1537 1542 1547 1552 1557 1562 1567 1572 1579 -
374. 1585 1591 1724 1729 1734 1739 1744 1749 1754 1759 1764 1769 1774 1779 -
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375. 1785 1791 1797 1809 1821 1845 2026 2029 2032 2035 2038 2041 2044 2047 -
376. 2050 2053 2056 2071 2074 2077 2083 2089 2092 2095 2116 2200 2203 2206 -
377. 2209 2212 2215 2218 2221 2224 2227 2230 2233 2248 2251 2254 2257 2260 -
378. 2263 2266 2269 2272 2275 2278 2281 2284 2287 2290 2293 2296 2299 2302 -
379. 2305 2308 2311 2314 2317 2320 2323 2332 2335 2413 2416 2419 2422 2425 -
380. 2428 2431 2434 2443 2461 2464 2467 2476 2488 2491 2500 2506 2509 2518 -
381. 2524 2527 2530 2533 2536 2539 2542 2545 2566 2774 2776 TO 2778 2856 -
382. 2858 2873 2874 2877 2878 2881 TO 2907 2909 2926 2928 2930 2932 2934 -
383. 2936 2939 2941 2943 2945 2947 2949 2951 2953 2972 2978 2996 2997 3001
384. _929.90_1 74 79 199 206 212 218 224 230 236 242 313 328 1209 1215 1221 -
385. 1227 1233 1239 1245 1251 1257 1263 1269 1275 1437 1442 1447 1452 1457 -
386. 1462 1467 1472 1477 1482 1489 1495 1501 1507 1597 1664 1669 1674 1679 -
387. 1684 1689 1694 1699 1704 1709 1714 1719 1815 1827 1833 1839 1851 1936 -
388. 1939 1942 1945 1948 1954 1957 1999 2002 2005 2008 2011 2014 2017 2020 -
389. 2023 2059 2086 2101 2104 2107 2110 2113 2119 2179 2182 2185 2188 2236 -
390. 2326 2329 2338 2341 2404 2407 2437 2440 2446 2455 2458 2470 2473 2479 -
391. 2482 2485 2494 2497 2503 2512 2515 2521 2548 2551 2554 2557 2560 2563 -
392. 2629 TO 2636 2645 TO 2656 2661 TO 2663 2665 2666 2668 TO 2670 2672 -
393. 2673 TO 2674 2676 TO 2678 2680 TO 2683 2685 2687 2703 2706 -
394. 2708 TO 2710 2716 2726 2729 TO 2732 2737 TO 2747 2749 2750 2752 2754 -
395. 2756 TO 2761 2763 TO 2767 2779 2781 2783 2785 2787 TO 2789 2791 2793 -
396. 2794 TO 2795 2797 2799 2801 2803 2980 2982 2983 3006 3007 3011 3023 -
397. 3035 TO 3048
398. _932.843_1 2711 2715 2717 2720 TO 2725 2733 TO 2736 2822 2984 2985
399. JOINT
400. _F1A 410 TO 412
401. _F1B 408
402. _F2 407 409 413 TO 415
403. _F3 405 406
404. MEMBER
405. _C1 3061 TO 3069
406. _C2 3059 3060
407. END GROUP DEFINITION
408. START USER TABLE
409. TABLE 1
410. UNIT MMS KN
411. GENERAL
412. L50X50X6SA
413. 1138 108 4.981 50 6 408000 645481 13536 11953 11953 569 569 -
414. 21053 21053 2.49175E+06 50
416. L65X65X6SA
417. 1506 138 5.261 65 6 928000 1.3129E+06 17856 19028 19028 753 -
418. 753 33132 33132 5.71987E+06 65
420. L75X75X8SA
421. 2300 158 6.962 75 8 1.866E+06 2.65021E+06 48469 33547 33547 -
422. 1150 1150 58190 58190 2.03612E+07 75
424. L90X90X8SA
425. 2780 188 7.128 90 8 3.32E+06 4.41798E+06 58709 47000 47000 -
426. 1390 1390 80620 80620 3.61845E+07 90
428. L90X90X10SA
429. 3420 188 8.617 90 10 4.02E+06 5.5771E+06 113333 59331 59331 -
430. 1710 1710 101916 101916 6.82361E+07 90
432. L100X100X8SA
433. 3100 208 7.212 100 8 4.6E+06 5.95648E+06 65536 57274 57274 -
434. 1550 1550 97340 97340 5.03317E+07 100
436. L100X100X10SA
```

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437. 3840 208 8.846 100 10 5.6E+06 7.52147E+06 126667 72322 72322 -
438. 1920 1920 123648 123648 9.52639E+07 100
440. L130X130X10SA
441. 5040 268 9.104 130 10 1.276E+07 1.58837E+07 166667 118535 118535 -
442. 2520 2520 199080 199080 2.17014E+08 130
444. L130X130X13SA
445. 6460 268 11.493 130 13 1.61E+07 2.08736E+07 361773 155773 155773 -
446. 3230 3230 263568 263568 4.59821E+08 130
448. L150X150X13SA
449. 7520 308 11.753 150 13 2.52E+07 3.15454E+07 420359 204841 204841 -
450. 3760 3760 343664 343664 7.21346E+08 150
452. L150X150X16SA
453. 9140 308 14.091 150 16 3.02E+07 3.91044E+07 775509 253925 253925 -
454. 4570 4570 428666 428666 1.30311E+09 150
456. L200X200X16SA
457. 12360 408 14.608 200 16 7.48E+07 9.19174E+07 1.04858E+06 450575 -
458. 450575 6180 6180 731712 731712 3.22123E+09 200
460. L200X200X20SA
461. 15260 408 17.794 200 20 9.06E+07 1.13411E+08 2.02667E+06 555935 -
462. 555935 7630 7630 927808 927808 6.09689E+09 200
464. IPE240T
465. 5865 240 9.3 120 14.7 4.032E+07 1.7146E+07 139207 336000 186267 -
466. 2542.48 3035.24 402040 259357 3.73912E+10 110.2
468. TABLE 2
469. UNIT MMS KN
470. WIDE FLANGE
471. W700X10X20X250
472. 16600 700 10 250 20 1.39591E+09 5.21383E+07 1.55333E+06 7000 6666.67
473. W700X10X20X200
474. 14600 700 10 200 20 1.16465E+09 2.67217E+07 1.28667E+06 7000 5333.33
475. W1100X16X32X350
476. 38976 1100 16 350 32 7.87199E+09 2.2902E+08 9.06035E+06 17600 14933.3
477. W900X16X20X250
478. 23760 900 16 250 20 2.78441E+09 5.23769E+07 2.50752E+06 14400 6666.67
479. W600X10X25X250
480. 18000 600 10 250 25 1.1725E+09 6.515E+07 2.7875E+06 6000 8333.33
481. W900X12X20X250
482. 20320 900 12 250 20 2.57239E+09 5.22072E+07 1.82869E+06 10800 6666.67
483. W900X12X25X250
484. 22700 900 12 250 25 3.00735E+09 6.52266E+07 3.09377E+06 10800 8333.33
485. W800X10X250X20
486. 17600 800 10 250 20 1.88715E+09 5.21467E+07 1.58667E+06 8000 -
487. 6666.67 250 20
488. END
489. UNIT METER MTON
490. MEMBER PROPERTY EUROPEAN
491. *
492. 249 255 261 267 272 277 282 288 293 298 303 318 323 333 343 348 352 -
493. 358 363 418 422 2748 2753 2780 2782 2784 2786 2790 2792 2796 2798 -
494. 2800 2802 2804 TO 2807 2814 TO 2817 2823 2824 2831 TO 2839 -
495. 2910 TO 2925 2927 2929 2931 2933 2935 2937 2940 2942 2944 2946 2948 -
496. 2950 2952 2954 TO 2970 2973 2974 2981 2986 TO 2988 2991 TO 2995 2999 -
497. 3000 3002 TO 3005 3009 3010 3012 3013 3024 3025 3055 TO 3057 -
498. 3058 TABLE ST L65X65X6
499. 2757 2758 2808 TO 2813 2860 TO 2863 3021 3022 3026 TABLE ST L90X90X8
500. 2849 TABLE ST L150X150X13
```



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501. MEMBER PROPERTY EUROPEAN
502. 2818 TO 2821 UPTABLE 1 L50X50X6SA
503. 192 921 961 991 1034 1065 1101 1142 1149 1161 1173 1185 1190 1647 1653 -
504. 2852 2854 3028 TO 3030 3032 3034 UPTABLE 1 L65X65X6SA
505. 898 914 968 984 1041 1052 1113 1125 UPTABLE 1 L90X90X8SA
506. 579 584 671 676 718 723 815 820 862 867 UPTABLE 1 L100X100X8SA
507. 567 591 603 615 622 636 648 660 707 730 742 754 766 780 792 804 851 -
508. 874 886 933 944 1015 1089 2600 2864 3014 TO 3018 -
509. 3019 UPTABLE 1 L130X130X10SA
510. MEMBER PROPERTY EUROPEAN
511. 313 328 1293 1299 1305 1311 1317 1323 1329 1335 1341 1347 1353 1359 -
512. 1512 1527 1542 1547 1557 1664 1669 1674 1684 1694 1699 1709 1719 1724 -
513. 1729 1734 1739 1744 1749 1754 1759 1764 1769 1774 1779 2788 2794 3035 -
514. 3036 3038 3040 3044 3046 TABLE ST UPN100
515. 2704 2705 2711 2714 2715 2717 2718 2720 2721 2723 2725 2727 2728 2733 -
516. 2734 TO 2736 2822 2825 TO 2830 3042 3048 TO 3050 3053 -
517. 3054 TABLE ST UPN120
518. 2722 2724 2984 2985 TABLE ST UPN240
519. 74 79 199 218 1197 1203 1215 1227 1263 1287 1366 1402 1437 1477 1489 -
520. 1495 1501 1507 1659 1679 1689 1704 1714 1827 2020 2023 2068 2194 2197 -
521. 2242 2245 2320 2323 2410 2629 2630 2661 2662 2708 2710 2730 2732 2737 -
522. 2738 TO 2739 2741 2746 2750 2760 2761 2763 TO 2767 2777 2778 2783 2795 -
523. 2801 2803 2856 2858 2936 2939 2951 2953 2972 2975 2976 2980 2982 2983 -
524. 2996 2997 3001 3006 3007 3011 3023 3027 TABLE ST IPE200
525. 1384 1420 1517 1522 1532 1537 1552 1562 1567 1572 1585 1610 1616 1622 -
526. 1628 1634 1641 1797 1809 1821 1845 2032 2035 2116 2215 2218 2221 2224 -
527. 2227 2230 2332 2335 2431 2443 2464 2476 2488 2500 2506 2518 2524 2530 -
528. 2533 2536 2539 2542 2545 2566 2873 2874 2877 2878 2881 2882 -
529. 2884 TO 2886 2888 TO 2891 2894 TO 2896 2898 TO 2900 2902 2905 TO 2907 -
530. 2909 2930 2945 TABLE ST IPE240
531. 1604 2643 2644 2690 2692 2844 TO 2848 TABLE ST IPE240
532. 224 230 236 1209 1239 1245 1251 1275 1378 1396 1408 1426 1442 1457 -
533. 1472 1482 1815 1851 1999 2008 2017 2086 2119 2338 2341 2446 2479 2503 -
534. 2521 2548 2551 2554 2557 2560 2563 2645 TO 2656 2666 2669 -
535. 2672 TO 2674 2677 2680 TO 2683 2685 2687 2703 2709 2716 2726 2731 -
536. 2740 2742 TO 2745 2779 2787 2789 2791 2799 2887 2892 2897 2904 2926 -
537. 2932 2941 2947 3039 3045 TABLE ST IPE300
538. 1372 1390 1414 1432 1591 2038 2041 2044 2047 2050 2053 2056 2077 2080 -
539. 2083 2089 2092 2095 2233 2434 2467 2491 2509 2527 2776 2883 2893 2901 -
540. 2903 2928 2934 2943 2949 TABLE ST IPE360
541. 1221 1233 1257 1269 1579 2026 2029 2200 2203 2206 2209 2212 2413 2416 -
542. 2419 2422 2425 2428 2461 2631 TO 2634 2641 2642 2663 2665 2668 2670 -
543. 2691 2693 TO 2700 2702 2749 2752 2754 2756 2781 2785 2793 -
544. 2797 TABLE ST IPE400
545. 1447 1452 1462 1467 1785 2002 2005 2011 2014 2071 2248 2251 2254 2257 -
546. 2260 2263 2266 2269 2272 2275 2278 2281 2635 2636 2676 2678 -
547. 2774 TABLE ST IPE450
548. 212 1791 1833 1939 1942 1945 1948 2074 2101 2104 2107 2110 2182 2185 -
549. 2284 2287 2290 2293 2296 2299 2302 2305 2308 2311 2314 2317 2326 2759 -
550. 2978 3041 3047 TABLE ST IPE500
551. MEMBER PROPERTY EUROPEAN
552. 206 242 1597 1839 1936 1954 1957 2059 2113 2179 2188 2236 2329 2404 -
553. 2407 2437 2440 2455 2458 2470 2473 2482 2485 2494 2497 2512 2515 2706 -
554. 2729 2747 3037 3043 UPTABLE 2 W800X10X250X20
555. 2170 2353 2356 2383 2386 2398 2401 2449 2452 2768 TABLE ST HE200A
556. 1857 1863 1869 1875 1881 1892 1898 1904 2125 2131 2134 2137 2140 2158 -
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557. 2167 2176 2344 2347 2365 2374 2377 2389 2392 2395 2607 2637 TO 2640 -
558. 2684 2686 2688 2689 2770 2772 2840 TO 2843 2851 TABLE ST HE260A
559. 2371 3020 TABLE TB HE260A WP 0.2 TH 0.012 BW 0.2 BT 0.012
560. MEMBER PROPERTY EUROPEAN
561. 3061 TO 3069 PRIS YD 0.7 ZD 0.7
562. 3059 3060 PRIS YD 0.5 ZD 0.5
563. *
564. DEFINE MATERIAL START
565. ISOTROPIC STEEL
566. E 2E+07
567. POISSON 0.3
568. DENSITY 7.75
569. ALPHA 1.2E-05
570. DAMP 0.03
571. ISOTROPIC CONCRETE
572. E 2.21467E+06
573. POISSON 0.17
574. DENSITY 2.40261
575. ALPHA 1E-05
576. DAMP 0.05
577. TYPE CONCRETE
578. STRENGTH FCU 2812.28
579. END DEFINE MATERIAL
580. CONSTANTS
581. BETA 90 MEMB 1857 1869 1875 1892 1904 2125 2137 2140 2158 2167 2176 -
582. 2344 2347 2365 2371 2374 2383 2386 2389 2392 2395 2398 2401 2452 2607 -
583. 2840 2842 2851 3020
584. *
585. MATERIAL STEEL MEMB 74 79 192 199 206 212 218 224 230 236 242 249 255 -
586. 261 267 272 277 282 288 293 298 303 313 318 323 328 333 343 348 352 -
587. 358 363 418 422 567 579 584 591 603 615 622 636 648 660 671 676 707 -
588. 718 723 730 742 754 766 780 792 804 815 820 851 862 867 874 886 898 -
589. 914 921 933 944 961 968 984 991 1015 1034 1041 1052 1065 1089 1101 -
590. 1113 1125 1142 1149 1161 1173 1185 1190 1197 1203 1209 1215 1221 1227 -
591. 1233 1239 1245 1251 1257 1263 1269 1275 1287 1293 1299 1305 1311 1317 -
592. 1323 1329 1335 1341 1347 1353 1359 1366 1372 1378 1384 1390 1396 1402 -
593. 1408 1414 1420 1426 1432 1437 1442 1447 1452 1457 1462 1467 1472 1477 -
594. 1482 1489 1495 1501 1507 1512 1517 1522 1527 1532 1537 1542 1547 1552 -
595. 1557 1562 1567 1572 1579 1585 1591 1597 1604 1610 1616 1622 1628 1634 -
596. 1641 1647 1653 1659 1664 1669 1674 1679 1684 1689 1694 1699 1704 1709 -
597. 1714 1719 1724 1729 1734 1739 1744 1749 1754 1759 1764 1769 1774 1779 -
598. 1785 1791 1797 1809 1815 1821 1827 1833 1839 1845 1851 1857 1863 1869 -
599. 1875 1881 1892 1898 1904 1936 1939 1942 1945 1948 1954 1957 1999 2002 -
600. 2005 2008 2011 2014 2017 2020 2023 2026 2029 2032 2035 2038 2041 2044 -
601. 2047 2050 2053 2056 2059 2068 2071 2074 2077 2080 2083 2086 2089 2092 -
602. 2095 2101 2104 2107 2110 2113 2116 2119 2125 2131 2134 2137 2140 2158 -
603. 2167 2170 2176 2179 2182 2185 2188 2194 2197 2200 2203 2206 2209 2212 -
604. 2215 2218 2221 2224 2227
605. MATERIAL STEEL MEMB 2230 2233 2236 2242 2245 2248 2251 2254 2257 2260 -
606. 2263 2266 2269 2272 2275 2278 2281 2284 2287 2290 2293 2296 2299 2302 -
607. 2305 2308 2311 2314 2317 2320 2323 2326 2329 2332 2335 2338 2341 2344 -
608. 2347 2353 2356 2365 2371 2374 2377 2383 2386 2389 2392 2395 2398 2401 -
609. 2404 2407 2410 2413 2416 2419 2422 2425 2428 2431 2434 2437 2440 2443 -
610. 2446 2449 2452 2455 2458 2461 2464 2467 2470 2473 2476 2479 2482 2485 -
611. 2488 2491 2494 2497 2500 2503 2506 2509 2512 2515 2518 2521 2524 2527 -
612. 2530 2533 2536 2539 2542 2545 2548 2551 2554 2557 2560 2563 2566 2600 -
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613. 2607 2629 TO 2656 2661 TO 2663 2665 2666 2668 TO 2670 2672 TO 2674 -
614. 2676 TO 2678 2680 TO 2700 2702 TO 2706 2708 TO 2711 2714 TO 2718 2720 -
615. 2721 TO 2750 2752 TO 2754 2756 TO 2761 2763 TO 2768 2770 2772 2774 -
616. 2776 TO 2849 2851 2852 2854 2856 2858 2860 TO 2864 2873 2874 2877 -
617. 2878 2881 TO 2907 2909 TO 2937 2939 TO 2970 2972 TO 2976 2978 2980 -
618. 2981 TO 2988 2991 TO 2997 2999 TO 3007 3009 TO 3030 3032 3034 TO 3050 -
619. 3053 TO 3058
620. MATERIAL CONCRETE MEMB 3059 TO 3069
621. MEMBER RELEASE
622. 74 79 236 249 255 261 267 272 277 282 358 363 418 422 567 584 591 603 -
623. 622 636 648 660 671 676 730 742 766 780 792 804 815 820 851 867 874 -
624. 886 898 914 921 933 944 961 968 984 991 1015 1034 1041 1052 1065 1089 -
625. 1101 1113 1125 1203 1209 1215 1221 1227 1233 1239 1245 1251 1257 1263 -
626. 1269 1275 1287 1293 1299 1305 1311 1317 1323 1329 1335 1341 1347 1353 -
627. 1359 1366 1372 1378 1384 1390 1396 1402 1408 1414 1420 1426 1432 1437 -
628. 1442 1447 1452 1457 1462 1467 1472 1477 1482 1501 1507 1512 1517 1522 -
629. 1527 1532 1537 1542 1547 1552 1557 1562 1567 1572 1641 1647 1653 1664 -
630. 1669 1674 1679 1684 1689 1694 1699 1704 1709 1714 1719 1724 1729 1734 -
631. 1739 1744 1749 1754 1759 1764 1769 1774 1779 2020 2023 2083 2095 2182 -
632. 2320 2323 2600 2637 TO 2640 2646 2648 2650 2652 2669 2673 2677 2681 -
633. 2684 2686 2688 2689 2704 2705 2708 2714 2715 2718 2727 2728 2734 2757 -
634. 2758 2765 2766 2780 2782 2784 2786 2788 2790 2792 2794 2796 2798 2800 -
635. 2802 2804 TO 2821 2823 2824 2831 TO 2839 2844 2846 2848 2849 2852 -
636. 2860 TO 2864 2882 2884 2891 2894 2898 2907 2909 TO 2925 2927 2929 -
637. 2931 2962 2981 2986 TO 2988 2991 TO 2995 2999 3000 3002 -
638. 3003 START MY MZ
639. 192 199 218 288 293 298 303 313 318 323 328 333 343 348 352 707 718 -
640. 723 1142 1149 1161 1173 1185 1604 1610 1616 1622 1628 1634 1809 1815 -
641. 1827 2041 2200 2248 2269 2284 2305 2332 2338 2482 2485 2512 2515 2527 -
642. 2566 2653 2711 2717 2720 2722 TO 2724 2733 2748 2753 2759 2896 2933 -
643. 2935 2937 2940 2942 2944 2946 2948 2950 2952 2954 TO 2961 -
644. 2963 TO 2970 2973 2974 3004 3005 3009 3010 3012 TO 3019 3021 3022 -
645. 3024 TO 3026 3028 TO 3030 3034 3038 3040 3042 3044 3046 3048 3055 -
646. 3056 TO 3058 START MY MZ
647. 206 242 249 255 261 267 272 277 282 288 293 298 303 313 318 323 328 -
648. 333 343 348 352 358 363 418 422 567 579 615 622 636 648 707 754 766 -
649. 780 792 851 961 1101 1142 1149 1161 1173 1495 1579 1585 1591 1597 -
650. 1610 1616 1622 1628 1634 1664 1669 1674 1684 1694 1699 1709 1785 1791 -
651. 1809 1833 1839 1845 1851 1999 2038 2266 2302 2629 2630 2661 2662 2710 -
652. 2732 2734 2748 2753 2757 2758 2760 2763 2764 2779 TO 2787 -
653. 2789 TO 2793 2795 TO 2802 2804 TO 2824 2849 2927 2929 2931 2933 2935 -
654. 2937 2940 2942 2944 2946 2948 2950 2952 2954 2981 3011 3014 TO 3019 -
655. 3023 3035 3036 END MY MZ
656. 192 591 603 730 742 874 886 968 984 1015 1034 1489 1604 1653 2086 2711 -
657. 2717 2720 2723 2733 2854 2858 2864 2874 2883 2887 2892 2893 2897 2901 -
658. 2903 2904 2910 2911 2930 2945 2955 TO 2968 2972 2987 2988 -
659. 2992 TO 2996 2999 3000 3002 3003 3026 3029 3030 3032 3034 3038 3040 -
660. 3042 3044 3046 3048 3055 TO 3058 END MY MZ
661. 660 804 862 898 914 921 933 944 991 1041 1052 1065 1089 1113 1125 1190 -
662. 1287 1293 1299 1305 1311 1317 1323 1329 1335 1341 1347 1353 1359 1501 -
663. 1507 1512 1527 1542 1547 1557 1641 1647 1719 1724 1729 1734 1739 1744 -
664. 1749 1754 1759 1764 1769 1774 1779 2077 2089 2600 2635 2636 2649 2651 -
665. 2654 2674 2676 2678 2680 2682 2683 2685 2687 2703 2730 2767 2777 2831 -
666. 2832 TO 2839 2845 2846 2848 2860 TO 2863 2873 2877 2878 2881 2885 2886 -
667. 2888 TO 2890 2895 2899 2900 2902 2905 2906 2912 TO 2925 2969 2970 -
668. 2973 2974 2982 TO 2986 2991 3001 3004 TO 3006 3009 3010 3012 3013 -
```

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669. 3021 3022 3024 3025 3039 END MY MZ
670. 212 END MZ
671. 3059 TO 3069 END MX MY MZ
672. SUPPORTS
673. *1 TO 11 PINNED
674. 405 TO 415 FIXED
675. ******************************************************************************
676. LOAD 1 DEAD LOAD (DL)
677. ******************************************************************************
678. SELFWEIGHT Y -1.05
679. *DL OF CHQD PLATE FLOOR
680. FLOOR LOAD
681. _922.15_1 FLOAD -0.035 GY
**NOTE** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

**NOTE** about FLOOR GROUP command
  All floor panels in a floor group should lie in a single plane.
  If not lying in a single plane break single FLOOR GROUP
  command into multiple FLOOR GROUP commands where each
  floor group lie in a single plane / height.

682. _929.90_1 FLOAD -0.035 GY
683. _932.843_1 FLOAD -0.035 GY
684. **LOAD ON STAIR
685. JOINT LOAD
686. 12 13 30 31 113 114 179 180 192 193 325 326 FY -0.26
687. **LOAD ON STAIR AT EL (+)929.9M
688. 374 TO 377 FY -0.145
689. ******************************************************************************
690. LOAD 2 LIVE LOAD (LL)
691. ******************************************************************************
692. **LL ON CHQD PLATE FLOOR
693. FLOOR LOAD
694. _922.15_1 FLOAD -0.5 GY
695. _929.90_1 FLOAD -0.5 GY
696. _932.843_1 FLOAD -0.5 GY
697. **LOAD ON STAIR
698. JOINT LOAD
699. 12 13 30 31 113 114 179 180 192 193 325 326 FY -0.84
700. **LOAD ON STAIR AT EL (+)929.9M
701. 374 TO 377 FY -0.313
702. ******************************************************************************
703. LOAD 3 OPERATING FLUID LOAD (FL1)
704. ******************************************************************************
705. ***LOAD FOR LAUNDER
706. **AT EL (+)922.15M
```



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707. MEMBER LOAD
708. 1579 1585 2026 2029 2032 2035 2200 2203 2206 2209 2212 2215 2218 2221 -
709. 2224 2227 2230 2413 2416 2419 2422 2425 2428 2431 2461 2464 2488 2506 -
710. 2524 2566 UNI GY -0.2
711. JOINT LOAD
712. *LOAD FOR PIPE
713. 195 199 203 207 211 FY -2
714. 181 183 184 FY -2 MX -1.7
715. ***AT EL(+)929.90
716. *P36.1
717. 353 354 FY -6
718. 353 354 FX 0.3
719. 353 354 FZ 0.3
720. *P36.2
721. 357 358 FY -5
722. 357 358 FX 0.3
723. 357 358 FZ 0.3
724. *P45.1
725. 351 352 FY -6
726. 351 352 FX 0.3
727. 351 352 FZ 0.3
728. *P45.2
729. 355 356 FY -5
730. 355 356 FX 0.3
731. 355 356 FZ 0.3
732. ******************************************************************************
733. LOAD 4 OPERATING FLUID LOAD (FL2)
734. ******************************************************************************
735. ***LOAD FOR LAUNDER
736. **AT EL (+)922.15M
737. MEMBER LOAD
738. 1579 1585 2026 2029 2032 2035 2200 2203 2206 2209 2212 2215 2218 2221 -
739. 2224 2227 2230 2413 2416 2419 2422 2425 2428 2431 2461 2464 2488 2506 -
740. 2524 2566 UNI GY -0.2
741. JOINT LOAD
742. *LOAD FOR PIPE
743. 195 199 203 207 211 FY -2
744. 181 183 184 FY -2 MX -1.7
745. ***AT EL(+)929.90
746. *P36.1
747. 353 354 FY -6
748. 353 354 FX -0.3
749. 353 354 FZ -0.3
750. *P36.2
751. 357 358 FY -5
752. 357 358 FX -0.3
753. 357 358 FZ -0.3
754. *P45.1
755. 351 352 FY -6
756. 351 352 FX -0.3
757. 351 352 FZ -0.3
758. *P45.2
759. 355 356 FY -5
760. 355 356 FX -0.3
761. 355 356 FZ -0.3
762. ******************************************************************************
```

```
763. LOAD 5 EQP DL (EQDL)
764. ********************************************************************************
765. ***AT EL(+)922.15
766. JOINT LOAD
767. **P23
768. 102 105 148 151 FY -6
769. **P24
770. 108 111 154 157 FY -6
771. ***AT EL(+)929.90
772. **P35.1/2/3/4
773. 214 215 238 239 FY -1.2
774. *P36.1
775. 353 354 FY -1.5
776. *P36.2
777. 357 358 FY -1.5
778. *P44.1/2/3/4
779. 212 213 236 237 FY -1.2
780. *P45.1
781. 351 352 FY -1.5
782. *P45.2
783. 355 356 FY -1.5
784. ********************************************************************************
785. LOAD 6 EQP LL (EQLL)
786. ********************************************************************************
787. JOINT LOAD
788. ***AT EL(+)929.90
789. *P36.1
790. 353 354 FY -12.5
791. *P36.2
792. 357 358 FY -9.5
793. *P45.1
794. 351 352 FY -12.5
795. *P45.2
796. 355 356 FY -9.5
797. ********************************************************************************
798. LOAD 7 WIND LOAD (+X) WL(+X)
799. ********************************************************************************
800. ***AT EL(+)922.15
801. JOINT LOAD
802. **P23
803. 102 105 148 151 FX 1.2
804. **P24
805. 108 111 154 157 FX 1.2
806. ***AT EL(+)929.90
807. *P35.1/2/3/4
808. 214 215 238 239 FX 0.1
809. *P36.1
810. 353 354 FX 0.2
811. *P36.2
812. 357 358 FX 0.2
813. *P44.1/2/3/4
814. 212 213 236 237 FX 0.05
815. *P45.1
816. 351 352 FX 0.2
817. *P45.2
818. 355 356 FX 0.2
```



```
819. *****LOAD ON GRID 3220/3
820. MEMBER LOAD
821. 2158 2389 2607 UNI GX 0.06
822. 1863 2131 2134 UNI GX 0.11
823. 1857 2125 UNI GX 0.05
824. 2383 2386 2452 UNI GX 0.07
825. 2353 2356 2449 UNI GX 0.12
826. 2344 2347 UNI GX 0.05
827. *****LOAD BETWEEN GRID 3220/5'
828. 1892 2167 2851 UNI GX 0.07
829. 1881 2841 2843 UNI GX 0.13
830. 1875 2840 2842 UNI GX 0.05
831. 2392 2395 UNI GX 0.08
832. 2377 2772 UNI GX 0.14
833. 2371 2374 3020 UNI GX 0.06
834. 2714 2718 2827 2828 UNI GX 0.03
835. *****LOAD ON GRID 3220/6'
836. 1904 2176 UNI GX 0.06
837. 1898 2770 UNI GX 0.11
838. 1869 2140 UNI GX 0.05
839. 2398 2401 UNI GX 0.07
840. 2170 2768 UNI GX 0.12
841. 2137 2365 UNI GX 0.05
842. *****LOAD ON FRAME NEAR GRID 3220/4
843. 2637 TO 2640 2684 2686 2688 2689 UNI GX 0.13
844. *****LOAD ON FRAME NEAR GRID 3220/5
845. 2704 2705 2727 2728 2825 2826 2829 2830 3049 3050 3053 3054 UNI GX 0.02
846. ********************************************************************************
847. LOAD 8 WIND LOAD (-X) WL(-X)
848. ********************************************************************************
849. ***AT EL(+)922.15
850. JOINT LOAD
851. **P23
852. 102 105 148 151 FX -1.2
853. **P24
854. 108 111 154 157 FX -1.2
855. ***AT EL(+)929.90
856. **P35.1/2/3/4
857. 214 215 238 239 FX -0.1
858. **36.1
859. 353 354 FX -0.2
860. **36.2
861. 357 358 FX -0.2
862. **44.1/2/3/4
863. 212 213 236 237 FX -0.05
864. **45.1
865. 351 352 FX -0.2
866. **45.2
867. 355 356 FX -0.2
868. *****LOAD ON GRID 3220/3
869. MEMBER LOAD
870. 2158 2389 2607 UNI GX -0.06
871. 1863 2131 2134 UNI GX -0.11
872. 1857 2125 UNI GX -0.05
873. 2383 2386 2452 UNI GX -0.07
874. 2353 2356 2449 UNI GX -0.12
```

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```
875. 2344 2347 UNI GX -0.05
876. *****LOAD BETWEEN GRID 3220/5'
877. 1892 2167 2851 UNI GX -0.07
878. 1881 2841 2843 UNI GX -0.13
879. 1875 2840 2842 UNI GX -0.05
880. 2392 2395 UNI GX -0.08
881. 2377 2772 UNI GX -0.14
882. 2371 2374 3020 UNI GX -0.06
883. 2714 2718 2827 2828 UNI GX -0.03
884. *****LOAD ON GRID 3220/6'
885. 1904 2176 UNI GX -0.06
886. 1898 2770 UNI GX -0.11
887. 1869 2140 UNI GX -0.05
888. 2398 2401 UNI GX -0.07
889. 2170 2768 UNI GX -0.12
890. 2137 2365 UNI GX -0.05
891. *****LOAD ON FRAME NEAR GRID 3220/4
892. 2637 TO 2640 2684 2686 2688 2689 UNI GX -0.13
893. *****LOAD ON FRAME NEAR GRID 3220/5
894. 2704 2705 2727 2728 2825 2826 2829 2830 3049 3050 3053 3054 UNI GX -0.02
895. ******************************************************************************
896. LOAD 9 WIND LOAD (+Z) WL(+Z)
897. ******************************************************************************
898. ***AT EL(+)922.15
899. JOINT LOAD
900. **P23
901. 102 105 148 151 FZ 1.2
902. **P24
903. 108 111 154 157 FZ 1.2
904. ***AT EL(+)929.90
905. **P35.1/2/3/4
906. 214 215 238 239 FZ 0.1
907. **36.1
908. 353 354 FZ 0.1
909. **36.2
910. 357 358 FZ 0.1
911. **44.1/2/3/4
912. 212 213 236 237 FZ 0.05
913. **45.1
914. 351 352 FZ 0.1
915. **45.2
916. 355 356 FZ 0.1
917. **WIND LOAD ALONG GRID 3220/B''
918. MEMBER LOAD
919. 1857 2125 UNI GZ 0.12
920. 1875 2840 2842 UNI GZ 0.23
921. 1869 2140 UNI GZ 0.12
922. 2344 2347 UNI GZ 0.13
923. 2371 2374 3020 UNI GZ 0.25
924. 2137 2365 UNI GZ 0.13
925. **WIND LOAD ALONG GRID 3220/B'
926. 1197 2197 2976 UNI GZ 0.03
927. 1863 2131 2134 UNI GZ 0.13
928. 1881 2841 2843 UNI GZ 0.2
929. 1898 2770 UNI GZ 0.1
930. 2194 2410 UNI GZ 0.03
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931. 2353 2356 2449 UNI GZ 0.14
932. 2377 2772 UNI GZ 0.21
933. 2170 2768 UNI GZ 0.11
934. ****WIND LOAD ALONG GRID 3220/A'
935. 1659 2068 2245 3027 UNI GZ 0.03
936. 2158 2389 2607 UNI GZ 0.14
937. 1892 2167 2851 UNI GZ 0.21
938. 1904 2176 UNI GZ 0.11
939. 2242 2975 UNI GZ 0.03
940. 2383 2386 2452 UNI GZ 0.15
941. 2392 2395 UNI GZ 0.22
942. 2398 2401 UNI GZ 0.11
943. **GRID NEAR GRID 3220/C-1
944. 2637 TO 2640 2684 2686 2688 2689 UNI GZ 0.07
945. **GRID NEAR GRID 3220/C-2
946. 2704 2705 2825 2826 3049 3050 UNI GZ 0.04
947. 2714 2718 2827 2828 UNI GZ 0.08
948. 2727 2728 2829 2830 3053 3054 UNI GZ 0.04
949. ******************************************************************************
950. LOAD 10 WIND LOAD (-Z) WL(-Z)
951. ******************************************************************************
952. ***AT EL(+)922.15
953. JOINT LOAD
954. **P23
955. 102 105 148 151 FZ -1.2
956. **P24
957. 108 111 154 157 FZ -1.2
958. ***AT EL(+)929.90
959. **P35.1/2/3/4
960. 214 215 238 239 FZ -0.1
961. **36.1
962. 353 354 FZ -0.1
963. **36.2
964. 357 358 FZ -0.1
965. **44.1/2/3/4
966. 212 213 236 237 FZ -0.05
967. **45.1
968. 351 352 FZ -0.1
969. **45.2
970. 355 356 FZ -0.1
971. **WIND LOAD ALONG GRID 3220/B''
972. MEMBER LOAD
973. 1857 2125 UNI GZ -0.12
974. 1875 2840 2842 UNI GZ -0.23
975. 1869 2140 UNI GZ -0.12
976. 2344 2347 UNI GZ -0.13
977. 2371 2374 3020 UNI GZ -0.25
978. 2137 2365 UNI GZ -0.13
979. **WIND LOAD ALONG GRID 3220/B'
980. 1197 2197 2976 UNI GZ -0.03
981. 1863 2131 2134 UNI GZ -0.13
982. 1881 2841 2843 UNI GZ -0.2
983. 1898 2770 UNI GZ -0.1
984. 2194 2410 UNI GZ -0.03
985. 2353 2356 2449 UNI GZ -0.14
986. 2377 2772 UNI GZ -0.21
```

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 987. 2170 2768 UNI GZ -0.11
 988. ****WIND LOAD ALONG GRID 3220/A'
 989. 1659 2068 2245 3027 UNI GZ -0.03
 990. 2158 2389 2607 UNI GZ -0.14
 991. 1892 2167 2851 UNI GZ -0.21
 992. 1904 2176 UNI GZ -0.11
 993. 2242 2975 UNI GZ -0.03
 994. 2383 2386 2452 UNI GZ -0.15
 995. 2392 2395 UNI GZ -0.22
 996. 2398 2401 UNI GZ -0.11
 997. **GRID NEAR GRID 3220/C-1
 998. 2637 TO 2640 2684 2686 2688 2689 UNI GZ -0.07
 999. **GRID NEAR GRID 3220/C-2
1000. 2704 2705 2825 2826 3049 3050 UNI GZ -0.04
1001. 2714 2718 2827 2828 UNI GZ -0.08
1002. 2727 2728 2829 2830 3053 3054 UNI GZ -0.04
1003. ********************************************************************************
1004. LOAD 11 SEISMIC LOAD SL(+X)
1005. ********************************************************************************
1006. *SL ON STRUCTURE
1007. JOINT LOAD
1008. 1 405 FX 0.038
1009. 2 406 FX 0.023
1010. 3 407 FX 0.025
1011. 4 408 FX 0.063
1012. 5 409 FX 0.014
1013. 6 410 FX 0.048
1014. 7 411 FX 0.124
1015. 8 412 FX 0.012
1016. 9 413 FX 0.05
1017. 10 414 FX 0.015
1018. 11 415 FX 0.027
1019. 53 FX 0.303
1020. 61 FX 0.586
1021. 68 FX 0.289
1022. 113 FX 0.119
1023. 114 FX 0.484
1024. 122 FX 0.965
1025. 129 FX 0.407
1026. 162 FX 0.299
1027. 170 FX 0.378
1028. 177 FX 0.227
1029. 219 FX 0.434
1030. 227 FX 0.956
1031. 235 FX 0.431
1032. 286 FX 0.478
1033. 293 FX 0.896
1034. 300 FX 0.447
1035. 325 FX 0.066
1036. 326 FX 0.274
1037. 333 FX 0.464
1038. 340 FX 0.209
1039. 388 FX 0.07
1040. *SL ON EQUIPMENTS
1041. JOINT LOAD
1042. *P23
```



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1043. 102 105 148 151 FX 0.6
1044. 102 105 148 151 FZ 0.6
1045. *P24
1046. 108 111 154 157 FX 0.6
1047. 108 111 154 157 FZ 0.6
1048. *P35.1/2/3/4
1049. 214 215 238 239 FX 0.3
1050. 214 215 238 239 FZ 0.3
1051. *P36.1
1052. 353 354 FX 0.7
1053. 353 354 FZ 0.7
1054. *P36.2
1055. 357 358 FX 0.5
1056. 357 358 FZ 0.5
1057. *P44.1/2/3/4
1058. 212 213 236 237 FX 0.3
1059. 212 213 236 237 FZ 0.3
1060. *P45.1
1061. 351 352 FX 0.6
1062. 351 352 FZ 0.6
1063. *P45.2
1064. 355 356 FX 0.5
1065. 355 356 FZ 0.5
1066. ********************************************************************************
1067. LOAD 12 SEISMIC LOAD SL(-X)
1068. ********************************************************************************
1069. *SL ON STRUCTURE
1070. JOINT LOAD
1071. 1 405 FX -0.038
1072. 2 406 FX -0.023
1073. 3 407 FX -0.025
1074. 4 408 FX -0.063
1075. 5 409 FX -0.014
1076. 6 410 FX -0.048
1077. 7 411 FX -0.124
1078. 8 412 FX -0.012
1079. 9 413 FX -0.05
1080. 10 414 FX -0.015
1081. 11 415 FX -0.027
1082. 53 FX -0.303
1083. 61 FX -0.586
1084. 68 FX -0.289
1085. 113 FX -0.119
1086. 114 FX -0.484
1087. 122 FX -0.965
1088. 129 FX -0.407
1089. 162 FX -0.299
1090. 170 FX -0.378
1091. 177 FX -0.227
1092. 219 FX -0.434
1093. 227 FX -0.956
1094. 235 FX -0.431
1095. 286 FX -0.478
1096. 293 FX -0.896
1097. 300 FX -0.447
1098. 325 FX -0.066
```

```
1099. 326 FX -0.274
1100. 333 FX -0.464
1101. 340 FX -0.209
1102. 388 FX -0.07
1103. *SL ON EQUIPMENTS
1104. JOINT LOAD
1105. *P23
1106. 102 105 148 151 FX -0.6
1107. 102 105 148 151 FZ -0.6
1108. *P24
1109. 108 111 154 157 FX -0.6
1110. 108 111 154 157 FZ -0.6
1111. *P35.1/2/3/4
1112. 214 215 238 239 FX -0.3
1113. 214 215 238 239 FZ -0.3
1114. *P36.1
1115. 353 354 FX -0.7
1116. 353 354 FZ -0.7
1117. *P36.2
1118. 357 358 FX -0.5
1119. 357 358 FZ -0.5
1120. *P44.1/2/3/4
1121. 212 213 236 237 FX -0.3
1122. 212 213 236 237 FZ -0.3
1123. *P45.1
1124. 351 352 FX -0.6
1125. 351 352 FZ -0.6
1126. *P45.2
1127. 355 356 FX -0.5
1128. 355 356 FZ -0.5
1129. ********************************************************************************
1130. LOAD 13 SEISMIC LOAD SL(+Z)
1131. ********************************************************************************
1132. *SL ON STRUCTURE
1133. JOINT LOAD
1134. 1 405 FZ 0.038
1135. 2 406 FZ 0.023
1136. 3 407 FZ 0.025
1137. 4 408 FZ 0.063
1138. 5 409 FZ 0.014
1139. 6 410 FZ 0.048
1140. 7 411 FZ 0.124
1141. 8 412 FZ 0.012
1142. 9 413 FZ 0.05
1143. 10 414 FZ 0.015
1144. 11 415 FZ 0.027
1145. 53 FZ 0.303
1146. 61 FZ 0.586
1147. 68 FZ 0.289
1148. 113 FZ 0.119
1149. 114 FZ 0.484
1150. 122 FZ 0.965
1151. 129 FZ 0.407
1152. 162 FZ 0.299
1153. 170 FZ 0.378
1154. 177 FZ 0.227
```



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1155. 219 FZ 0.434
1156. 227 FZ 0.956
1157. 235 FZ 0.431
1158. 286 FZ 0.478
1159. 293 FZ 0.896
1160. 300 FZ 0.447
1161. 325 FZ 0.066
1162. 326 FZ 0.274
1163. 333 FZ 0.464
1164. 340 FZ 0.209
1165. 388 FZ 0.07
1166. *SL ON EQUIPMENTS
1167. JOINT LOAD
1168. *P23
1169. 102 105 148 151 FX 0.6
1170. 102 105 148 151 FZ 0.6
1171. *P24
1172. 108 111 154 157 FX 0.6
1173. 108 111 154 157 FZ 0.6
1174. *P35.1/2/3/4
1175. 214 215 238 239 FX 0.3
1176. 214 215 238 239 FZ 0.3
1177. *P36.1
1178. 353 354 FX 0.7
1179. 353 354 FZ 0.7
1180. *P36.2
1181. 357 358 FX 0.5
1182. 357 358 FZ 0.5
1183. *P44.1/2/3/4
1184. 212 213 236 237 FX 0.3
1185. 212 213 236 237 FZ 0.3
1186. *P45.1
1187. 351 352 FX 0.6
1188. 351 352 FZ 0.6
1189. *P45.2
1190. 355 356 FX 0.5
1191. 355 356 FZ 0.5
1192. ********************************************************************************
1193. LOAD 14 SEISMIC LOAD SL(-Z)
1194. ********************************************************************************
1195. *SL ON STRUCTURE
1196. JOINT LOAD
1197. 1 405 FZ -0.038
1198. 2 406 FZ -0.023
1199. 3 407 FZ -0.025
1200. 4 408 FZ -0.063
1201. 5 409 FZ -0.014
1202. 6 410 FZ -0.048
1203. 7 411 FZ -0.124
1204. 8 412 FZ -0.012
1205. 9 413 FZ -0.05
1206. 10 414 FZ -0.015
1207. 11 415 FZ -0.027
1208. 53 FZ -0.303
1209. 61 FZ -0.586
1210. 68 FZ -0.289
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1211. 113 FZ -0.119
1212. 114 FZ -0.484
1213. 122 FZ -0.965
1214. 129 FZ -0.407
1215. 162 FZ -0.299
1216. 170 FZ -0.378
1217. 177 FZ -0.227
1218. 219 FZ -0.434
1219. 227 FZ -0.956
1220. 235 FZ -0.431
1221. 286 FZ -0.478
1222. 293 FZ -0.896
1223. 300 FZ -0.447
1224. 325 FZ -0.066
1225. 326 FZ -0.274
1226. 333 FZ -0.464
1227. 340 FZ -0.209
1228. 388 FZ -0.07
1229. *SL ON EQUIPMENTS
1230. JOINT LOAD
1231. *P23
1232. 102 105 148 151 FX -0.6
1233. 102 105 148 151 FZ -0.6
1234. *P24
1235. 108 111 154 157 FX -0.6
1236. 108 111 154 157 FZ -0.6
1237. *P35.1/2/3/4
1238. 214 215 238 239 FX -0.3
1239. 214 215 238 239 FZ -0.3
1240. *P36.1
1241. 353 354 FX -0.7
1242. 353 354 FZ -0.7
1243. *P36.2
1244. 357 358 FX -0.5
1245. 357 358 FZ -0.5
1246. *P44.1/2/3/4
1247. 212 213 236 237 FX -0.3
1248. 212 213 236 237 FZ -0.3
1249. *P45.1
1250. 351 352 FX -0.6
1251. 351 352 FZ -0.6
1252. *P45.2
1253. 355 356 FX -0.5
1254. 355 356 FZ -0.5
1255. ********************************************************************************
1256. LOAD 15 TEMPERATURE LOAD (TL)
1257. ********************************************************************************
1258. TEMPERATURE LOAD
1259. 74 79 192 199 206 212 218 224 230 236 242 249 255 261 267 272 277 282 -
1260. 288 293 298 303 313 318 323 328 333 343 348 352 358 363 418 422 567 -
1261. 579 584 591 603 615 622 636 648 660 671 676 707 718 723 730 742 754 -
1262. 766 780 792 804 815 820 851 862 867 874 886 898 914 921 933 944 961 -
1263. 968 984 991 1015 1034 1041 1052 1065 1089 1101 1113 1125 1142 1149 -
1264. 1161 1173 1185 1190 1197 1203 1209 1215 1221 1227 1233 1239 1245 1251 -
1265. 1257 1263 1269 1275 1287 1293 1299 1305 1311 1317 1323 1329 1335 1341 -
1266. 1347 1353 1359 1366 1372 1378 1384 1390 1396 1402 1408 1414 1420 1426 -
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1267. 1432 1437 1442 1447 1452 1457 1462 1467 1472 1477 1482 1489 1495 1501 -
1268. 1507 1512 1517 1522 1527 1532 1537 1542 1547 1552 1557 1562 1567 1572 -
1269. 1579 1585 1591 1597 1604 1610 1616 1622 1628 1634 1641 1647 1653 1659 -
1270. 1664 1669 1674 1679 1684 1689 1694 1699 1704 1709 1714 1719 1724 1729 -
1271. 1734 1739 1744 1749 1754 1759 1764 1769 1774 1779 1785 1791 1797 1809 -
1272. 1815 1821 1827 1833 1839 1845 1851 1857 1863 1869 1875 1881 1892 1898 -
1273. 1904 1936 1939 1942 1945 1948 1954 1957 1999 2002 2005 2008 2011 2014 -
1274. 2017 2020 2023 2026 2029 2032 2035 2038 2041 2044 2047 2050 2053 2056 -
1275. 2059 2068 2071 2074 2077 2080 2083 2086 2089 2092 2095 2101 2104 2107 -
1276. 2110 2113 2116 2119 2125 2131 2134 2137 2140 2158 2167 2170 2176 2179 -
1277. 2182 2185 2188 2194 2197 2200 2203 2206 2209 2212 2215 2218 2221 2224 -
1278. 2227 TEMP 25
1279. 2230 2233 2236 2242 2245 2248 2251 2254 2257 2260 2263 2266 2269 2272 -
1280. 2275 2278 2281 2284 2287 2290 2293 2296 2299 2302 2305 2308 2311 2314 -
1281. 2317 2320 2323 2326 2329 2332 2335 2338 2341 2344 2347 2353 2356 2365 -
1282. 2371 2374 2377 2383 2386 2389 2392 2395 2398 2401 2404 2407 2410 2413 -
1283. 2416 2419 2422 2425 2428 2431 2434 2437 2440 2443 2446 2449 2452 2455 -
1284. 2458 2461 2464 2467 2470 2473 2476 2479 2482 2485 2488 2491 2494 2497 -
1285. 2500 2503 2506 2509 2512 2515 2518 2521 2524 2527 2530 2533 2536 2539 -
1286. 2542 2545 2548 2551 2554 2557 2560 2563 2566 2600 2607 2629 TO 2656 -
1287. 2661 TO 2663 2665 2666 2668 TO 2670 2672 TO 2674 2676 TO 2678 2680 -
1288. 2681 TO 2700 2702 TO 2706 2708 TO 2711 2714 TO 2718 2720 TO 2750 2752 -
1289. 2753 TO 2754 2756 TO 2761 2763 TO 2768 2770 2772 2774 2776 TO 2849 -
1290. 2851 2852 2854 2856 2858 2860 TO 2864 2873 2874 2877 2878 -
1291. 2881 TO 2907 2909 TO 2937 2939 TO 2970 2972 TO 2976 2978 2980 TO 2988 -
1292. 2991 TO 2997 2999 TO 3007 3009 TO 3030 3032 3034 TO 3050 3053 TO 3057 -
1293. 3058 TEMP 25
1294. ********************************************************************************
1295. LOAD 16 PIPE LATERAL LOAD (PL)
1296. ********************************************************************************
1297. JOINT LOAD
1298. *LOAD FOR PIPE
1299. 195 199 203 207 211 FX 0.6 FZ 0.6
1300. 181 183 184 FX 0.5 FZ 0.5
1301. ********************************************************************************
1302. LOAD 17 EQP TEST LOAD (EQTL)
1303. ********************************************************************************
1304. ***AT EL(+)922.15
1305. JOINT LOAD
1306. **P23
1307. 102 105 148 151 FY -12
1308. **P24
1309. 108 111 154 157 FY -12
1310. ***AT EL(+)929.90
1311. **P35.1/2/3/4
1312. 214 215 238 239 FY -6
1313. **44.1/2/3/4
1314. 212 213 236 237 FY -6
1315. ********************************************************************************
1316. ***LOAD COMBINATION FOR FOUNDATION DESIGM***
1317. ******************************************
1318. LOAD COMB 101 DL+EQDL+FL1
1319. 1 1.0 5 1.0 3 1.0
1320. LOAD COMB 102 DL+EQDL+FL2
1321. 1 1.0 5 1.0 4 1.0
1322. LOAD COMB 103 DL+EQDL+FL1+LL+EQLL+TL
```

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1323. 1 1.0 5 1.0 3 1.0 2 1.0 6 1.0 15 1.0
1324. LOAD COMB 104 DL+EQDL+FL2+LL+EQLL+TL
1325. 1 1.0 5 1.0 4 1.0 2 1.0 6 1.0 15 1.0
1326. LOAD COMB 105 DL+EQDL+FL1+0.75(LL+EQLL)+0.75TL
1327. 1 1.0 5 1.0 3 1.0 2 0.75 6 0.75 15 0.75
1328. LOAD COMB 106 DL+EQDL+FL2+0.75(LL+EQLL)+0.75TL
1329. 1 1.0 5 1.0 4 1.0 2 0.75 6 0.75 15 0.75
1330. LOAD COMB 107 DL+EQDL+FL1+PL
1331. 1 1.0 5 1.0 3 1.0 16 1.0
1332. LOAD COMB 108 DL+EQDL+FL2+PL
1333. 1 1.0 5 1.0 4 1.0 16 1.0
1334. LOAD COMB 109 DL+EQDL+FL1+PL+LL+EQLL+TL
1335. 1 1.0 5 1.0 3 1.0 16 1.0 2 1.0 6 1.0 15 1.0
1336. LOAD COMB 110 DL+EQDL+FL2+PL+LL+EQLL+TL
1337. 1 1.0 5 1.0 4 1.0 16 1.0 2 1.0 6 1.0 15 1.0
1338. LOAD COMB 111 DL+EQDL+FL1+PL+0.75(LL+EQLL)+0.75TL
1339. 1 1.0 5 1.0 3 1.0 16 1.0 2 0.75 6 0.75 15 0.75
1340. LOAD COMB 112 DL+EQDL+FL2+PL+0.75(LL+EQLL)+0.75TL
1341. 1 1.0 5 1.0 4 1.0 16 1.0 2 0.75 6 0.75 15 0.75
1342. LOAD COMB 113 DL+EQDL+FL1+WL(+X)
1343. 1 1.0 5 1.0 3 1.0 7 1.0
1344. LOAD COMB 114 DL+EQDL+FL1+WL(-X)
1345. 1 1.0 5 1.0 3 1.0 8 1.0
1346. LOAD COMB 115 DL+EQDL+FL1+WL(+Z)
1347. 1 1.0 5 1.0 3 1.0 9 1.0
1348. LOAD COMB 116 DL+EQDL+FL1+WL(-Z)
1349. 1 1.0 5 1.0 3 1.0 10 1.0
1350. LOAD COMB 117 DL+EQDL+FL2+WL(+X)
1351. 1 1.0 5 1.0 4 1.0 7 1.0
1352. LOAD COMB 118 DL+EQDL+FL2+WL(-X)
1353. 1 1.0 5 1.0 4 1.0 8 1.0
1354. LOAD COMB 119 DL+EQDL+FL2+WL(+Z)
1355. 1 1.0 5 1.0 4 1.0 9 1.0
1356. LOAD COMB 120 DL+EQDL+FL2+WL(-Z)
1357. 1 1.0 5 1.0 4 1.0 10 1.0
1358. LOAD COMB 121 DL+EQDL+FL1+0.7SL(+X)
1359. 1 1.0 5 1.0 3 1.0 11 0.7
1360. LOAD COMB 122 DL+EQDL+FL1+0.7SL(-X)
1361. 1 1.0 5 1.0 3 1.0 12 0.7
1362. LOAD COMB 123 DL+EQDL+FL1+0.7SL(+Z)
1363. 1 1.0 5 1.0 3 1.0 13 0.7
1364. LOAD COMB 124 DL+EQDL+FL1+0.7SL(-Z)
1365. 1 1.0 5 1.0 3 1.0 14 0.7
1366. LOAD COMB 125 DL+EQDL+FL2+0.7SL(+X)
1367. 1 1.0 5 1.0 4 1.0 11 0.7
1368. LOAD COMB 126 DL+EQDL+FL2+0.7SL(-X)
1369. 1 1.0 5 1.0 4 1.0 12 0.7
1370. LOAD COMB 127 DL+EQDL+FL2+0.7SL(+Z)
1371. 1 1.0 5 1.0 4 1.0 13 0.7
1372. LOAD COMB 128 DL+EQDL+FL2+0.7SL(-Z)
1373. 1 1.0 5 1.0 4 1.0 14 0.7
1374. LOAD COMB 129 DL+EQDL+FL1+0.75WL(+X)+0.75(LL+EQLL)
1375. 1 1.0 5 1.0 3 1.0 7 0.75 2 0.75 6 0.75
1376. LOAD COMB 130 DL+EQDL+FL1+0.75WL(-X)+0.75(LL+EQLL)
1377. 1 1.0 5 1.0 3 1.0 8 0.75 2 0.75 6 0.75
1378. LOAD COMB 131 DL+EQDL+FL1+0.75WL(+Z)+0.75(LL+EQLL)
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1379. 1 1.0 5 1.0 3 1.0 9 0.75 2 0.75 6 0.75
1380. LOAD COMB 132 DL+EQDL+FL1+0.75WL(-Z)+0.75(LL+EQLL)
1381. 1 1.0 5 1.0 3 1.0 10 0.75 2 0.75 6 0.75
1382. LOAD COMB 133 DL+EQDL+FL2+0.75WL(+X)+0.75(LL+EQLL)
1383. 1 1.0 5 1.0 4 1.0 7 0.75 2 0.75 6 0.75
1384. LOAD COMB 134 DL+EQDL+FL2+0.75WL(-X)+0.75(LL+EQLL)
1385. 1 1.0 5 1.0 4 1.0 8 0.75 2 0.75 6 0.75
1386. LOAD COMB 135 DL+EQDL+FL2+0.75WL(+Z)+0.75(LL+EQLL)
1387. 1 1.0 5 1.0 4 1.0 9 0.75 2 0.75 6 0.75
1388. LOAD COMB 136 DL+EQDL+FL2+0.75WL(-Z)+0.75(LL+EQLL)
1389. 1 1.0 5 1.0 4 1.0 10 0.75 2 0.75 6 0.75
1390. LOAD COMB 137 DL+EQDL+FL1+0.525SL(+X)+0.75(LL+EQLL)
1391. 1 1.0 5 1.0 3 1.0 11 0.525 2 0.75 6 0.75
1392. LOAD COMB 138 DL+EQDL+FL1+0.525SL(-X)+0.75(LL+EQLL)
1393. 1 1.0 5 1.0 3 1.0 12 0.525 2 0.75 6 0.75
1394. LOAD COMB 139 DL+EQDL+FL1+0.525SL(+Z)+0.75(LL+EQLL)
1395. 1 1.0 5 1.0 3 1.0 13 0.525 2 0.75 6 0.75
1396. LOAD COMB 140 DL+EQDL+FL1+0.525SL(-Z)+0.75(LL+EQLL)
1397. 1 1.0 5 1.0 3 1.0 14 0.525 2 0.75 6 0.75
1398. LOAD COMB 141 DL+EQDL+FL2+0.525SL(+X)+0.75(LL+EQLL)
1399. 1 1.0 5 1.0 4 1.0 11 0.525 2 0.75 6 0.75
1400. LOAD COMB 142 DL+EQDL+FL2+0.525SL(-X)+0.75(LL+EQLL)
1401. 1 1.0 5 1.0 4 1.0 12 0.525 2 0.75 6 0.75
1402. LOAD COMB 143 DL+EQDL+FL2+0.525SL(+Z)+0.75(LL+EQLL)
1403. 1 1.0 5 1.0 4 1.0 13 0.525 2 0.75 6 0.75
1404. LOAD COMB 144 DL+EQDL+FL2+0.525SL(-Z)+0.75(LL+EQLL)
1405. 1 1.0 5 1.0 4 1.0 14 0.525 2 0.75 6 0.75
1406. LOAD COMB 145 0.6(DL+EQDL)+WL(+X)
1407. 1 0.6 5 0.6 7 1.0
1408. LOAD COMB 146 0.6(DL+EQDL)+WL(-X)
1409. 1 0.6 5 0.6 8 1.0
1410. LOAD COMB 147 0.6(DL+EQDL)+WL(+Z)
1411. 1 0.6 5 0.6 9 1.0
1412. LOAD COMB 148 0.6(DL+EQDL)+WL(-Z)
1413. 1 0.6 5 0.6 10 1.0
1414. LOAD COMB 149 0.6(DL+EQDL)+0.7SL(+X)
1415. 1 0.6 5 0.6 11 0.7
1416. LOAD COMB 150 0.6(DL+EQDL)+0.7SL(-X)
1417. 1 0.6 5 0.6 12 0.7
1418. LOAD COMB 151 0.6(DL+EQDL)+0.7SL(+Z)
1419. 1 0.6 5 0.6 13 0.7
1420. LOAD COMB 152 0.6(DL+EQDL)+0.7SL(-Z)
1421. 1 0.6 5 0.6 14 0.7
1422. LOAD COMB 153 DL+EQTL
1423. 1 1.0 17 1.0
1424. ******************************************
1425. *LOAD COMBINATION FOR STRENGTH CALCULATION*
1426. ******************************************
1427. LOAD COMB 201 1.4(DL+EQDL+FL1)
1428. 1 1.4 5 1.4 3 1.4
1429. LOAD COMB 202 1.4(DL+EQDL+FL2)
1430. 1 1.4 5 1.4 4 1.4
1431. LOAD COMB 203 1.4(DL+EQDL+FL1)+1.7(LL+EQLL)
1432. 1 1.4 5 1.4 3 1.4 2 1.7 6 1.7
1433. LOAD COMB 204 1.4(DL+EQDL+FL2)+1.7(LL+EQLL)
1434. 1 1.4 5 1.4 4 1.4 2 1.7 6 1.7
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1435. LOAD COMB 205 1.4(DL+EQDL+FL1+TL)+1.7(LL+EQLL)
1436. 1 1.4 5 1.4 3 1.4 15 1.4 2 1.7 6 1.7
1437. LOAD COMB 206 1.4(DL+EQDL+FL2+TL)+1.7(LL+EQLL)
1438. 1 1.4 5 1.4 4 1.4 15 1.4 2 1.7 6 1.7
1439. LOAD COMB 207 1.4(DL+EQDL+FL1+PL)
1440. 1 1.4 5 1.4 3 1.4 16 1.4
1441. LOAD COMB 208 1.4(DL+EQDL+FL2+PL)
1442. 1 1.4 5 1.4 4 1.4 16 1.4
1443. LOAD COMB 209 1.4(DL+EQDL+FL1+PL)+1.7(LL+EQLL)
1444. 1 1.4 5 1.4 3 1.4 16 1.4 2 1.7 6 1.7
1445. LOAD COMB 210 1.4(DL+EQDL+FL2+PL)+1.7(LL+EQLL)
1446. 1 1.4 5 1.4 4 1.4 16 1.4 2 1.7 6 1.7
1447. LOAD COMB 211 1.4(DL+EQDL+FL1+PL+TL)+1.7(LL+EQLL)
1448. 1 1.4 5 1.4 3 1.4 16 1.4 15 1.4 2 1.7 6 1.7
1449. LOAD COMB 212 1.4(DL+EQDL+FL2+PL+TL)+1.7(LL+EQLL)
1450. 1 1.4 5 1.4 4 1.4 16 1.4 15 1.4 2 1.7 6 1.7
1451. LOAD COMB 213 1.2(DL+EQDL)+0.8WL(+X)
1452. 1 1.2 5 1.2 7 0.8
1453. LOAD COMB 214 1.2(DL+EQDL)+0.8WL(-X)
1454. 1 1.2 5 1.2 8 0.8
1455. LOAD COMB 215 1.2(DL+EQDL)+0.8WL(+Z)
1456. 1 1.2 5 1.2 9 0.8
1457. LOAD COMB 216 1.2(DL+EQDL)+0.8WL(-Z)
1458. 1 1.2 5 1.2 10 0.8
1459. LOAD COMB 217 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
1460. 1 1.2 5 1.2 7 1.6 2 0.5 6 0.5
1461. LOAD COMB 218 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
1462. 1 1.2 5 1.2 8 1.6 2 0.5 6 0.5
1463. LOAD COMB 219 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
1464. 1 1.2 5 1.2 9 1.6 2 0.5 6 0.5
1465. LOAD COMB 220 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
1466. 1 1.2 5 1.2 10 1.6 2 0.5 6 0.5
1467. LOAD COMB 221 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL1
1468. 1 1.2 5 1.2 11 1.0 2 0.5 6 0.5 3 0.5
1469. LOAD COMB 222 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL1
1470. 1 1.2 5 1.2 12 1.0 2 0.5 6 0.5 3 0.5
1471. LOAD COMB 223 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL1
1472. 1 1.2 5 1.2 13 1.0 2 0.5 6 0.5 3 0.5
1473. LOAD COMB 224 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL1
1474. 1 1.2 5 1.2 14 1.0 2 0.5 6 0.5 3 0.5
1475. LOAD COMB 225 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL2
1476. 1 1.2 5 1.2 11 1.0 2 0.5 6 0.5 4 0.5
1477. LOAD COMB 226 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL2
1478. 1 1.2 5 1.2 12 1.0 2 0.5 6 0.5 4 0.5
1479. LOAD COMB 227 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL2
1480. 1 1.2 5 1.2 13 1.0 2 0.5 6 0.5 4 0.5
1481. LOAD COMB 228 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL2
1482. 1 1.2 5 1.2 14 1.0 2 0.5 6 0.5 4 0.5
1483. LOAD COMB 229 0.9(DL+EQDL)+1.6WL(+X)
1484. 1 0.9 5 0.9 7 1.6
1485. LOAD COMB 230 0.9(DL+EQDL)+1.6WL(-X)
1486. 1 0.9 5 0.9 8 1.6
1487. LOAD COMB 231 0.9(DL+EQDL)+1.6WL(+Z)
1488. 1 0.9 5 0.9 9 1.6
1489. LOAD COMB 232 0.9(DL+EQDL)+1.6WL(-Z)
1490. 1 0.9 5 0.9 10 1.6
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1491. LOAD COMB 233 0.9(DL+EQDL)+1.6SL(+X)
1492. 1 0.9 5 0.9 11 1.6
1493. LOAD COMB 234 0.9(DL+EQDL)+1.6SL(-X)
1494. 1 0.9 5 0.9 12 1.6
1495. LOAD COMB 235 0.9(DL+EQDL)+1.6SL(+Z)
1496. 1 0.9 5 0.9 13 1.6
1497. LOAD COMB 236 0.9(DL+EQDL)+1.6SL(-Z)
1498. 1 0.9 5 0.9 14 1.6
1499. LOAD COMB 237 1.2DL+EQTL
1500. 1 1.2 17 1.0
1501. ***
1502. PERFORM ANALYSIS PRINT STATICS CHECK


         P R O B L E M   S T A T I S T I C S
         -----------------------------------

   NUMBER OF JOINTS        408  NUMBER OF MEMBERS     779
   NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
   NUMBER OF SURFACES        0  NUMBER OF SUPPORTS     11

        Using 64-bit analysis engine.

        SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

 TOTAL      PRIMARY LOAD CASES =    17, TOTAL DEGREES OF FREEDOM =    2382
 TOTAL LOAD COMBINATION  CASES =    90  SO FAR.
```

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STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
DEAD LOAD (DL)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.442259770E+03
Y = 0.920071710E+03
Z = -0.371248898E+03

TOTAL APPLIED LOAD 1

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -104.33
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -38733.52 MY= 0.00 MZ= -46142.40

TOTAL REACTION LOAD 1

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 104.33
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 38733.52 MY= -0.00 MZ= 46142.40

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	4.18343E-02	347
Y =	-2.86145E-01	135
Z =	7.83688E-02	183
RX=	-9.47521E-04	215
RY=	-1.03121E-04	61
RZ=	8.64958E-04	83

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LIVE LOAD (LL)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.442780757E+03
Y = 0.926213882E+03
Z = -0.371473058E+03

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TOTAL APPLIED LOAD 2

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -163.03
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -60559.92 MY= 0.00 MZ= -72184.96

TOTAL REACTION LOAD 2

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 163.03
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 60559.92 MY= 0.00 MZ= 72184.96

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	1.38158E-01	347
Y =	-1.43242E+00	141
Z =	2.64139E-01	183
RX=	4.66058E-03	144
RY=	-3.84235E-04	61
RZ=	-4.24014E-03	70

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
OPERATING FLUID LOAD (FL1)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443325009E+03
Y = 0.931400030E+03
Z = -0.372276993E+03

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443338862E+03
Y = 0.929781049E+03
Z = -0.373491080E+03

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443325009E+03
Y = 0.929275030E+03
Z = -0.372276993E+03

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TOTAL APPLIED LOAD 3

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 2.40
SUMMATION FORCE-Y = -66.40
SUMMATION FORCE-Z = 2.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -22564.45 MY= -1957.44 MZ= -31673.06

TOTAL REACTION LOAD 3

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -2.40
SUMMATION FORCE-Y = 66.40
SUMMATION FORCE-Z = -2.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 22564.45 MY= 1957.44 MZ= 31673.06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	3.79205E-01	358
Y =	-1.11992E+00	199
Z =	5.70349E-01	183
RX=	-6.91848E-03	212
RY=	9.34534E-04	352
RZ=	-2.65566E-03	70

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
OPERATING FLUID LOAD (FL2)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443325009E+03
Y = 0.931400030E+03
Z = -0.372276993E+03

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443338862E+03
Y = 0.929781049E+03
Z = -0.373491080E+03

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443325009E+03
Y = 0.933525030E+03
Z = -0.372276993E+03

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TOTAL APPLIED LOAD 4

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -2.40
SUMMATION FORCE-Y = -66.40
SUMMATION FORCE-Z = -2.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -27035.17 MY= 1957.44 MZ= -27202.34

TOTAL REACTION LOAD 4

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 2.40
SUMMATION FORCE-Y = 66.40
SUMMATION FORCE-Z = 2.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 27035.17 MY= -1957.44 MZ= 27202.34

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	-3.21271E-01	371
Y =	-1.15072E+00	199
Z =	-5.72804E-01	370
RX=	-6.96472E-03	212
RY=	-1.02309E-03	353
RZ=	2.62944E-03	83

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
EQP DL (EQDL)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443324999E+03
Y = 0.924813780E+03
Z = -0.371810651E+03

TOTAL APPLIED LOAD 5

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -69.60
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -25878.02 MY= 0.00 MZ= -30855.42

TOTAL REACTION LOAD 5

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 69.60
SUMMATION FORCE-Z = 0.00

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SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 25878.02 MY= 0.00 MZ= 30855.42

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X	-9.93190E-02	365
Y	-1.14186E+00	149
Z	-2.30272E-01	366
RX	1.41540E-03	134
RY	-6.41767E-04	61
RZ	-3.02527E-03	86

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
EQP LL (EQLL)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443325009E+03
Y = 0.931400030E+03
Z = -0.372478129E+03

TOTAL APPLIED LOAD 6

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -88.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -32778.07 MY= 0.00 MZ= -39012.60

TOTAL REACTION LOAD 6

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 88.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 32778.07 MY= -0.00 MZ= 39012.60

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X	-3.78644E-01	371
Y	-1.09854E+00	358
Z	4.13960E-01	373
RX	4.49076E-03	215
RY	-1.44881E-03	377
RZ	2.92643E-03	394

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
WIND LOAD (+X) WL(+X)

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CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443377752E+03
Y = 0.923497856E+03
Z = -0.371349581E+03

TOTAL APPLIED LOAD 7

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 25.53
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -9479.45 MZ= -23574.16

TOTAL REACTION LOAD 7

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = -25.53
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.00 MY= 9479.45 MZ= 23574.16

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)

	MAXIMUMS	AT NODE
X	7.10207E-01	187
Y	5.50327E-02	195
Z	-5.46424E-02	339
RX	-1.38416E-04	214
RY	7.85587E-04	139
RZ	1.58760E-03	300

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
WIND LOAD (-X) WL(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443377752E+03
Y = 0.923497856E+03
Z = -0.371349581E+03

TOTAL APPLIED LOAD 8

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = -25.53
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 9479.45 MZ= 23574.16

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TOTAL REACTION LOAD 8

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = 25.53
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -9479.45 MZ= -23574.16

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 8)

	MAXIMUMS	AT NODE
X	-7.10207E-01	187
Y	-5.50327E-02	195
Z	5.46424E-02	339
RX	1.38416E-04	214
RY	-7.85587E-04	139
RZ	-1.58760E-03	300

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 9
WIND LOAD (+Z) WL(+Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.442871470E+03
Y = 0.922737669E+03
Z = -0.371379204E+03

TOTAL APPLIED LOAD 9

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 36.48

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 33661.07 MY= -16155.76 MZ= 0.00

TOTAL REACTION LOAD 9

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -36.48

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -33661.07 MY= 16155.76 MZ= 0.00

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 9)

	MAXIMUMS	AT NODE
X	5.70957E-02	158
Y	1.37319E-01	92
Z	1.14813E+00	399
RX	2.39712E-03	4
RY	-1.01177E-03	102
RZ	1.80934E-04	86

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 10
WIND LOAD (-Z) WL(-Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.442871470E+03
Y = 0.922737669E+03
Z = -0.371379204E+03

TOTAL APPLIED LOAD 10

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -36.48

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -33661.07 MY= 16155.76 MZ= 0.00

TOTAL REACTION LOAD 10

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 36.48

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 33661.07 MY= -16155.76 MZ= -0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 10)

	MAXIMUMS	AT NODE
X	-5.70957E-02	158
Y	-1.37319E-01	92
Z	-1.14813E+00	399
RX	-2.39712E-03	4
RY	1.01177E-03	102
RZ	-1.80934E-04	86

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 11
SEISMIC LOAD SL(+X)

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CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443035841E+03
Y = 0.926268384E+03
Z = -0.372139917E+03

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443388561E+03
Y = 0.927332209E+03
Z = -0.372443520E+03

TOTAL APPLIED LOAD 11

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = 21.46
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 11.80

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 10942.52 MY= -13218.11 MZ= -19877.72

TOTAL REACTION LOAD 11

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = -21.46
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -11.80

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -10942.52 MY= 13218.11 MZ= 19877.72

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 11)

	MAXIMUMS	AT NODE
X =	8.96466E-01	355
Y =	1.19079E-01	195
Z =	7.25230E-01	356
RX=	7.41773E-04	217
RY=	1.76597E-03	350
RZ=	-8.57123E-04	348

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 12
SEISMIC LOAD SL(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443035841E+03
Y = 0.926268384E+03
Z = -0.372139917E+03

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 38

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443388561E+03
Y = 0.927332209E+03
Z = -0.372443520E+03

TOTAL APPLIED LOAD 12

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = -21.46
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -11.80

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -10942.52 MY= 13218.11 MZ= 19877.72

TOTAL REACTION LOAD 12

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = 21.46
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 11.80

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 10942.52 MY= -13218.11 MZ= -19877.72

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 12)

	MAXIMUMS	AT NODE
X =	-8.96466E-01	355
Y =	-1.19079E-01	195
Z =	-7.25230E-01	356
RX=	-7.41773E-04	217
RY=	-1.76597E-03	350
RZ=	8.57123E-04	348

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 13
SEISMIC LOAD SL(+Z)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443388561E+03
Y = 0.927332209E+03
Z = -0.372443520E+03

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443035841E+03
Y = 0.926268384E+03
Z = -0.372139917E+03

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TOTAL APPLIED LOAD 13

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = 11.80
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 21.46

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 19877.72 MY= -13902.38 MZ= -10942.52

TOTAL REACTION LOAD 13

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = -11.80
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -21.46

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -19877.72 MY= 13902.38 MZ= 10942.52

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 13)

	MAXIMUMS	AT NODE
X =	7.03508E-01	357
Y =	-1.21231E-01	336
Z =	1.08845E+00	356
RX=	1.02534E-03	217
RY=	1.84466E-03	350
RZ=	-7.98891E-04	348

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 14
SEISMIC LOAD SL(-Z)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443388561E+03
Y = 0.927332209E+03
Z = -0.372443520E+03

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443035841E+03
Y = 0.926268384E+03
Z = -0.372139917E+03

TOTAL APPLIED LOAD 14

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = -11.80
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -21.46

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -19877.72 MY= 13902.38 MZ= 10942.52

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TOTAL REACTION LOAD 14

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = 11.80
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 21.46

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 19877.72 MY= -13902.38 MZ= -10942.52

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 14)

	MAXIMUMS	AT NODE
X =	-7.03508E-01	357
Y =	1.21231E-01	336
Z =	-1.08845E+00	356
RX=	-1.02534E-03	217
RY=	-1.84466E-03	350
RZ=	7.98891E-04	348

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 15
TEMPERATURE LOAD (TL)

TOTAL APPLIED LOAD 15

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 15)
SUMMATION FORCE-X = 1.8532065E-14
SUMMATION FORCE-Y = -4.0770542E-13
SUMMATION FORCE-Z = -4.1898581E-14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.8777494E-10 MY= 4.2417018E-11 MZ= -3.5207802E-11

TOTAL REACTION LOAD 15

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 15)
SUMMATION FORCE-X = 1.3124466E-12
SUMMATION FORCE-Y = -9.6449514E-13
SUMMATION FORCE-Z = 6.4320869E-14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.9887861E-10 MY= -5.2198710E-10 MZ= -1.7472310E-09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 15)

	MAXIMUMS	AT NODE
X =	3.64227E-01	177
Y =	5.86488E-01	363
Z =	-1.56123E-01	43
RX=	4.19278E-04	11
RY=	4.27498E-04	8
RZ=	-1.05242E-03	5

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 16
PIPE LATERAL LOAD (PL)

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CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443361668E+03
Y = 0.928550038E+03
Z = -0.376166670E+03

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443361668E+03
Y = 0.928550038E+03
Z = -0.376166670E+03

TOTAL APPLIED LOAD 16

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 16)
SUMMATION FORCE-X = 4.50
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 4.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4178.48 MY= -3687.88 MZ= -4178.48

TOTAL REACTION LOAD 16

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 16)
SUMMATION FORCE-X = -4.50
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -4.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4178.48 MY= 3687.88 MZ= 4178.48

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 16)

	MAXIMUMS	AT NODE
X =	2.76115E-01	203
Y =	9.17367E-02	38
Z =	3.79773E-01	184
RX=	6.45930E-04	68
RY=	-6.76842E-04	217
RZ=	-1.85207E-04	39

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 17
EQP TEST LOAD (EQTL)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.443324997E+03
Y = 0.924733324E+03
Z = -0.372426995E+03

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TOTAL APPLIED LOAD 17

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 17)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -144.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -53629.49 MY= 0.00 MZ= -63838.80

TOTAL REACTION LOAD 17

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 17)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 144.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 53629.49 MY= 0.00 MZ= 63838.80

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 17)

	MAXIMUMS	AT NODE
X =	-3.27990E-01	364
Y =	-2.27352E+00	149
Z =	-7.93263E-01	372
RX=	-5.17931E-03	215
RY=	-1.24328E-03	61
RZ=	-6.05434E-03	86

********** END OF DATA FROM INTERNAL STORAGE **********

1503. LOAD LIST 101 TO 153
1504. PRINT SUPPORT REACTIONS
SUPPORT REACTION

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SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
405	101	0.01	3.74	0.11	0.36	0.00	-0.03
	102	0.02	3.74	0.11	0.36	0.00	-0.05
	103	0.06	6.26	0.70	2.20	0.00	-0.18
	104	0.06	6.26	0.70	2.20	0.00	-0.19
	105	0.05	5.63	0.55	1.74	0.00	-0.14
	106	0.05	5.63	0.55	1.74	0.00	-0.16
	107	0.01	3.74	0.11	0.36	0.00	-0.02
	108	0.01	3.74	0.11	0.36	0.00	-0.04
	109	0.05	6.26	0.70	2.20	0.00	-0.17
	110	0.06	6.26	0.70	2.20	0.00	-0.18
	111	0.04	5.63	0.55	1.74	0.00	-0.13
	112	0.05	5.63	0.55	1.74	0.00	-0.14
	113	-0.02	3.74	0.11	0.36	0.00	0.07
	114	0.04	3.74	0.11	0.36	0.00	-0.14
	115	0.01	2.36	-0.34	-1.06	0.00	-0.03
	116	0.01	5.13	0.57	1.78	0.00	-0.04
	117	-0.02	3.74	0.11	0.36	0.00	0.06
	118	0.05	3.74	0.11	0.36	0.00	-0.16
	119	0.02	2.36	-0.34	-1.06	0.00	-0.05
	120	0.02	5.13	0.57	1.78	0.00	-0.05
	121	-0.05	3.74	0.11	0.36	0.00	0.09
	122	0.08	3.74	0.11	0.36	0.00	-0.16
	123	0.00	3.36	-0.02	0.01	0.00	-0.01
	124	0.02	4.13	0.25	0.72	0.00	-0.06
	125	-0.05	3.74	0.11	0.36	0.00	0.07
	126	0.08	3.74	0.11	0.36	0.00	-0.17
	127	0.01	3.36	-0.02	0.01	0.00	-0.02
	128	0.02	4.13	0.25	0.72	0.00	-0.07
	129	-0.00	5.63	0.23	0.71	0.00	0.02
	130	0.05	5.63	0.23	0.71	0.00	-0.14
	131	0.02	4.59	-0.11	-0.35	0.00	-0.06
	132	0.02	6.67	0.56	1.78	0.00	-0.07
	133	-0.00	5.63	0.23	0.71	0.00	0.00
	134	0.05	5.63	0.23	0.71	0.00	-0.16
	135	0.02	4.59	-0.11	-0.35	0.00	-0.08
	136	0.03	6.68	0.56	1.78	0.00	-0.08
	137	-0.03	5.63	0.23	0.71	0.00	0.03
	138	0.07	5.63	0.23	0.71	0.00	-0.16
	139	0.01	5.35	0.12	0.45	0.00	-0.05
	140	0.03	5.92	0.33	0.98	0.00	-0.08
	141	-0.02	5.63	0.23	0.71	0.00	0.01
	142	0.07	5.63	0.23	0.71	0.00	-0.17
	143	0.02	5.35	0.12	0.45	0.00	-0.06
	144	0.03	5.92	0.33	0.98	0.00	-0.10
	145	-0.03	2.25	0.07	0.22	0.00	0.09
	146	0.04	2.25	0.07	0.22	0.00	-0.12

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SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	147	0.01	0.86	-0.38	-1.20	0.00	-0.02
	148	0.01	3.63	0.52	1.64	0.00	-0.02
	149	-0.06	2.25	0.07	0.22	0.00	0.10
	150	0.07	2.25	0.07	0.22	0.00	-0.14
	151	-0.00	1.86	-0.07	-0.14	0.00	0.01
	152	0.01	2.63	0.21	0.57	0.00	-0.04
	153	0.01	3.74	0.11	0.36	0.00	-0.05
406	101	0.00	3.80	-0.11	-0.36	0.00	-0.00
	102	0.00	3.80	-0.11	-0.36	0.00	-0.00
	103	0.01	6.32	-0.70	-2.20	0.00	-0.03
	104	0.01	6.32	-0.70	-2.20	0.00	-0.03
	105	0.01	5.69	-0.55	-1.74	0.00	-0.02
	106	0.01	5.69	-0.55	-1.74	0.00	-0.02
	107	0.00	3.80	-0.11	-0.36	0.00	-0.00
	108	0.00	3.80	-0.11	-0.36	0.00	-0.00
	109	0.01	6.32	-0.70	-2.20	0.00	-0.03
	110	0.01	6.32	-0.70	-2.20	0.00	-0.03
	111	0.01	5.69	-0.55	-1.74	0.00	-0.02
	112	0.01	5.69	-0.55	-1.74	0.00	-0.02
	113	-0.00	3.80	-0.11	-0.36	0.00	0.01
	114	0.00	3.80	-0.11	-0.36	0.00	-0.01
	115	0.00	5.19	-0.56	-1.78	0.00	-0.00
	116	0.00	2.41	0.33	1.05	0.00	-0.00
	117	-0.00	3.80	-0.11	-0.36	0.00	0.01
	118	0.00	3.80	-0.11	-0.36	0.00	-0.01
	119	0.00	5.19	-0.56	-1.78	0.00	-0.00
	120	0.00	2.41	0.33	1.05	0.00	-0.00
	121	-0.03	3.80	-0.11	-0.36	0.00	0.04
	122	0.03	3.80	-0.11	-0.36	0.00	-0.04
	123	0.00	4.18	-0.24	-0.70	0.00	-0.00
	124	0.00	3.42	0.01	-0.02	0.00	-0.00
	125	-0.03	3.80	-0.11	-0.36	0.00	0.04
	126	0.03	3.80	-0.11	-0.36	0.00	-0.04
	127	0.00	4.18	-0.24	-0.70	0.00	-0.00
	128	0.00	3.42	0.01	-0.02	0.00	-0.00
	129	-0.00	5.69	-0.23	-0.71	0.00	0.00
	130	0.00	5.69	-0.23	-0.71	0.00	-0.01
	131	0.00	6.73	-0.56	-1.78	0.00	-0.00
	132	0.00	4.65	0.11	0.35	0.00	-0.00
	133	-0.00	5.69	-0.23	-0.71	0.00	0.00
	134	0.00	5.69	-0.23	-0.71	0.00	-0.01
	135	0.00	6.73	-0.56	-1.78	0.00	-0.00
	136	0.00	4.65	0.11	0.35	0.00	-0.00
	137	-0.02	5.69	-0.23	-0.71	0.00	0.03
	138	0.02	5.69	-0.23	-0.71	0.00	-0.04
	139	0.00	5.98	-0.32	-0.97	0.00	-0.00

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 45

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	140	0.00	5.40	-0.13	-0.46	0.00	-0.00
	141	-0.02	5.69	-0.23	-0.71	0.00	0.03
	142	0.02	5.69	-0.23	-0.71	0.00	-0.04
	143	0.00	5.98	-0.32	-0.97	0.00	-0.00
	144	0.00	5.40	-0.13	-0.46	0.00	-0.00
	145	-0.00	2.28	-0.07	-0.22	0.00	0.01
	146	0.00	2.28	-0.07	-0.22	0.00	-0.01
	147	0.00	3.67	-0.52	-1.63	0.00	-0.00
	148	0.00	0.89	0.38	1.20	0.00	-0.00
	149	-0.03	2.28	-0.07	-0.22	0.00	0.04
	150	0.03	2.28	-0.07	-0.22	0.00	-0.04
	151	0.00	2.66	-0.19	-0.56	0.00	-0.00
	152	0.00	1.90	0.06	0.12	0.00	-0.00
	153	0.00	3.80	-0.11	-0.36	0.00	-0.00
407	101	1.86	19.93	0.08	0.26	0.00	-5.85
	102	3.02	26.01	0.08	0.25	0.00	-9.52
	103	6.97	39.74	0.05	0.17	0.00	-21.95
	104	8.13	45.83	0.05	0.17	0.00	-25.62
	105	5.69	34.79	0.06	0.19	0.00	-17.92
	106	6.85	40.87	0.06	0.19	0.00	-21.59
	107	0.31	13.17	0.12	0.39	0.00	-0.98
	108	1.48	19.25	0.12	0.38	0.00	-4.66
	109	5.42	32.98	0.10	0.30	0.00	-17.08
	110	6.59	39.06	0.09	0.30	0.00	-20.76
	111	4.15	28.03	0.10	0.33	0.00	-13.06
	112	5.31	34.11	0.10	0.32	0.00	-16.73
	113	-2.39	7.15	0.08	0.26	0.00	7.52
	114	6.10	32.71	0.08	0.26	0.00	-19.21
	115	1.77	19.78	-0.32	-1.01	0.00	-5.59
	116	1.94	20.07	0.49	1.53	0.00	-6.11
	117	-1.22	13.23	0.08	0.25	0.00	3.85
	118	7.26	38.79	0.08	0.26	0.00	-22.88
	119	2.94	25.86	-0.32	-1.02	0.00	-9.26
	120	3.10	26.16	0.48	1.52	0.00	-9.78
	121	-1.24	7.51	0.08	0.26	0.00	3.85
	122	4.95	32.35	0.08	0.26	0.00	-15.55
	123	-0.02	12.09	0.05	0.20	0.00	0.06
	124	3.73	27.77	0.12	0.32	0.00	-11.75
	125	-0.08	13.59	0.08	0.25	0.00	0.18
	126	6.12	38.43	0.08	0.25	0.00	-19.22
	127	1.15	18.17	0.04	0.19	0.00	-3.61
	128	4.90	33.85	0.12	0.31	0.00	-15.42
	129	0.88	25.19	0.06	0.18	0.00	-2.79
	130	7.25	44.35	0.06	0.18	0.00	-22.84
	131	4.01	34.66	-0.25	-0.77	0.00	-12.62
	132	4.13	34.88	0.36	1.13	0.00	-13.01

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 46

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	133	2.05	31.27	0.05	0.17	0.00	-6.46
	134	8.41	50.43	0.06	0.18	0.00	-26.51
	135	5.17	40.74	-0.25	-0.78	0.00	-16.29
	136	5.29	40.96	0.36	1.13	0.00	-16.68
	137	1.74	25.46	0.06	0.18	0.00	-5.54
	138	6.39	44.08	0.06	0.18	0.00	-20.09
	139	2.66	28.89	0.03	0.14	0.00	-8.38
	140	5.47	40.65	0.08	0.22	0.00	-17.24
	141	2.91	31.54	0.05	0.17	0.00	-9.21
	142	7.55	50.17	0.05	0.17	0.00	-23.76
	143	3.83	34.97	0.03	0.13	0.00	-12.05
	144	6.64	46.73	0.08	0.22	0.00	-20.91
	145	-3.51	-4.80	-0.00	-0.01	0.00	11.05
	146	4.98	20.76	-0.00	-0.00	0.00	-15.69
	147	0.65	7.83	-0.40	-1.28	0.00	-2.06
	148	0.82	8.13	0.40	1.27	0.00	-2.58
	149	-2.36	-4.44	-0.00	-0.00	0.00	7.38
	150	3.83	20.40	-0.00	-0.00	0.00	-12.02
	151	-1.14	0.14	-0.04	-0.06	0.00	3.58
	152	2.61	15.82	0.03	0.05	0.00	-8.23
	153	2.44	21.59	0.02	0.05	0.00	-7.67
408	101	-3.11	42.64	0.16	0.49	0.00	9.79
	102	-2.05	38.07	0.19	0.59	0.00	6.46
	103	-7.94	88.16	0.20	0.63	0.00	25.02
	104	-6.88	83.60	0.23	0.73	0.00	21.69
	105	-6.73	76.78	0.19	0.60	0.00	21.21
	106	-5.68	72.22	0.22	0.69	0.00	17.88
	107	-4.63	48.76	0.15	0.46	0.00	14.57
	108	-3.57	44.20	0.18	0.55	0.00	11.24
	109	-9.46	94.28	0.19	0.60	0.00	29.80
	110	-8.40	89.72	0.22	0.69	0.00	26.47
	111	-8.25	82.90	0.18	0.56	0.00	25.99
	112	-7.19	78.34	0.21	0.66	0.00	22.66
	113	-7.29	55.22	0.15	0.47	0.00	22.98
	114	1.08	30.05	0.16	0.52	0.00	-3.40
	115	-3.04	40.62	-0.61	-1.93	0.00	9.56
	116	-3.18	44.65	0.93	2.92	0.00	10.02
	117	-6.24	50.66	0.18	0.56	0.00	19.65
	118	2.14	25.49	0.19	0.61	0.00	-6.73
	119	-1.98	36.06	-0.58	-1.84	0.00	6.23
	120	-2.12	40.09	0.96	3.01	0.00	6.69
	121	-6.14	53.66	0.10	0.33	0.00	19.19
	122	-0.08	31.61	0.21	0.66	0.00	0.39
	123	-4.84	48.70	-0.00	0.14	0.00	15.24
	124	-1.38	36.58	0.31	0.85	0.00	4.33
	125	-5.08	49.10	0.13	0.42	0.00	15.86

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 47

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	126	0.98	27.05	0.24	0.75	0.00	-2.94
	127	-3.78	44.14	0.03	0.23	0.00	11.92
	128	-0.32	32.01	0.34	0.94	0.00	1.01
	129	-8.31	86.20	0.17	0.55	0.00	26.17
	130	-2.03	67.33	0.19	0.58	0.00	6.40
	131	-5.12	75.25	-0.40	-1.26	0.00	16.11
	132	-5.22	78.28	0.76	2.39	0.00	16.46
	133	-7.25	81.64	0.20	0.64	0.00	22.84
	134	-0.97	62.77	0.21	0.68	0.00	3.07
	135	-4.06	70.69	-0.37	-1.16	0.00	12.78
	136	-4.17	73.72	0.79	2.48	0.00	13.13
	137	-7.44	85.03	0.14	0.44	0.00	23.34
	138	-2.90	68.50	0.22	0.69	0.00	9.23
	139	-6.47	81.31	0.06	0.30	0.00	20.38
	140	-3.87	72.22	0.30	0.83	0.00	12.19
	141	-6.38	80.47	0.17	0.53	0.00	20.01
	142	-1.84	63.93	0.25	0.78	0.00	5.90
	143	-5.41	76.75	0.09	0.39	0.00	17.05
	144	-2.81	67.66	0.33	0.92	0.00	8.86
	145	-5.04	26.61	0.05	0.14	0.00	15.88
	146	3.33	1.45	0.06	0.19	0.00	-10.49
	147	-0.78	12.01	-0.72	-2.26	0.00	2.46
	148	-0.93	16.05	0.82	2.60	0.00	2.92
	149	-3.88	25.06	0.00	0.00	0.00	12.09
	150	2.17	3.00	0.11	0.34	0.00	-6.71
	151	-2.59	20.09	-0.10	-0.19	0.00	8.15
	152	0.88	7.97	0.21	0.52	0.00	-2.76
	153	-2.60	41.79	0.17	0.53	0.00	8.19
409	101	0.00	21.77	0.08	0.25	0.00	-0.01
	102	0.01	21.86	0.08	0.25	0.00	-0.02
	103	-0.04	40.75	0.05	0.15	0.00	0.12
	104	-0.04	40.83	0.05	0.15	0.00	0.11
	105	-0.03	36.00	0.06	0.18	0.00	0.08
	106	-0.02	36.09	0.06	0.18	0.00	0.08
	107	0.00	21.77	0.12	0.38	0.00	-0.00
	108	0.00	21.86	0.12	0.38	0.00	-0.01
	109	-0.04	40.75	0.09	0.29	0.00	0.13
	110	-0.04	40.84	0.09	0.28	0.00	0.12
	111	-0.03	36.00	0.10	0.31	0.00	0.09
	112	-0.03	36.09	0.10	0.31	0.00	0.09
	113	-0.09	21.88	0.08	0.25	0.00	0.28
	114	0.10	21.66	0.08	0.25	0.00	-0.31
	115	0.00	21.68	-0.32	-1.01	0.00	-0.01
	116	0.01	21.87	0.48	1.51	0.00	-0.02
	117	-0.09	21.97	0.08	0.25	0.00	0.27
	118	0.10	21.75	0.08	0.25	0.00	-0.31

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 48

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	119	0.00	21.77	-0.32	-1.01	0.00	-0.01
	120	0.01	21.95	0.48	1.51	0.00	-0.02
	121	-0.02	21.73	0.08	0.25	0.00	0.03
	122	0.03	21.82	0.08	0.25	0.00	-0.06
	123	0.00	21.72	0.06	0.22	0.00	-0.01
	124	0.01	21.83	0.10	0.29	0.00	-0.02
	125	-0.02	21.82	0.08	0.25	0.00	0.02
	126	0.03	21.91	0.08	0.25	0.00	-0.06
	127	0.00	21.81	0.06	0.21	0.00	-0.01
	128	0.01	21.92	0.10	0.28	0.00	-0.03
	129	-0.06	36.09	0.05	0.17	0.00	0.18
	130	0.08	35.92	0.05	0.16	0.00	-0.26
	131	0.01	35.93	-0.25	-0.78	0.00	-0.04
	132	0.01	36.07	0.35	1.11	0.00	-0.05
	133	-0.05	36.17	0.05	0.16	0.00	0.17
	134	0.08	36.01	0.05	0.16	0.00	-0.27
	135	0.01	36.02	-0.25	-0.79	0.00	-0.04
	136	0.02	36.16	0.35	1.11	0.00	-0.05
	137	-0.00	35.97	0.05	0.16	0.00	-0.01
	138	0.03	36.04	0.05	0.17	0.00	-0.08
	139	0.01	35.96	0.04	0.14	0.00	-0.04
	140	0.02	36.05	0.07	0.19	0.00	-0.05
	141	-0.00	36.06	0.05	0.16	0.00	-0.01
	142	0.03	36.13	0.05	0.16	0.00	-0.08
	143	0.01	36.05	0.04	0.14	0.00	-0.04
	144	0.02	36.13	0.07	0.19	0.00	-0.05
	145	-0.09	7.58	-0.00	-0.00	0.00	0.29
	146	0.09	7.37	-0.00	-0.01	0.00	-0.30
	147	-0.00	7.38	-0.40	-1.27	0.00	0.00
	148	0.00	7.57	0.40	1.26	0.00	-0.01
	149	-0.02	7.43	-0.00	-0.01	0.00	0.04
	150	0.03	7.52	-0.00	-0.00	0.00	-0.05
	151	-0.00	7.42	-0.02	-0.04	0.00	0.00
	152	0.00	7.53	0.02	0.03	0.00	-0.01
	153	0.01	20.39	0.02	0.05	0.00	-0.02
410	101	2.97	23.57	-0.62	-1.94	0.00	-9.36
	102	3.98	30.20	0.85	2.66	0.00	-12.55
	103	8.99	52.64	-0.28	-0.90	0.00	-28.33
	104	10.01	59.28	1.18	3.71	0.00	-31.52
	105	7.49	45.38	-0.37	-1.16	0.00	-23.59
	106	8.50	52.01	1.09	3.45	0.00	-26.78
	107	2.03	19.12	-1.51	-4.76	0.00	-6.40
	108	3.04	25.76	-0.05	-0.15	0.00	-9.59
	109	8.06	48.20	-1.18	-3.71	0.00	-25.38
	110	9.07	54.83	0.28	0.89	0.00	-28.57
	111	6.55	40.93	-1.26	-3.97	0.00	-20.63

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 49

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	112	7.56	47.56	0.20	0.63	0.00	-23.82
	113	-2.14	12.74	-1.09	-3.43	0.00	6.74
	114	8.08	34.40	-0.14	-0.45	0.00	-25.45
	115	2.00	2.87	-10.87	-34.23	0.00	-6.31
	116	3.94	44.26	9.63	30.35	0.00	-12.40
	117	-1.13	19.37	0.37	1.17	0.00	3.55
	118	9.09	41.03	1.32	4.15	0.00	-28.64
	119	3.02	9.50	-9.40	-29.63	0.00	-9.50
	120	4.95	50.90	11.10	34.95	0.00	-15.59
	121	-0.08	11.37	-2.99	-9.41	0.00	0.15
	122	6.02	35.76	1.76	5.54	0.00	-18.87
	123	0.71	8.72	-4.96	-15.53	0.00	-2.23
	124	5.23	38.41	3.73	11.65	0.00	-16.48
	125	0.93	18.01	-1.53	-4.81	0.00	-3.04
	126	7.04	42.39	3.22	10.14	0.00	-22.06
	127	1.72	15.36	-3.50	-10.92	0.00	-5.42
	128	6.25	45.04	5.19	16.25	0.00	-19.67
	129	2.02	37.26	-0.76	-2.40	0.00	-6.36
	130	9.68	53.51	-0.05	-0.17	0.00	-30.50
	131	5.13	29.86	-8.10	-25.50	0.00	-16.15
	132	6.58	60.91	7.28	22.93	0.00	-20.71
	133	3.03	43.90	0.70	2.20	0.00	-9.55
	134	10.70	60.14	1.41	4.43	0.00	-33.69
	135	6.14	36.50	-6.63	-20.90	0.00	-19.34
	136	7.59	67.54	8.74	27.53	0.00	-23.90
	137	3.56	36.24	-2.19	-6.89	0.00	-11.30
	138	8.14	54.53	1.37	4.32	0.00	-25.56
	139	4.15	34.25	-3.67	-11.47	0.00	-13.08
	140	7.55	56.52	2.85	8.91	0.00	-23.77
	141	4.57	42.87	-0.73	-2.29	0.00	-14.49
	142	9.15	61.16	2.83	8.92	0.00	-28.75
	143	5.17	40.89	-2.21	-6.87	0.00	-16.27
	144	8.56	63.15	4.31	13.51	0.00	-26.96
	145	-3.63	1.29	-0.36	-1.14	0.00	11.43
	146	6.59	22.95	0.58	1.84	0.00	-20.76
	147	0.51	-8.58	-10.14	-31.94	0.00	-1.62
	148	2.45	32.81	10.36	32.64	0.00	-7.71
	149	-1.57	-0.08	-2.26	-7.13	0.00	4.85
	150	4.53	24.31	2.48	7.82	0.00	-14.17
	151	-0.78	-2.72	-4.24	-13.24	0.00	2.46
	152	3.74	26.96	4.46	13.94	0.00	-11.79
	153	3.72	27.41	0.42	1.32	0.00	-11.73
411	101	-3.14	52.28	-1.05	-3.32	0.00	9.91
	102	-2.81	55.90	0.72	2.28	0.00	8.86
	103	-8.94	106.63	-1.00	-3.16	0.00	28.15
	104	-8.60	110.25	0.77	2.44	0.00	27.10

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 50

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	105	-7.49	93.05	-1.02	-3.20	0.00	23.59
	106	-7.16	96.66	0.76	2.40	0.00	22.54
	107	-3.64	49.75	-2.73	-8.58	0.00	11.46
	108	-3.31	53.37	-0.95	-2.98	0.00	10.42
	109	-9.43	104.10	-2.68	-8.43	0.00	29.70
	110	-9.10	107.72	-0.90	-2.82	0.00	28.66
	111	-7.98	90.52	-2.69	-8.47	0.00	25.14
	112	-7.65	94.13	-0.91	-2.86	0.00	24.10
	113	-8.14	62.32	-1.00	-3.14	0.00	25.63
	114	1.85	42.24	-1.11	-3.50	0.00	-5.82
	115	-2.14	29.75	-14.51	-45.71	0.00	6.74
	116	-4.15	74.81	12.40	39.06	0.00	13.07
	117	-7.80	65.94	0.78	2.46	0.00	24.59
	118	2.18	45.86	0.67	2.10	0.00	-6.86
	119	-1.81	33.37	-12.73	-40.10	0.00	5.70
	120	-3.82	78.43	14.18	44.66	0.00	12.03
	121	-5.43	50.46	-4.32	-13.61	0.00	16.83
	122	-0.86	54.10	2.21	6.97	0.00	2.99
	123	-3.99	42.78	-6.86	-21.33	0.00	12.58
	124	-2.30	61.78	4.75	14.69	0.00	7.24
	125	-5.10	54.08	-2.54	-8.01	0.00	15.78
	126	-0.53	57.72	3.99	12.57	0.00	1.94
	127	-3.66	46.40	-5.08	-15.73	0.00	11.53
	128	-1.97	65.39	6.53	20.29	0.00	6.19
	129	-9.66	100.61	-1.02	-3.20	0.00	30.42
	130	-2.17	85.55	-1.10	-3.46	0.00	6.84
	131	-5.16	76.19	-11.15	-35.12	0.00	16.25
	132	-6.67	109.98	9.03	28.46	0.00	21.01
	133	-9.33	104.23	0.76	2.41	0.00	29.38
	134	-1.84	89.17	0.68	2.14	0.00	5.80
	135	-4.83	79.80	-9.37	-29.52	0.00	15.21
	136	-6.34	113.60	10.81	34.06	0.00	19.96
	137	-7.63	91.72	-3.51	-11.05	0.00	23.82
	138	-4.20	94.45	1.39	4.39	0.00	13.44
	139	-6.55	85.96	-5.41	-16.84	0.00	20.63
	140	-5.28	100.20	3.30	10.18	0.00	16.63
	141	-7.30	95.33	-1.73	-5.45	0.00	22.78
	142	-3.87	98.06	3.17	9.99	0.00	12.40
	143	-6.22	89.58	-3.63	-11.23	0.00	19.59
	144	-4.95	103.82	5.07	15.78	0.00	15.59
	145	-6.23	33.49	-0.09	-0.29	0.00	19.62
	146	3.75	13.41	-0.20	-0.64	0.00	-11.82
	147	-0.23	0.92	-13.60	-42.85	0.00	0.73
	148	-2.24	45.98	13.31	41.92	0.00	7.07
	149	-3.52	21.63	-3.42	-10.76	0.00	10.82
	150	1.05	25.27	3.12	9.83	0.00	-3.02

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 51

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	151	-2.08	13.95	-5.95	-18.47	0.00	6.57
	152	-0.39	32.94	5.66	17.54	0.00	1.23
	153	-3.14	59.39	-0.57	-1.79	0.00	9.89
412	101	0.03	21.84	-0.82	-2.58	0.00	-0.08
	102	0.03	26.15	0.69	2.17	0.00	-0.10
	103	-0.08	47.06	-0.83	-2.63	0.00	0.27
	104	-0.08	51.36	0.67	2.12	0.00	0.25
	105	-0.06	40.76	-0.83	-2.62	0.00	0.18
	106	-0.05	45.06	0.68	2.13	0.00	0.16
	107	0.02	16.26	-2.80	-8.81	0.00	-0.06
	108	0.03	20.57	-1.29	-4.06	0.00	-0.08
	109	-0.09	41.48	-2.81	-8.86	0.00	0.29
	110	-0.09	45.78	-1.30	-4.11	0.00	0.27
	111	-0.06	35.17	-2.81	-8.85	0.00	0.20
	112	-0.06	39.48	-1.30	-4.10	0.00	0.18
	113	-0.23	22.78	-0.39	-1.24	0.00	0.73
	114	0.29	20.91	-1.24	-3.92	0.00	-0.90
	115	0.03	2.77	-10.11	-31.84	0.00	-0.08
	116	0.03	40.92	8.47	26.68	0.00	-0.08
	117	-0.23	27.09	1.11	3.51	0.00	0.72
	118	0.29	25.21	0.26	0.83	0.00	-0.92
	119	0.03	7.08	-8.60	-27.09	0.00	-0.10
	120	0.03	45.22	9.98	31.43	0.00	-0.10
	121	-0.01	15.33	-3.34	-10.52	0.00	0.00
	122	0.06	28.36	1.70	5.36	0.00	-0.17
	123	0.01	10.91	-5.01	-15.74	0.00	-0.05
	124	0.04	32.78	3.37	10.58	0.00	-0.12
	125	-0.00	19.64	-1.83	-5.77	0.00	-0.01
	126	0.07	32.66	3.21	10.11	0.00	-0.19
	127	0.02	15.22	-3.50	-10.99	0.00	-0.06
	128	0.04	37.08	4.88	15.33	0.00	-0.14
	129	-0.15	41.47	-0.54	-1.71	0.00	0.47
	130	0.24	40.06	-1.18	-3.72	0.00	-0.76
	131	0.05	26.46	-7.83	-24.66	0.00	-0.15
	132	0.05	55.07	6.10	19.23	0.00	-0.15
	133	-0.14	45.77	0.96	3.04	0.00	0.45
	134	0.25	44.37	0.33	1.03	0.00	-0.78
	135	0.05	30.77	-6.32	-19.91	0.00	-0.16
	136	0.05	59.38	7.61	23.98	0.00	-0.16
	137	0.02	35.88	-2.75	-8.68	0.00	-0.08
	138	0.07	45.65	1.03	3.24	0.00	-0.21
	139	0.04	32.57	-4.00	-12.59	0.00	-0.12
	140	0.06	48.96	2.28	7.15	0.00	-0.17
	141	0.03	40.18	-1.25	-3.93	0.00	-0.10
	142	0.08	49.96	2.54	7.99	0.00	-0.23
	143	0.04	36.87	-2.50	-7.84	0.00	-0.14

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 52

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	144	0.06	53.27	3.78	11.90	0.00	-0.19
	145	-0.25	11.68	0.50	1.58	0.00	0.78
	146	0.27	9.80	-0.35	-1.10	0.00	-0.85
	147	0.01	-8.33	-9.21	-29.02	0.00	-0.04
	148	0.01	29.81	9.37	29.50	0.00	-0.04
	149	-0.02	4.22	-2.44	-7.70	0.00	0.05
	150	0.05	17.25	2.60	8.19	0.00	-0.12
	151	0.00	-0.19	-4.11	-12.92	0.00	-0.00
	152	0.02	21.67	4.26	13.40	0.00	-0.07
	153	0.03	24.20	0.20	0.64	0.00	-0.10
413	101	0.80	12.56	-0.04	-0.11	0.00	-2.51
	102	1.02	10.04	-0.03	-0.09	0.00	-3.20
	103	4.53	25.82	-0.13	-0.42	0.00	-14.25
	104	4.74	23.31	-0.13	-0.40	0.00	-14.95
	105	3.59	22.51	-0.11	-0.34	0.00	-11.32
	106	3.81	19.99	-0.10	-0.32	0.00	-12.01
	107	1.05	15.39	-0.04	-0.13	0.00	-3.31
	108	1.27	12.87	-0.03	-0.11	0.00	-4.00
	109	4.78	28.65	-0.14	-0.43	0.00	-15.06
	110	5.00	26.14	-0.13	-0.41	0.00	-15.75
	111	3.85	25.34	-0.11	-0.36	0.00	-12.12
	112	4.07	22.82	-0.11	-0.33	0.00	-12.81
	113	-2.44	4.99	-0.03	-0.11	0.00	7.70
	114	4.04	20.12	-0.04	-0.12	0.00	-12.72
	115	2.03	33.38	-0.35	-1.09	0.00	-6.40
	116	-0.44	-8.27	0.27	0.86	0.00	1.38
	117	-2.23	2.47	-0.03	-0.09	0.00	7.01
	118	4.26	17.61	-0.03	-0.10	0.00	-13.41
	119	2.25	30.87	-0.34	-1.07	0.00	-7.09
	120	-0.22	-10.79	0.28	0.88	0.00	0.69
	121	-0.44	13.67	-0.05	-0.14	0.00	1.28
	122	2.03	11.44	-0.03	-0.08	0.00	-6.30
	123	0.83	21.42	-0.13	-0.30	0.00	-2.62
	124	0.76	3.69	0.06	0.07	0.00	-2.40
	125	-0.22	11.16	-0.04	-0.12	0.00	0.59
	126	2.25	8.92	-0.02	-0.06	0.00	-6.99
	127	1.05	18.90	-0.12	-0.28	0.00	-3.31
	128	0.98	1.18	0.07	0.10	0.00	-3.09
	129	-0.45	16.84	-0.07	-0.21	0.00	1.43
	130	4.41	28.20	-0.07	-0.22	0.00	-13.89
	131	2.90	38.14	-0.30	-0.95	0.00	-9.15
	132	1.05	6.90	0.16	0.52	0.00	-3.31
	133	-0.23	14.33	-0.06	-0.19	0.00	0.73
	134	4.63	25.68	-0.06	-0.20	0.00	-14.58
	135	3.12	35.63	-0.29	-0.92	0.00	-9.84
	136	1.27	4.38	0.17	0.54	0.00	-4.01

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 53

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	137	1.05	23.36	-0.08	-0.24	0.00	-3.39
	138	2.91	21.68	-0.06	-0.19	0.00	-9.07
	139	2.00	29.16	-0.14	-0.36	0.00	-6.31
	140	1.95	15.87	0.00	-0.07	0.00	-6.15
	141	1.27	20.84	-0.07	-0.22	0.00	-4.08
	142	3.13	19.17	-0.05	-0.17	0.00	-9.76
	143	2.22	26.65	-0.13	-0.33	0.00	-7.00
	144	2.17	13.36	0.01	-0.05	0.00	-6.84
	145	-2.70	-0.72	-0.01	-0.04	0.00	8.52
	146	3.78	14.42	-0.01	-0.05	0.00	-11.90
	147	1.77	27.68	-0.32	-1.02	0.00	-5.58
	148	-0.70	-13.98	0.30	0.93	0.00	2.20
	149	-0.70	7.97	-0.02	-0.07	0.00	2.10
	150	1.77	5.73	-0.00	-0.01	0.00	-5.48
	151	0.57	15.71	-0.11	-0.23	0.00	-1.80
	152	0.50	-2.01	0.08	0.15	0.00	-1.58
	153	1.02	11.93	-0.03	-0.10	0.00	-3.20
414	101	-1.82	25.15	-0.15	-0.48	0.00	5.72
	102	-0.81	16.34	-0.14	-0.45	0.00	2.56
	103	-5.91	52.54	-0.30	-0.95	0.00	18.62
	104	-4.91	43.73	-0.29	-0.93	0.00	15.46
	105	-4.89	45.69	-0.27	-0.84	0.00	15.39
	106	-3.88	36.89	-0.26	-0.81	0.00	12.24
	107	-2.06	29.93	-0.16	-0.52	0.00	6.50
	108	-1.06	21.12	-0.16	-0.49	0.00	3.35
	109	-6.16	57.31	-0.31	-0.99	0.00	19.41
	110	-5.16	48.51	-0.31	-0.96	0.00	16.25
	111	-5.14	50.47	-0.28	-0.87	0.00	16.18
	112	-4.13	41.66	-0.27	-0.85	0.00	13.02
	113	-5.07	33.45	-0.16	-0.51	0.00	15.98
	114	1.44	16.85	-0.14	-0.45	0.00	-4.55
	115	-3.08	49.72	-0.60	-1.90	0.00	9.70
	116	-0.55	0.57	0.30	0.94	0.00	1.73
	117	-4.07	24.64	-0.15	-0.48	0.00	12.82
	118	2.45	8.04	-0.14	-0.43	0.00	-7.70
	119	-2.08	40.92	-0.59	-1.87	0.00	6.54
	120	0.45	-8.23	0.31	0.96	0.00	-1.43
	121	-3.95	39.12	-0.17	-0.54	0.00	12.39
	122	0.31	11.18	-0.13	-0.42	0.00	-0.96
	123	-3.37	42.10	-0.20	-0.60	0.00	10.62
	124	-0.26	8.20	-0.10	-0.36	0.00	0.81
	125	-2.94	30.31	-0.16	-0.51	0.00	9.23
	126	1.32	2.38	-0.13	-0.40	0.00	-4.12
	127	-2.37	33.29	-0.19	-0.58	0.00	7.46
	128	0.74	-0.60	-0.10	-0.33	0.00	-2.35
	129	-5.74	51.89	-0.22	-0.71	0.00	18.10

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 54

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	130	-0.86	39.44	-0.21	-0.66	0.00	2.70
	131	-4.25	64.09	-0.56	-1.75	0.00	13.39
	132	-2.35	27.23	0.12	0.38	0.00	7.41
	133	-4.74	43.08	-0.22	-0.68	0.00	14.94
	134	0.15	30.63	-0.20	-0.64	0.00	-0.46
	135	-3.25	55.29	-0.55	-1.72	0.00	10.23
	136	-1.35	18.43	0.13	0.41	0.00	4.25
	137	-4.90	56.14	-0.23	-0.73	0.00	15.40
	138	-1.70	35.19	-0.20	-0.64	0.00	5.39
	139	-4.47	58.37	-0.25	-0.78	0.00	14.08
	140	-2.13	32.95	-0.18	-0.60	0.00	6.72
	141	-3.90	47.33	-0.22	-0.70	0.00	12.25
	142	-0.70	26.38	-0.20	-0.61	0.00	2.23
	143	-3.47	49.57	-0.25	-0.75	0.00	10.92
	144	-1.13	24.15	-0.17	-0.57	0.00	3.56
	145	-3.94	19.41	-0.07	-0.23	0.00	12.42
	146	2.57	2.81	-0.05	-0.17	0.00	-8.11
	147	-1.95	35.68	-0.51	-1.62	0.00	6.14
	148	0.58	-13.47	0.39	1.22	0.00	-1.83
	149	-2.81	25.07	-0.08	-0.26	0.00	8.83
	150	1.45	-2.86	-0.05	-0.14	0.00	-4.52
	151	-2.24	28.05	-0.11	-0.32	0.00	7.06
	152	0.87	-5.84	-0.01	-0.08	0.00	-2.75
	153	-1.50	22.62	-0.19	-0.61	0.00	4.71
415	101	0.00	13.05	-0.04	-0.13	0.00	-0.01
	102	0.00	8.21	-0.03	-0.09	0.00	-0.01
	103	-0.04	25.43	-0.15	-0.46	0.00	0.13
	104	-0.04	20.58	-0.13	-0.42	0.00	0.13
	105	-0.03	22.33	-0.12	-0.37	0.00	0.10
	106	-0.03	17.49	-0.11	-0.34	0.00	0.10
	107	0.00	18.64	-0.05	-0.17	0.00	-0.01
	108	0.00	13.80	-0.04	-0.14	0.00	-0.01
	109	-0.04	31.01	-0.16	-0.50	0.00	0.13
	110	-0.04	26.17	-0.15	-0.47	0.00	0.13
	111	-0.03	27.92	-0.13	-0.42	0.00	0.10
	112	-0.03	23.08	-0.12	-0.38	0.00	0.09
	113	-0.11	12.26	-0.03	-0.11	0.00	0.33
	114	0.11	13.85	-0.05	-0.15	0.00	-0.35
	115	0.01	32.20	-0.29	-0.91	0.00	-0.02
	116	0.00	-6.10	0.21	0.66	0.00	-0.00
	117	-0.11	7.41	-0.02	-0.07	0.00	0.33
	118	0.11	9.01	-0.04	-0.11	0.00	-0.35
	119	0.01	27.36	-0.28	-0.88	0.00	-0.02
	120	0.00	-10.94	0.22	0.69	0.00	-0.00
	121	-0.04	19.94	-0.06	-0.18	0.00	0.05
	122	0.04	6.17	-0.02	-0.08	0.00	-0.07

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 55

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	123	0.00	24.35	-0.10	-0.27	0.00	-0.01
	124	0.00	1.76	0.02	0.01	0.00	-0.01
	125	-0.04	15.09	-0.04	-0.14	0.00	0.05
	126	0.04	1.33	-0.01	-0.04	0.00	-0.08
	127	0.00	19.51	-0.09	-0.23	0.00	-0.01
	128	0.00	-3.09	0.03	0.05	0.00	-0.01
	129	-0.07	21.73	-0.07	-0.22	0.00	0.23
	130	0.09	22.92	-0.08	-0.25	0.00	-0.28
	131	0.01	36.69	-0.26	-0.82	0.00	-0.03
	132	0.01	7.96	0.11	0.35	0.00	-0.02
	133	-0.07	16.88	-0.06	-0.18	0.00	0.23
	134	0.09	18.08	-0.07	-0.22	0.00	-0.28
	135	0.01	31.84	-0.25	-0.79	0.00	-0.03
	136	0.01	3.12	0.12	0.39	0.00	-0.02
	137	-0.02	27.49	-0.09	-0.27	0.00	0.02
	138	0.04	17.16	-0.06	-0.20	0.00	-0.07
	139	0.01	30.80	-0.12	-0.34	0.00	-0.03
	140	0.01	13.85	-0.03	-0.13	0.00	-0.02
	141	-0.02	22.64	-0.08	-0.24	0.00	0.02
	142	0.04	12.32	-0.05	-0.16	0.00	-0.07
	143	0.01	25.95	-0.11	-0.31	0.00	-0.03
	144	0.01	9.01	-0.02	-0.10	0.00	-0.02
	145	-0.11	5.29	-0.01	-0.02	0.00	0.34
	146	0.11	6.89	-0.02	-0.06	0.00	-0.35
	147	0.00	25.24	-0.26	-0.83	0.00	-0.01
	148	-0.00	-13.06	0.24	0.74	0.00	0.00
	149	-0.04	12.97	-0.03	-0.09	0.00	0.06
	150	0.04	-0.79	0.00	0.01	0.00	-0.07
	151	0.00	17.38	-0.08	-0.18	0.00	-0.01
	152	0.00	-5.21	0.05	0.10	0.00	-0.00
	153	0.00	11.47	-0.03	-0.10	0.00	-0.01

********** END OF LATEST ANALYSIS RESULT **********

1505. FINISH

TRIMBLE SOLUTIONS CORPORATION -- PAGE NO. 56

********** END OF THE STAAD.Pro RUN **********

**** DATE= FEB 29,2020 TIME= 18:10:55 ****


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*   For technical assistance on STAAD.Pro, please visit    *
*   http://www.bentley.com/en/support/                     *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2017 Bentley Systems, Inc.       *
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ATTACHMENT - 5

COMMENTS RESOLUTIONS SHEET

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-001787, 25 March 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 26 December 2019	Comment by	EPC Response DATE: 02 January 2020	Status	Remarks
			General Comments (cc)				
MD-216-3230-EG-CV-CAL-1167-A01	3230	1	Wrong, but..., on the safe side	Mr. Doron Golan - Bechtel	Load Combination for strength calculation we consider seismics load factor 1.6 for safe side purpose.	Closed	
MD-216-3230-EG-CV-CAL-1167-A01	3230	1	This document has not been issued on Aconex	Mr. Doron Golan - Bechtel	Structural input design drawing submitted through Aconex on date 13.03.2020	Closed	